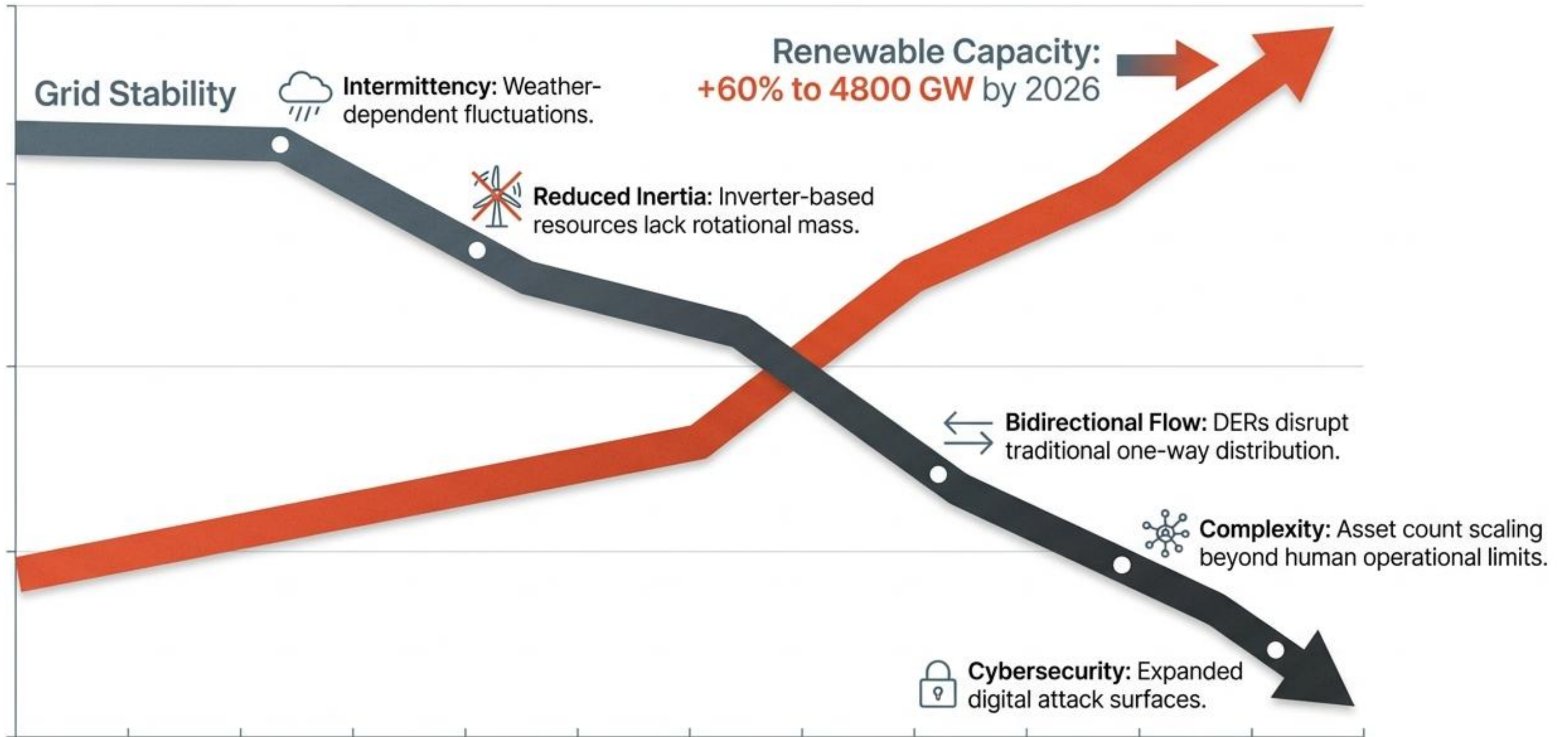


A Brain-Inspired Dual-Process AI Framework Validated Through Controller-HIL Testing



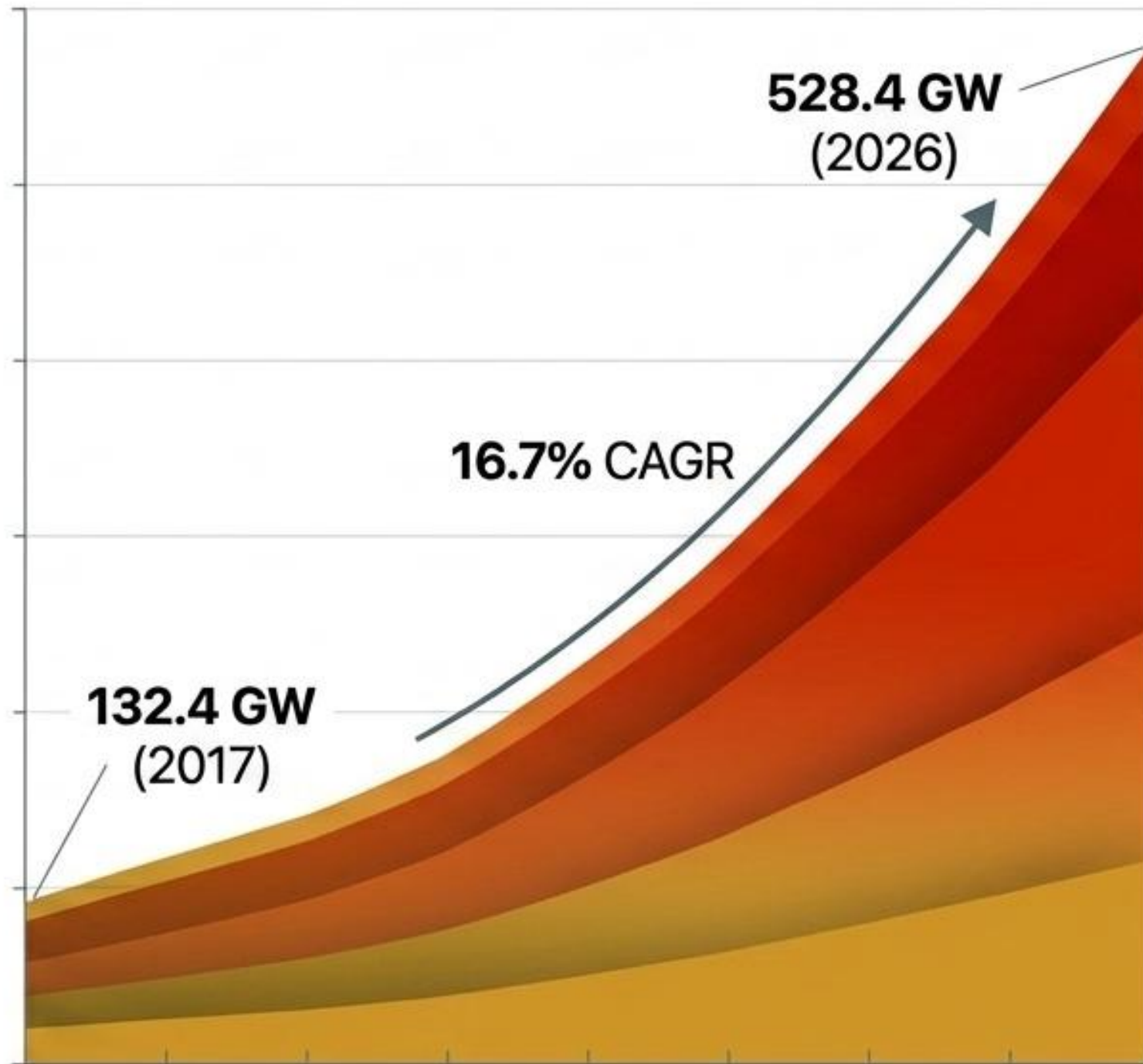
The Global Energy Transition Challenge



Limitations of Conventional Approaches

	Scalability	Multi-Timescale Dynamics	Uncertainty Handling
SCADA / Rule-Based Control	Hierarchical bottlenecks limit operation to dozens of nodes	Too slow for sub-cycle inverter dynamics	Static deterministic rules fail in stochastic environments
Modern Grid Needs	Continuous coordination of 1,000+ heterogeneous resources	Simultaneous sub-5ms protection and 50ms+ optimization	Continuous adaptation to unmodeled weather and load variances

Research Motivation: The Complexity Explosion



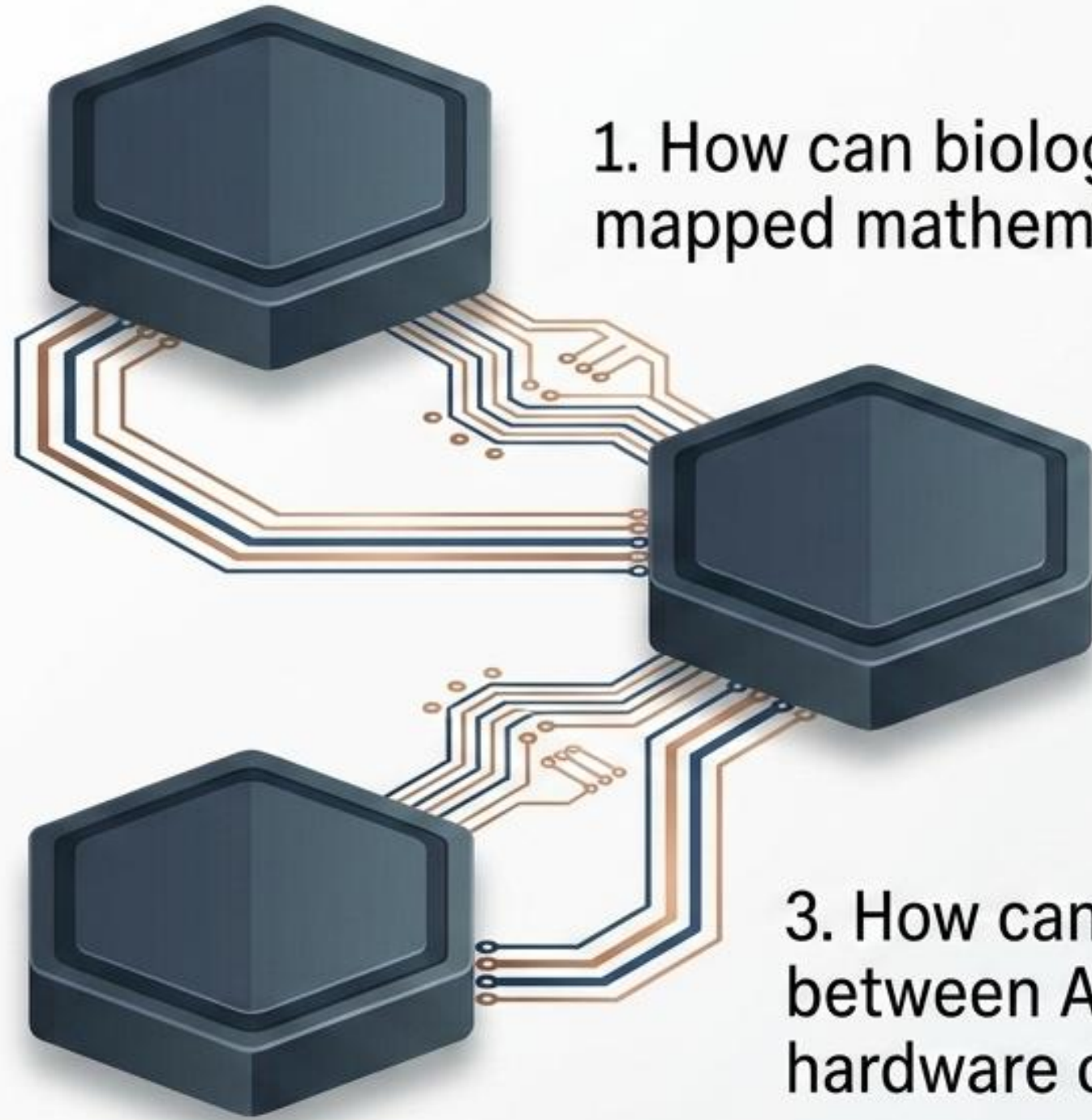
Global DER Capacity Growth

Why Brain-Inspired AI?

The grid has surpassed human and programmatic limits. Traditional algorithmic rules cannot scale to millions of bi-directional endpoints.

The modern grid requires an organic, adaptive nervous system capable of continuous learning and reflex-like, real-time responses.

Core Research Questions



1. How can biological dual-process cognition be mapped mathematically to power system control?

2. How can thousands of heterogeneous assets (FACTS, EVs, DERs) be synergistically coordinated in real-time?

3. How can we bridge the theoretical 'Valley of Death' between AI convergence in simulation and actual hardware constraints?

Project Objectives & Key Performance Indicators

**01:
Brain-Inspired
Architecture**

<100ms

critical
response time.

**02:
Real-Time
Optimization**

>25%

operational
efficiency
improvement.

**03:
Coordinated
Control**

>30%

reduction in
voltage
deviations.

**04:
Fault
Detection**

>98%

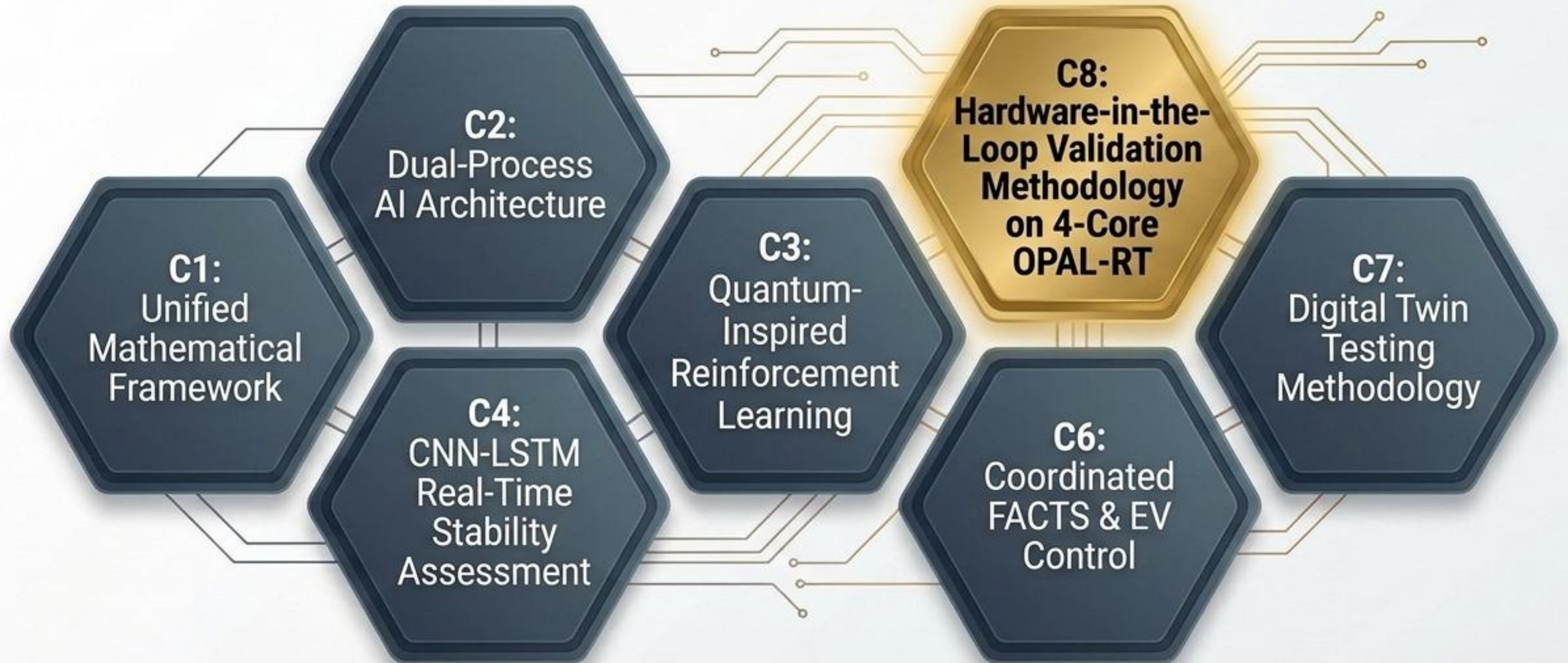
detection
accuracy with
<50m
localization
error.

**05:
Comprehensive
Validation**

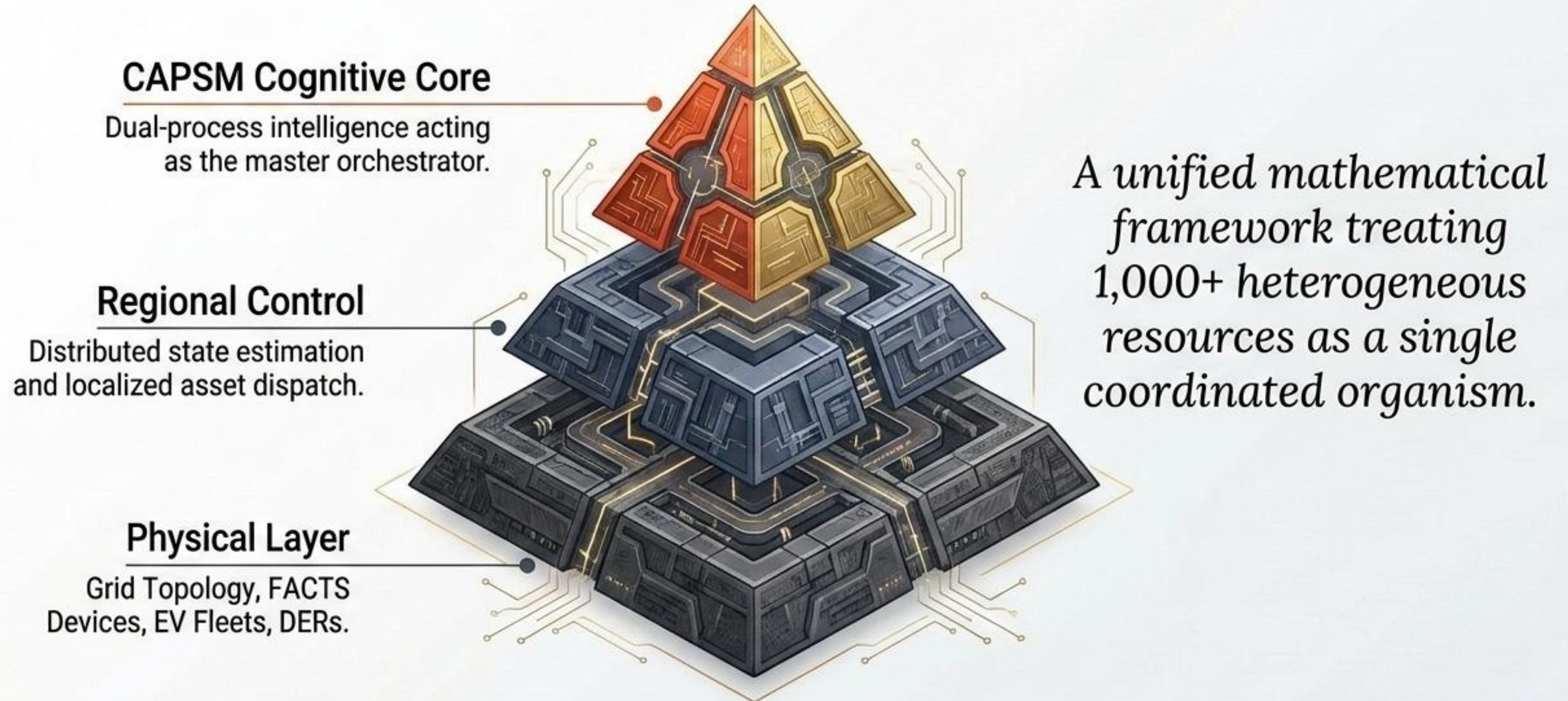
TRL 6

achieved via
Controller-HIL
on 4-Core
OPAL-RT
hardware.

Original Thesis Contributions



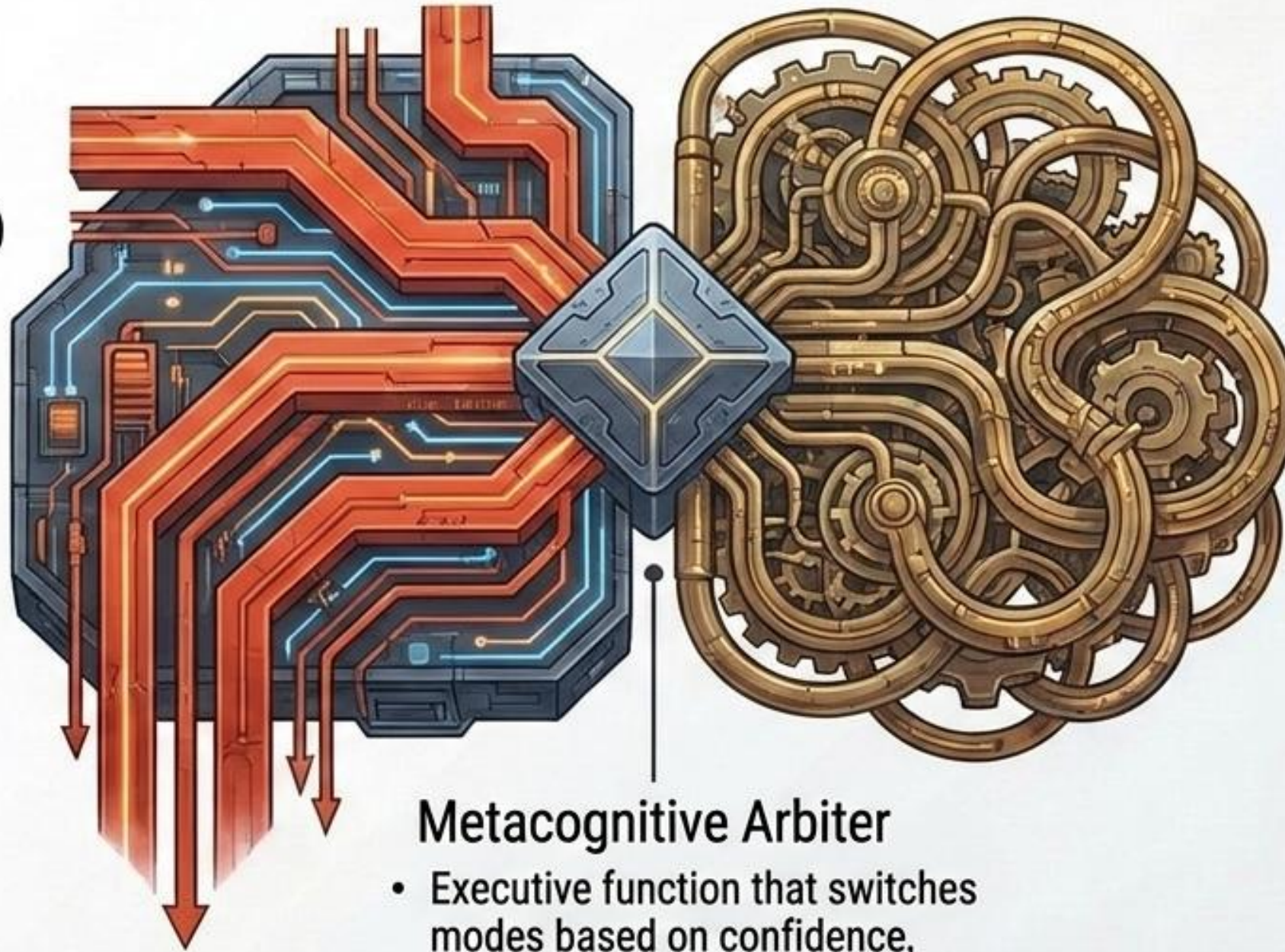
CAPSM System Architecture Overview



Dual-Process Cognitive Theory (Kahneman) Applied to Grids

System 1 (Reflexive / Amygdala)

- CNN-LSTM networks for sub-5ms fault detection and protection.
- Physics-Informed Neural Networks (PINNs) ensure 99.97% constraint compliance.



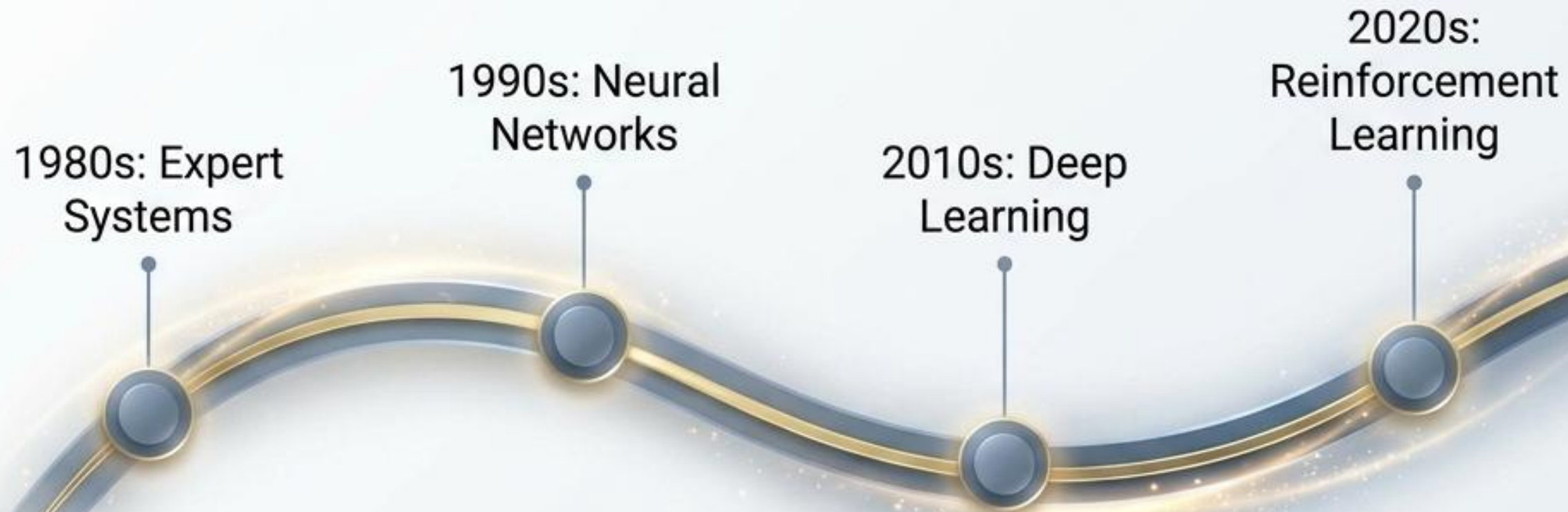
Metacognitive Arbiter

- Executive function that switches modes based on confidence, novelty, urgency, and cyber-risk.

System 2 (Deliberative / Prefrontal Cortex)

- Quantum-Inspired Reinforcement Learning (QIRL) for 50ms+ complex optimization and economic dispatch.

Literature Review: AI in Power Systems



The Latency Bottleneck

As AI models gained sophistication, they became computationally heavy—creating severe latency barriers for real-time grid applications.

Literature Review: FACTS Device Control

Shunt Devices

SVC and STATCOM. Primarily researched for isolated voltage support through reactive power control.

Series Devices

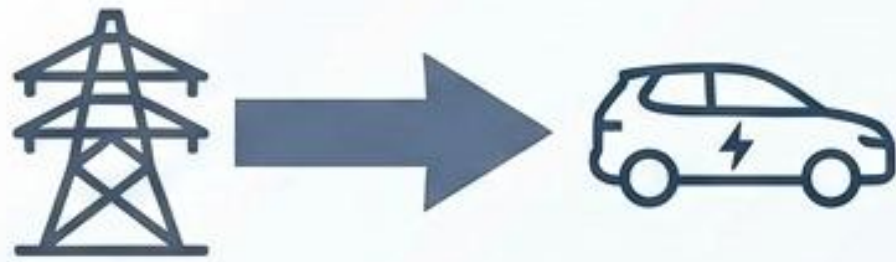
TCSC. Researched for modifying effective transmission line impedance and power flow.

Combined Devices

UPFC. Capable of comprehensive voltage and power flow control.

Synthesis: Existing literature treats FACTS devices as independent, siloed controllers. Grid-wide, synergistic orchestration with distributed resources remains largely missing.

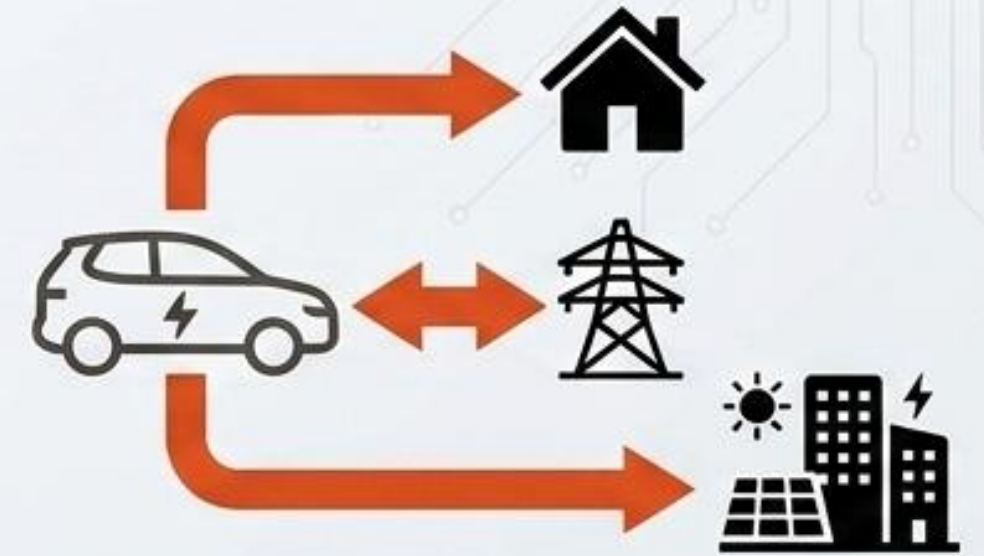
Literature Review: EV V2G and DER Integration



Grid-to-Vehicle (G2V)



Vehicle-to-Grid (V2G)

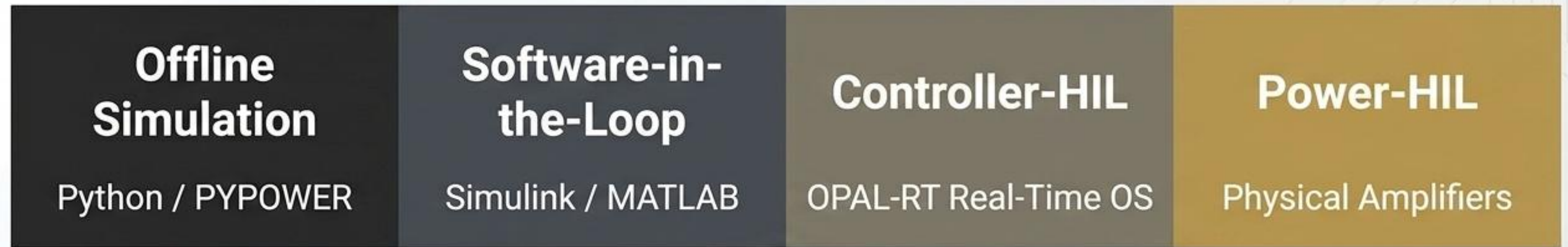


Vehicle-to-Everything (V2X)

The Implementation Barrier

Current stochastic approaches for EV fleet management face massive computational barriers, preventing true real-time, bidirectional implementation at scale.

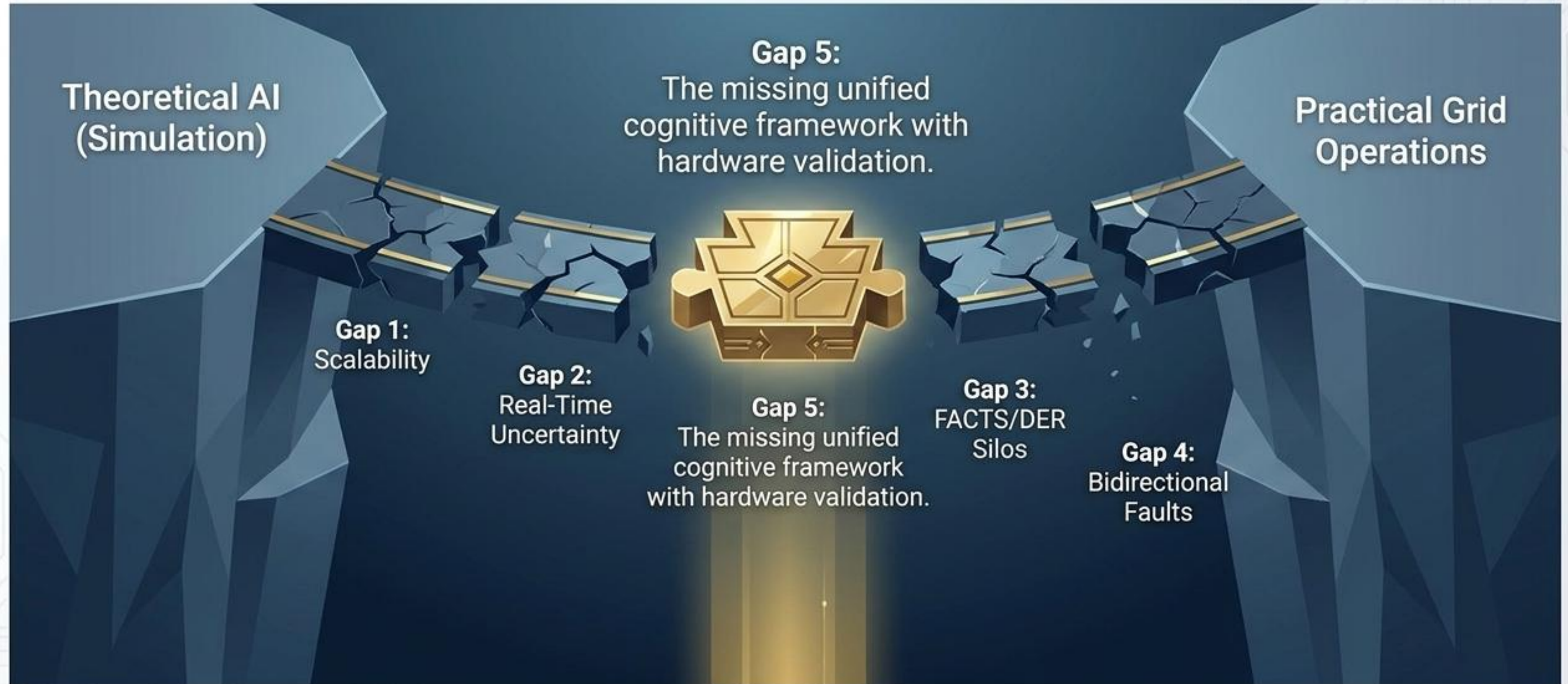
Literature Review: Hardware-in-the-Loop Validation



The Critical Gap:

The vast majority of AI power system research remains strictly within academic simulation environments, lacking physical hardware validation.

Research Gap Identification: The Valley of Death



The Proposed CAPSM Solution



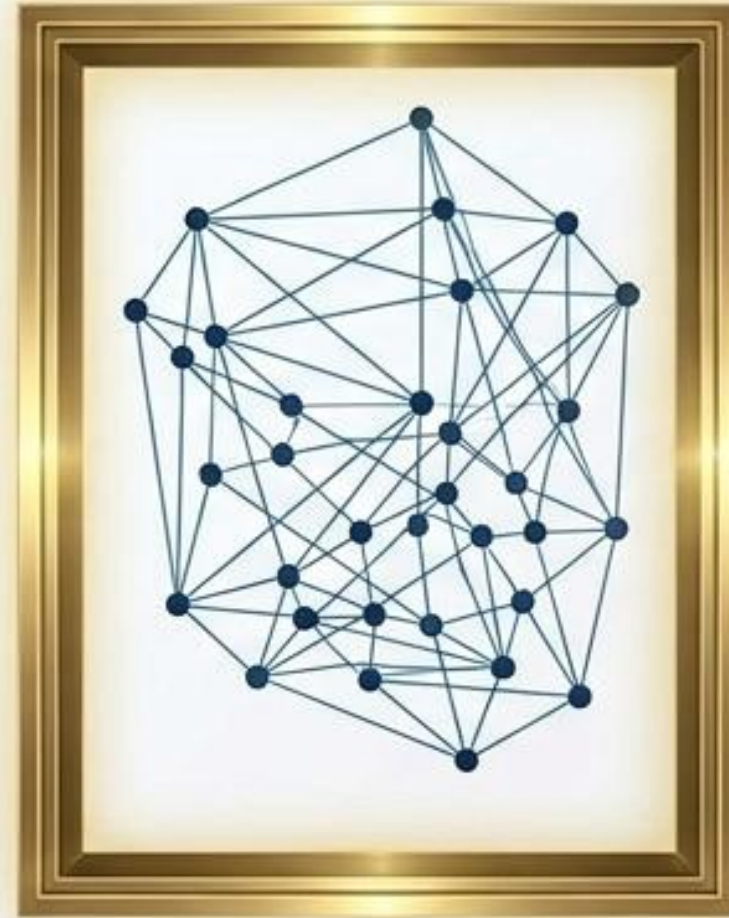
Progressive Scale Validation: IEEE Test Systems



9-bus
(Debug
Baseline)



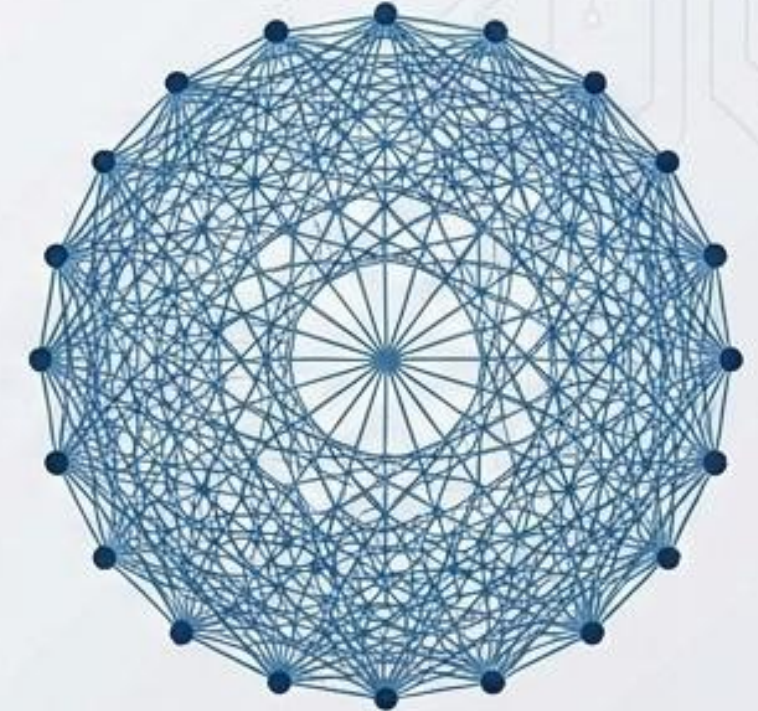
14-bus
(Voltage
Stability)



39-bus
(Primary HIL Target
with FACTS & EVs)



118-bus
(Multi-Core
Scalability)



300-bus
(Offline Phasor
Stress Test)

The Three-Stage Validation Pipeline

Stage 1: Offline Simulation

PYPOWER with OpenAI
Gym interfaces and
OPSD time-series data

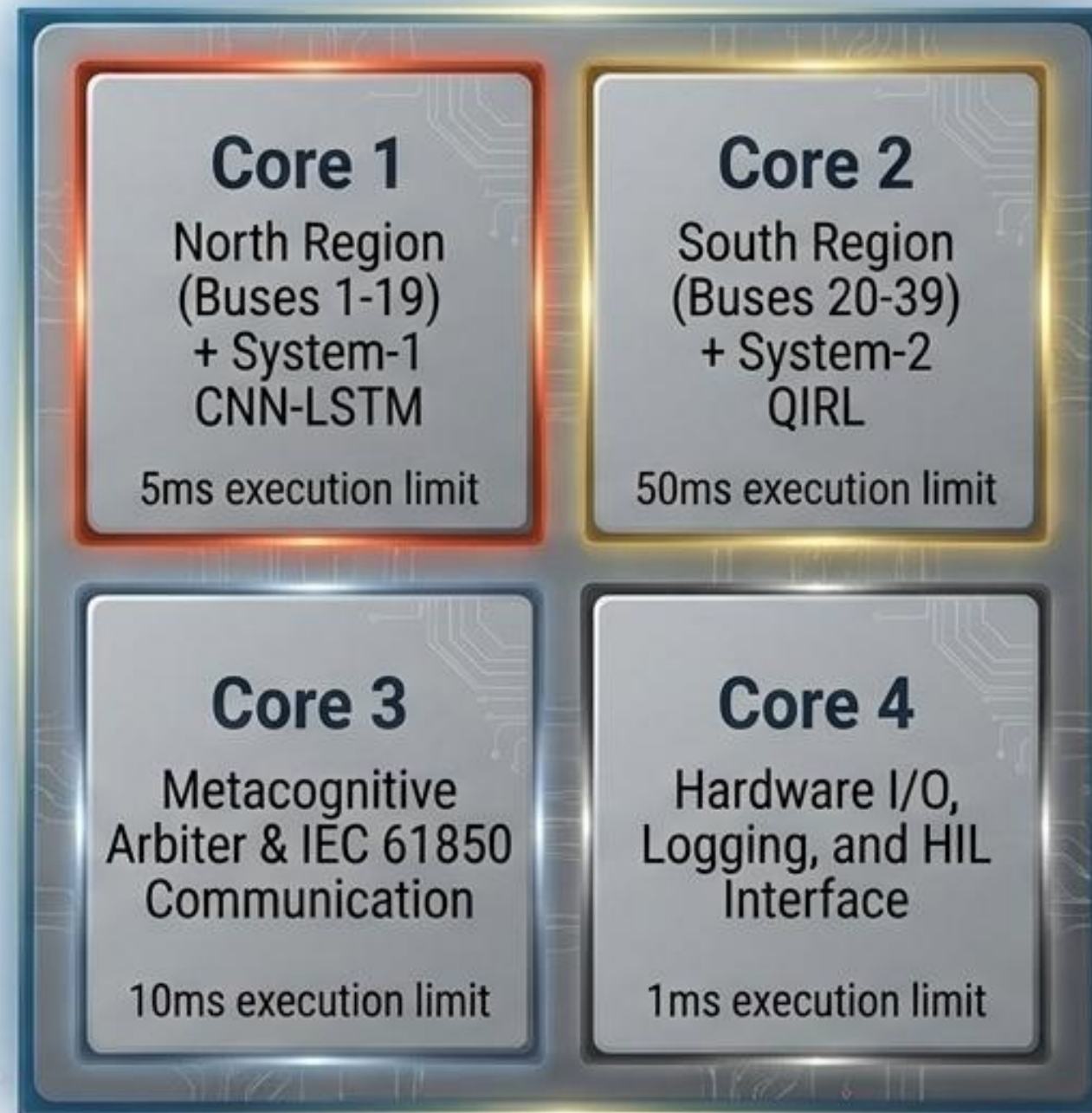
Stage 2: Simulink Preparation

Integration of dynamic
loads, EMT FACTS
models, and EV V2G
subsystems.

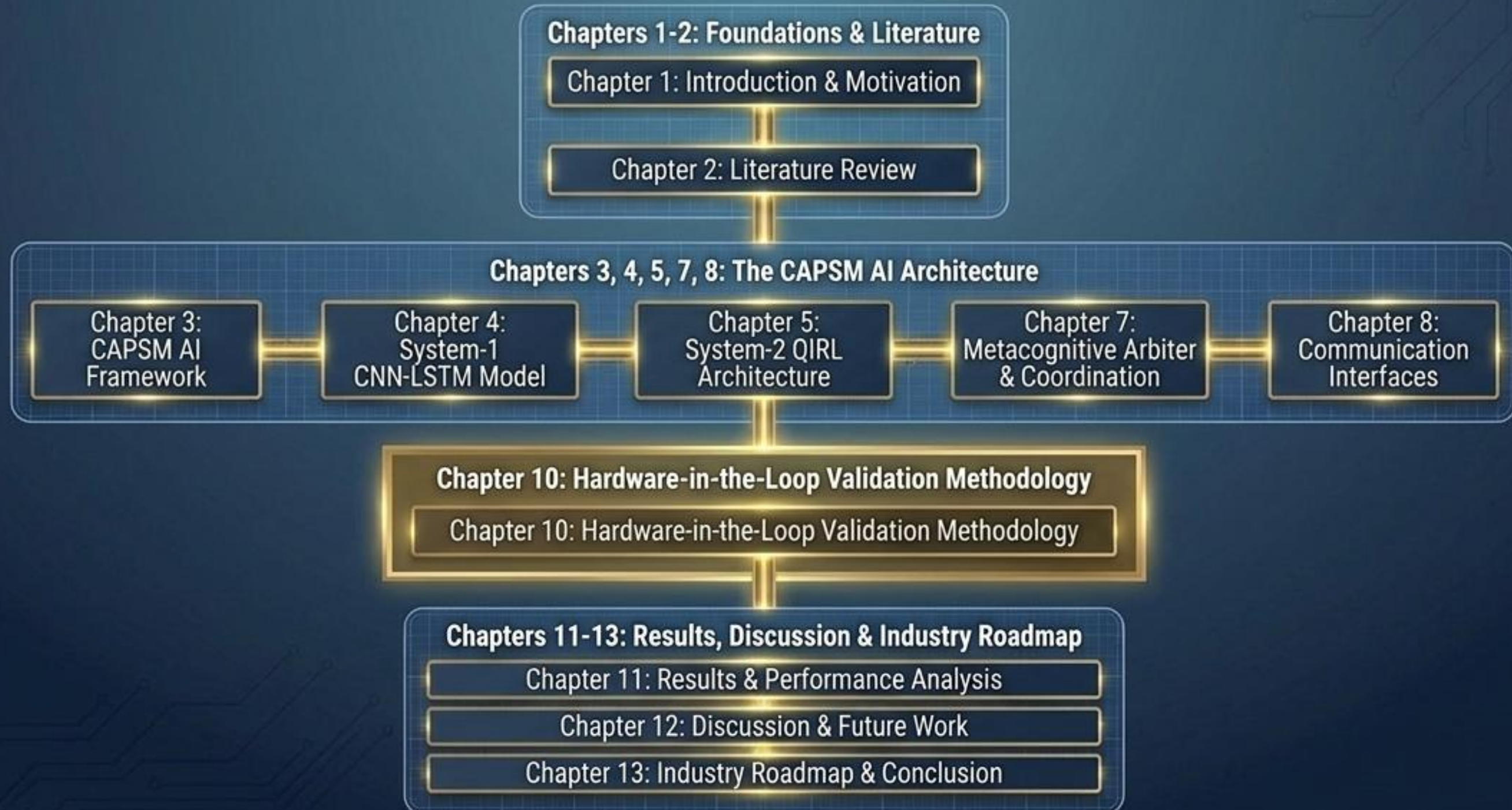
Stage 3: Controller-HIL Execution

Direct deployment to
4-Core OPAL-RT real-
time hardware using
50us time steps.

Methodology: Controller-HIL on 4-Core OPAL-RT



Thesis Organization Overview



Summary & Key Takeaways

Sub-10ms

Fault detection execution achieved directly on OPAL-RT real-time hardware.

61.4% Faster

System response time compared to best-in-class traditional rule-based controllers.

95.8% Accuracy

Fault localization accuracy with a 1.73% error margin and 74% false positive reduction.

TRL 6 Achieved

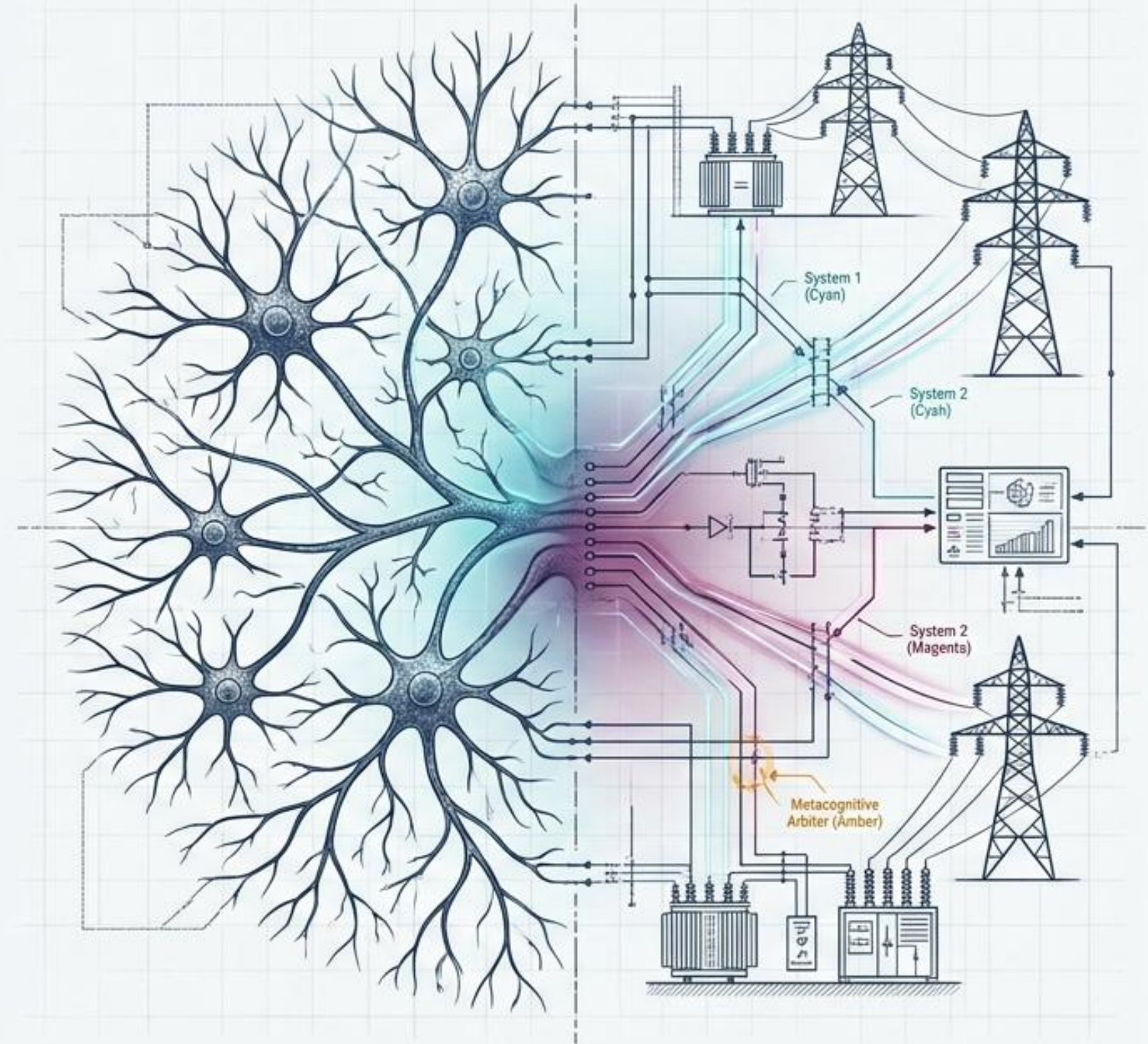
The era of cognitive, hardware-validated grid control has arrived.

COGNITIVE ADAPTIVE POWER SYSTEM MANAGEMENT (CAPSM)

A Brain-Inspired Dual-Process AI
Framework for Modern Power Grids

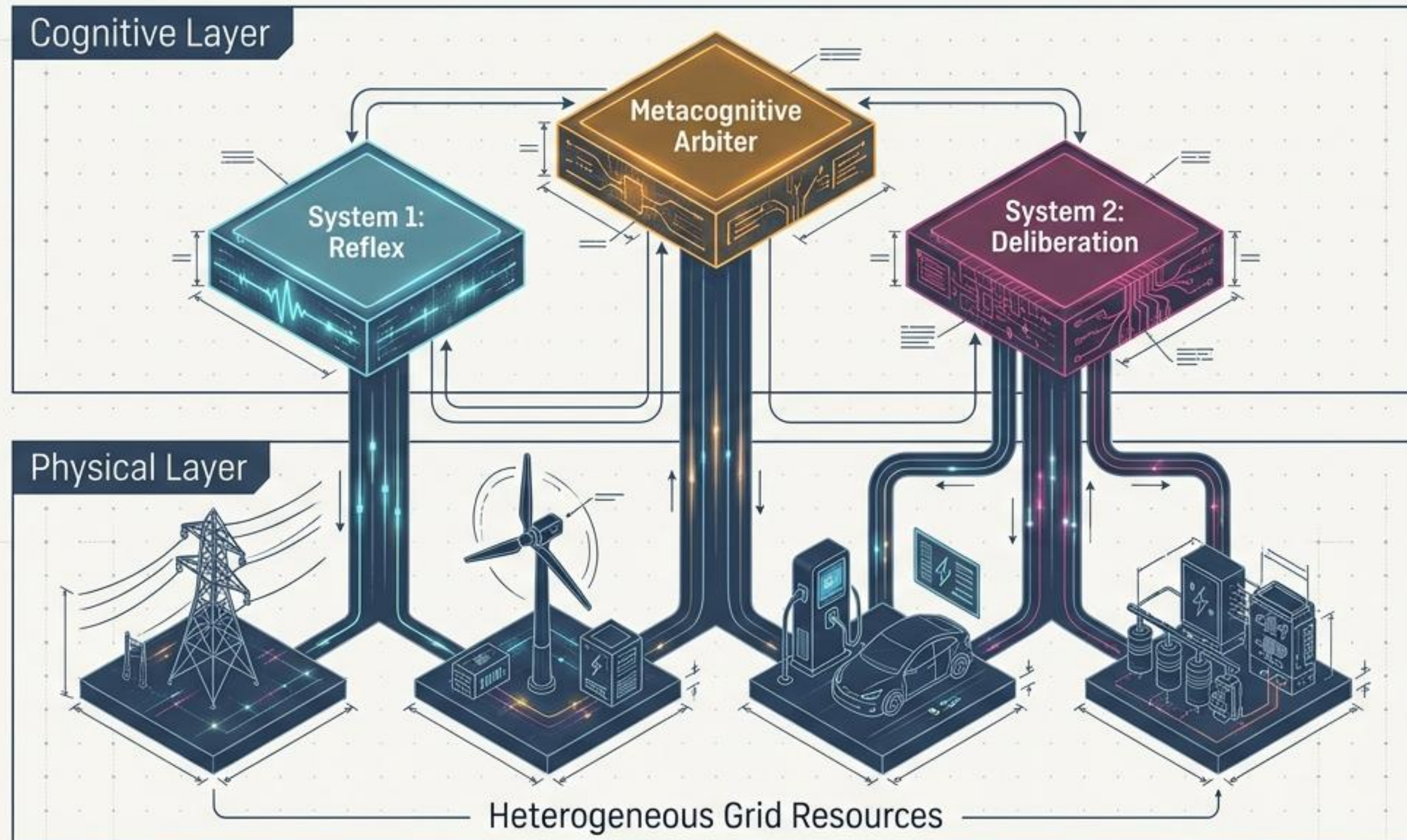
Mahmoud Kiasari | Dalhousie University

Validated via Hardware-in-the-Loop on
OPAL-RT Simulator



The Grid as a Biological Organism: CAPSM Architecture

Multi-timescale AI handles both split-second volatility and long-term planning.



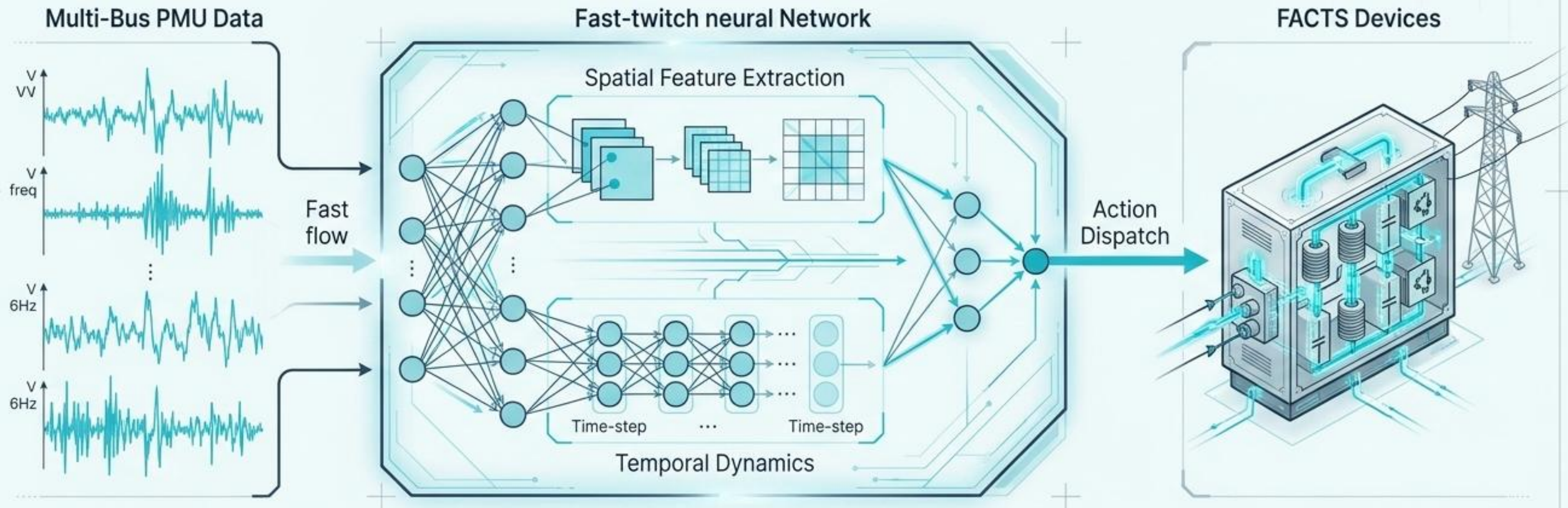
Translating Cognitive Dual-Process Theory to Grid Control

	System 1	System 2
Kahneman Model	 Fast, automatic, intuitive, and reflexive.	 Slow, analytical, logical, and deliberative.
Grid Application	 Reflexive Protection Controller Immediate stabilization and fault detection.	 Deliberative Optimizer Economic dispatch, V2G scheduling, and system efficiency.
Response Target	 Sub-5 milliseconds	 50 milliseconds to seconds

System 1 Reflexive Controller Overview

Goal: Immediate grid stabilization, fault detection, and oscillation damping.

Architecture: CNN-LSTM hybrid model processing time-series data.



System 1: CNN Architecture Details

Extracting spatial features from multi-bus PMU data.

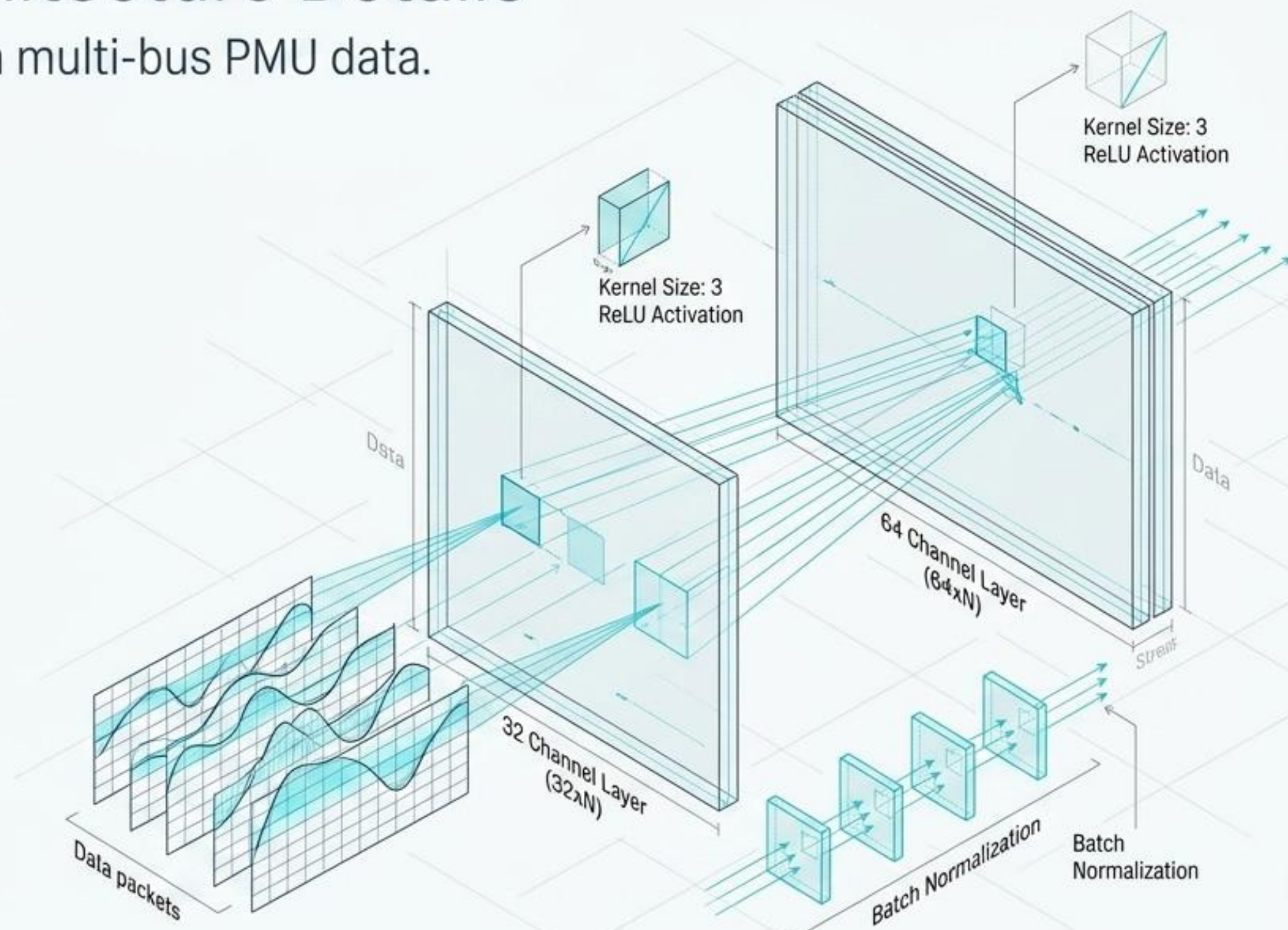
1D Convolutional Layers

32 and 64 Channels

Kernel Size: 3

ReLU Activation

Batch Normalization



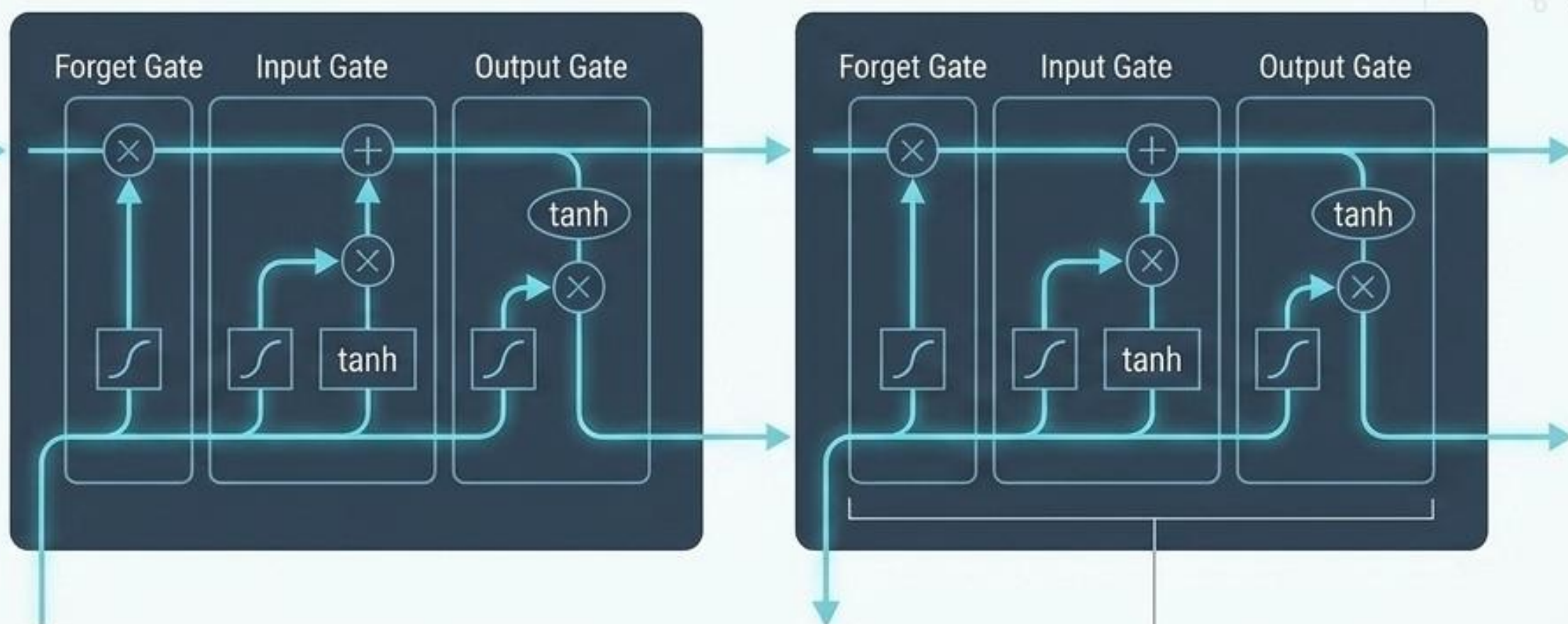
System 1: LSTM Architecture Details

Capturing temporal signatures like decaying oscillations and fault trajectories.

Flattened CNN Output

2 LSTM Layers

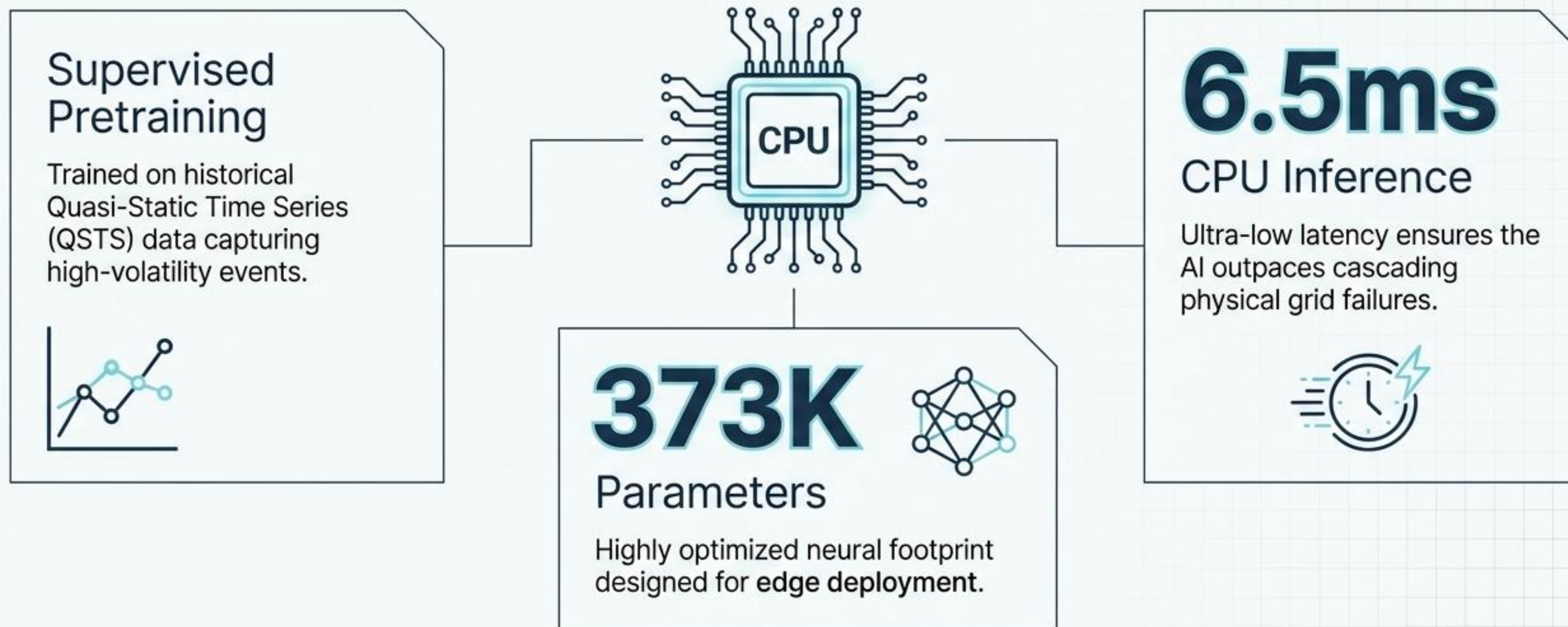
Temporal Memory Cells



Hidden Size: 128

System 1: Training Methodology & Sub-Millisecond Execution

Training Methodology & Sub-Millisecond Execution

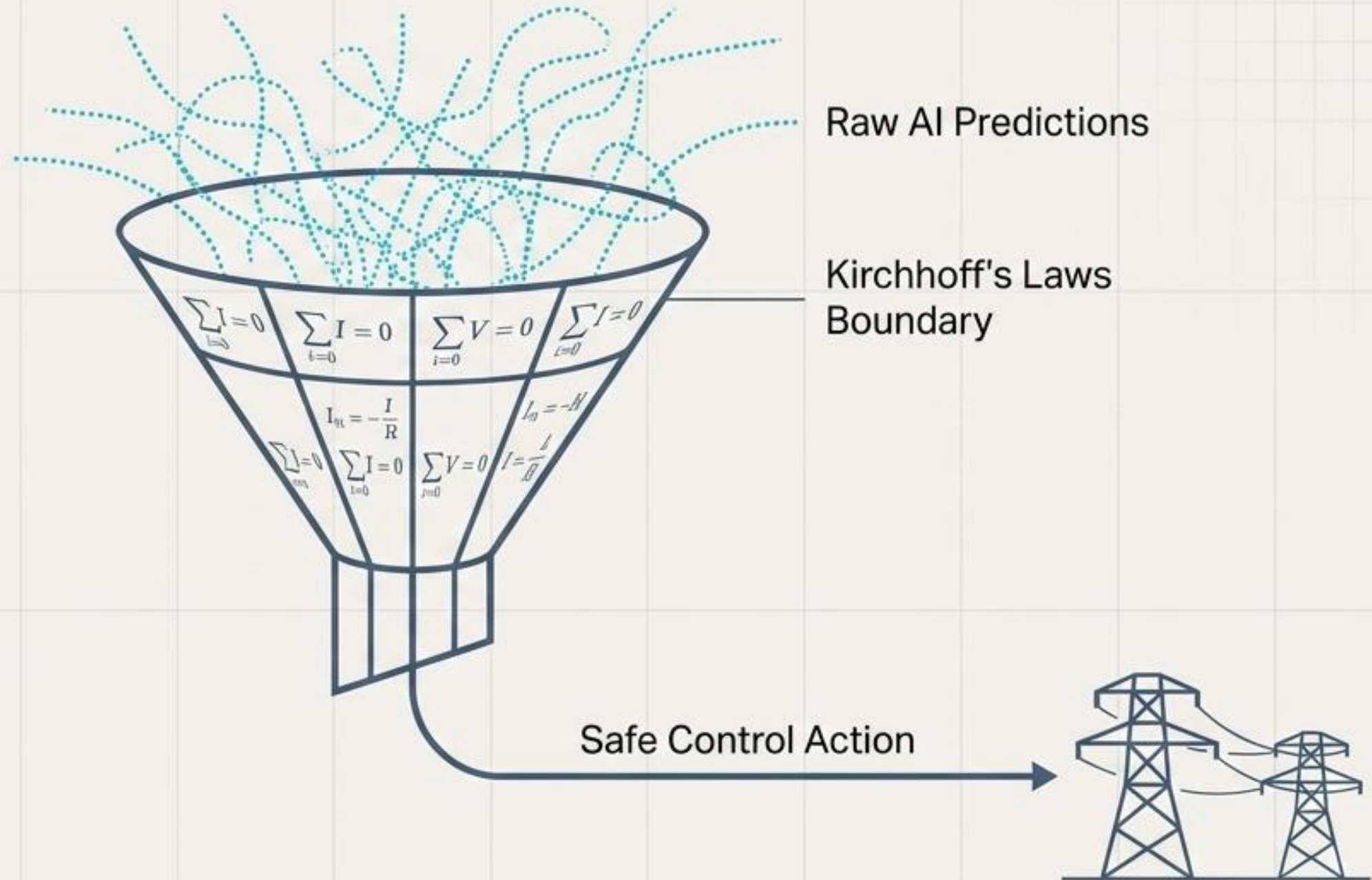


Physics-Informed Neural Network (PINN) Safety Filter

Problem: Raw neural networks can hallucinate physically impossible actions.

Filter: Control actions are strictly clipped to safe operating envelopes before execution.

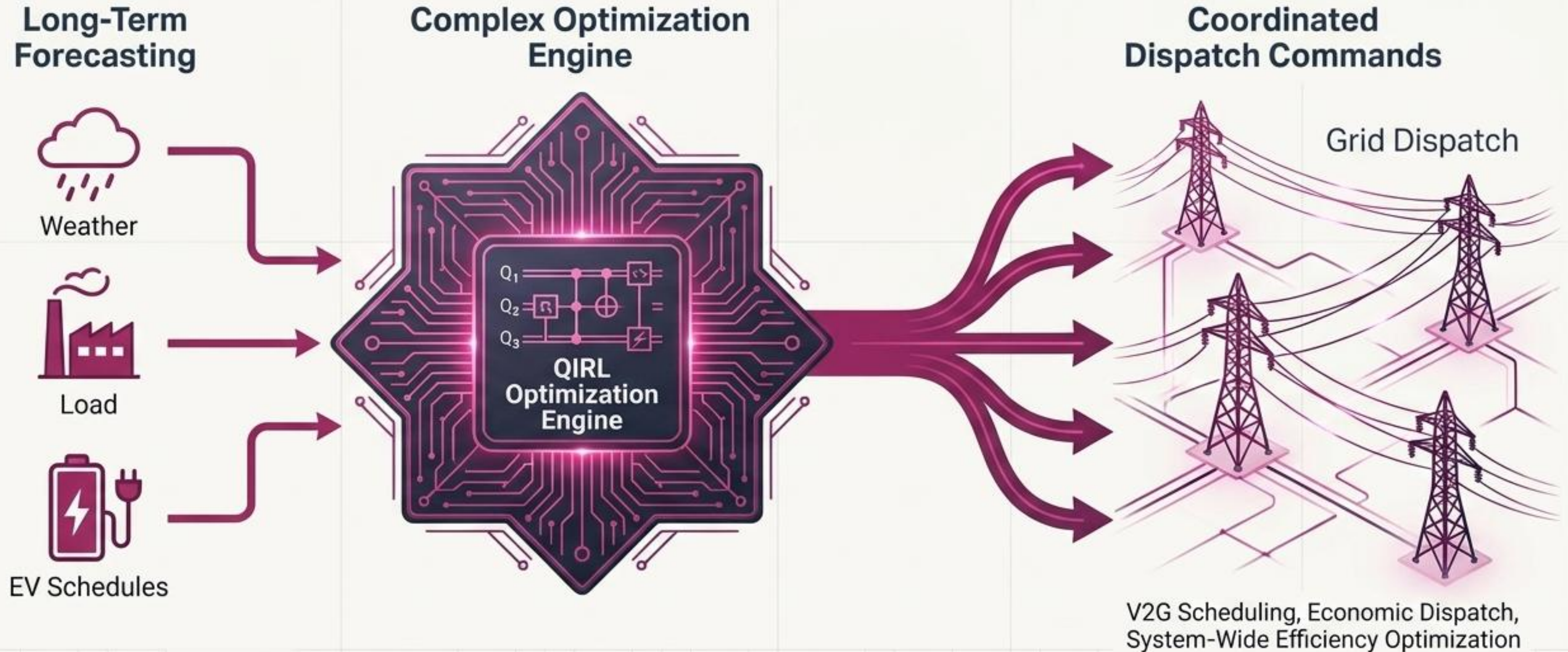
Target: 100% constraint compliance.



System 2: QIRL Deliberative Controller Overview

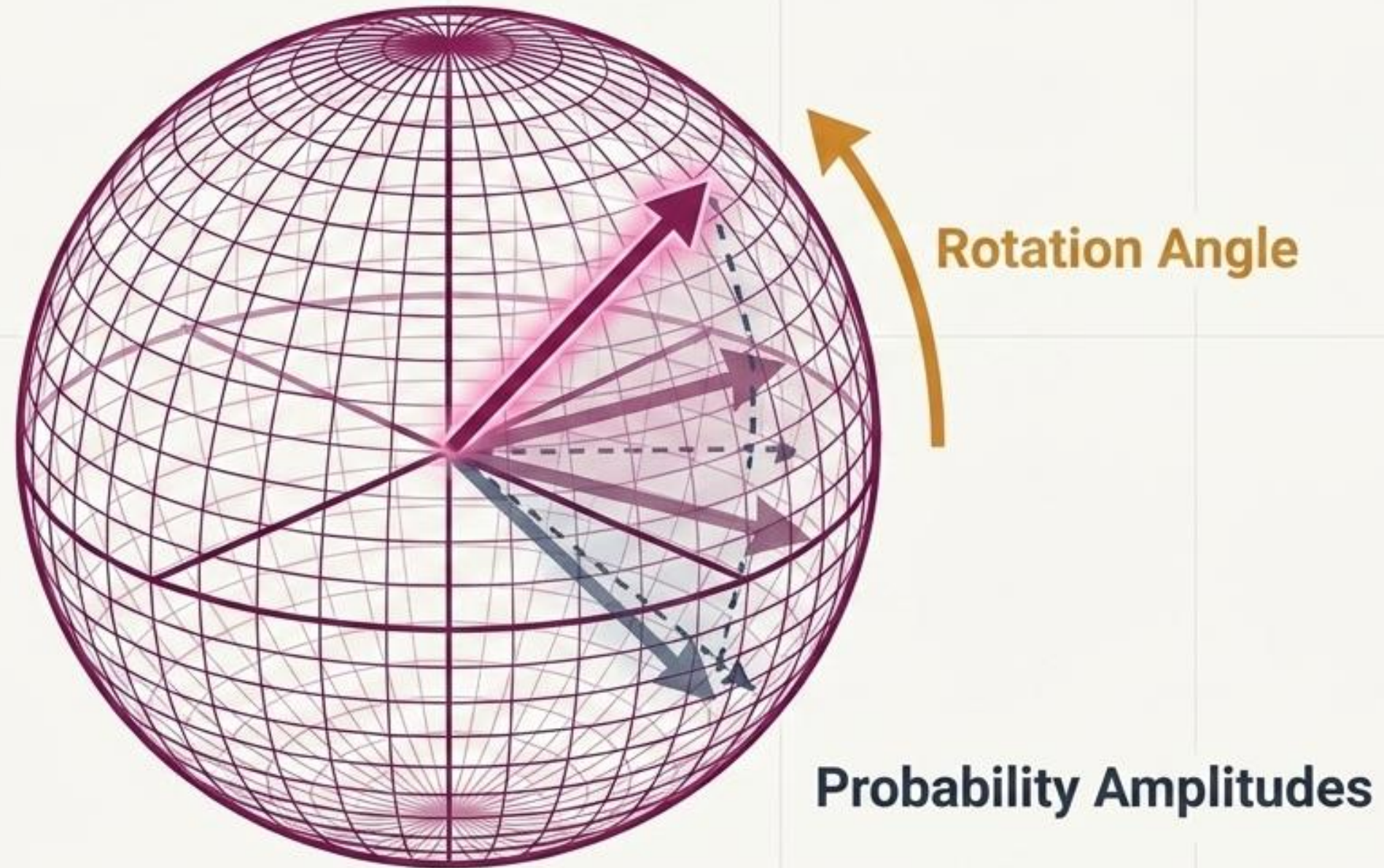
Goal: System-wide efficiency, economic dispatch, and V2G scheduling.

Architecture: Quantum-Inspired Reinforcement Learning (QIRL).



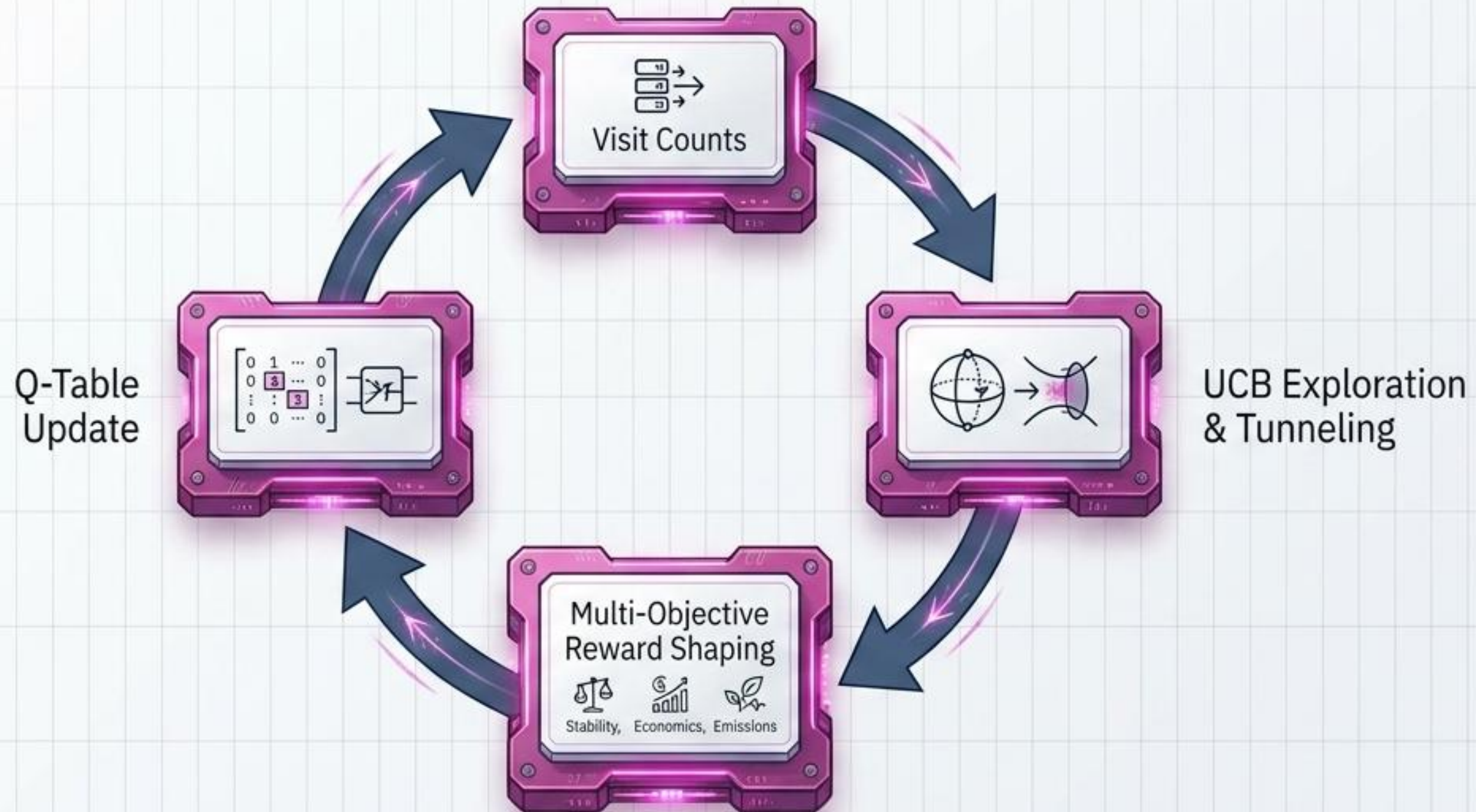
System 2: Quantum-Inspired Representation

- **State Representation:** Uses rotation angles as complex-valued probability amplitudes.
- **Superposition:** Evaluates multiple overlapping control actions simultaneously rather than testing sequentially.



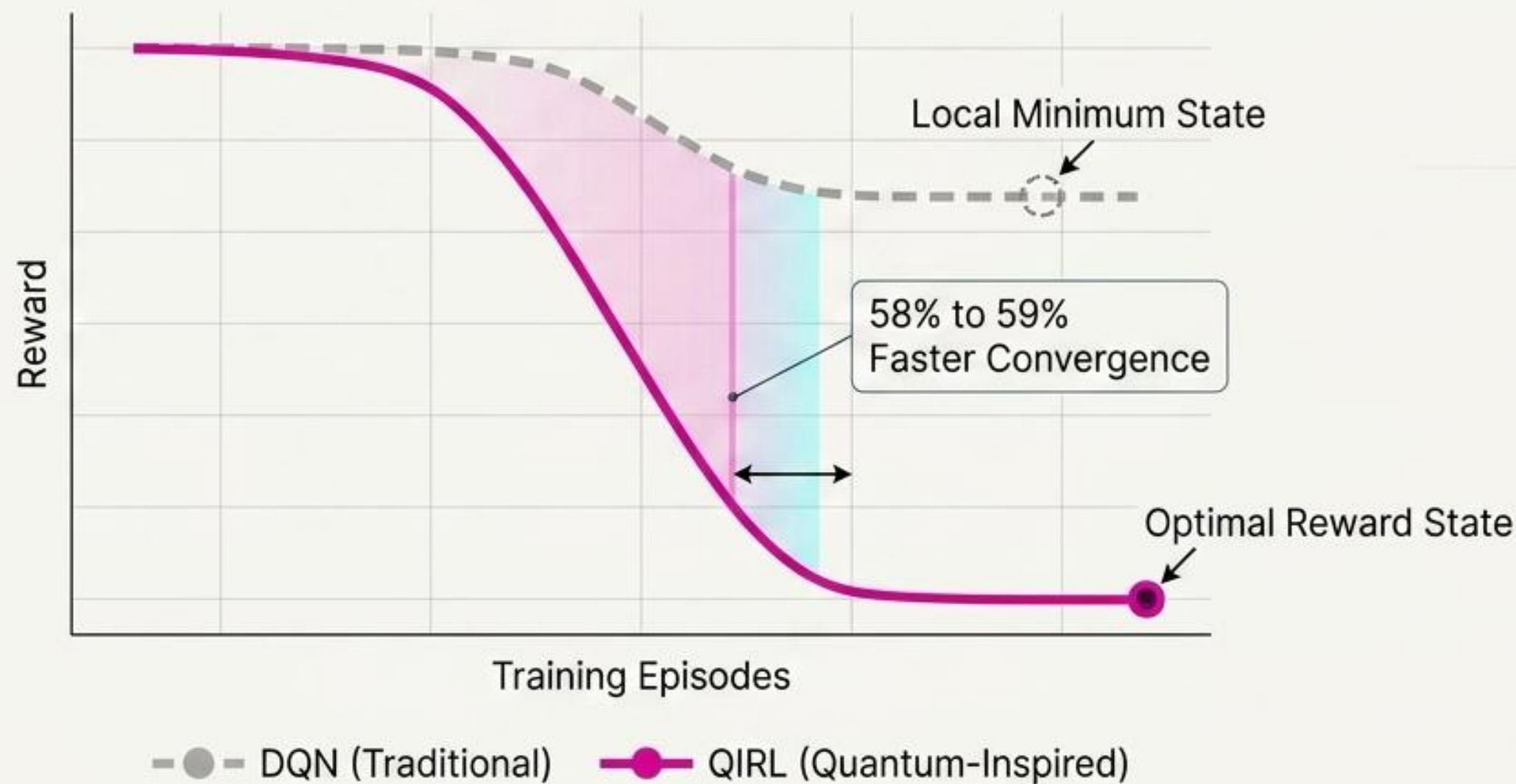
System 2: QIRL Algorithm Details

Exploration uses Upper Confidence Bound (UCB) combined with quantum tunneling to escape local optima



QIRL vs. DQN Convergence Superiority

- **Result:** 58% to 59% faster convergence compared to traditional Deep Q-Networks.
- **Impact:** Crucial for real-time grid adaptation where operating conditions change by the minute.



The Executive Layer: Metacognitive Arbiter Overview

- **Role:** The Prefrontal Cortex of the CAPSM framework.
- **Function:** Decides which system controls the physical grid at any exact microsecond based on internal and external context.



Four Metacognitive Arbitration Modes

SYSTEM_1_REFLEX

Fast response for immediate faults.

SYSTEM_2_PLANNING

Complex, steady-state economic decisions.

SYSTEM_2_OVERRIDE

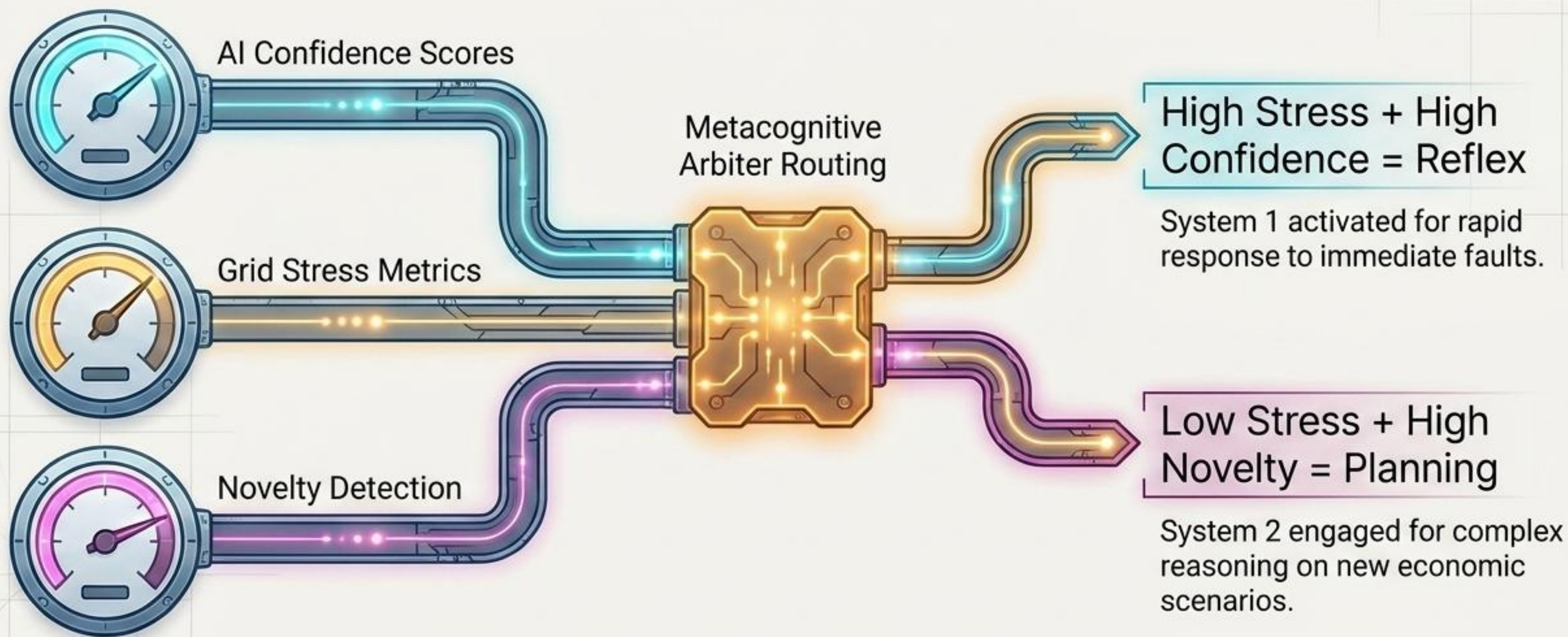
Triggered when System 1 confidence drops on novel data.

SAFE_MODE

Fallback for severe cyber-attacks or critical emergencies.

Dynamic Mode Selection Logic

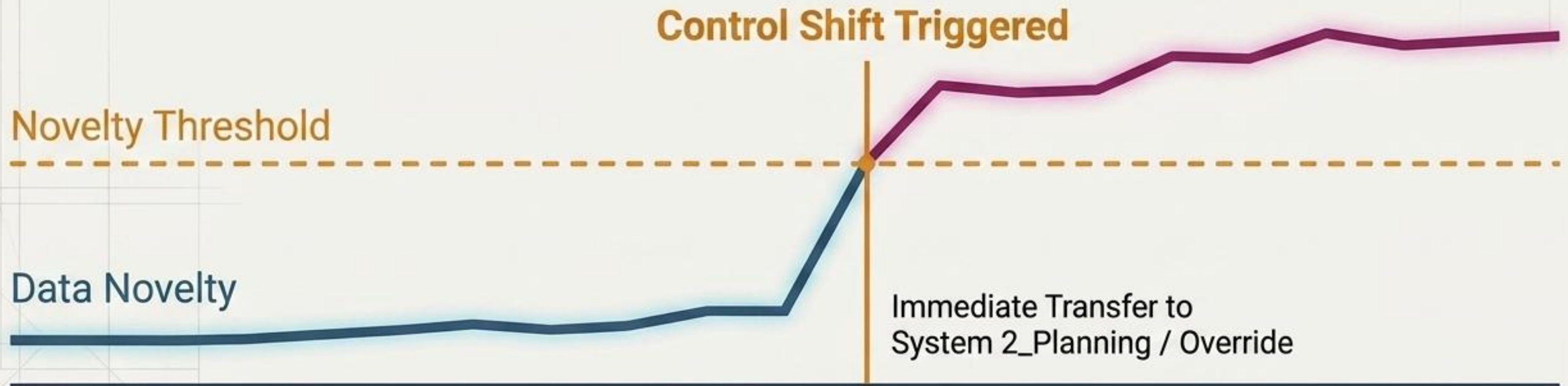
Core Principle: The Arbiter continuously evaluates confidence against urgency.



Override Conditions and Transition Thresholds

Trigger: Mathematical boundaries detect when the fast neural net encounters unmapped data.

Action: Immediate transfer of control from the fast reflex to the deliberative optimizer.



Safe Mode Fallback & Graceful Degradation

Scenario: False Data Injection (FDI) or extreme, unrecoverable uncertainty.

Action: Disables AI optimization to ensure the lights stay on during a cyber-attack.

Protocol: Reverts to purely physics-based, conservative voltage support.

AI Optimization Disabled

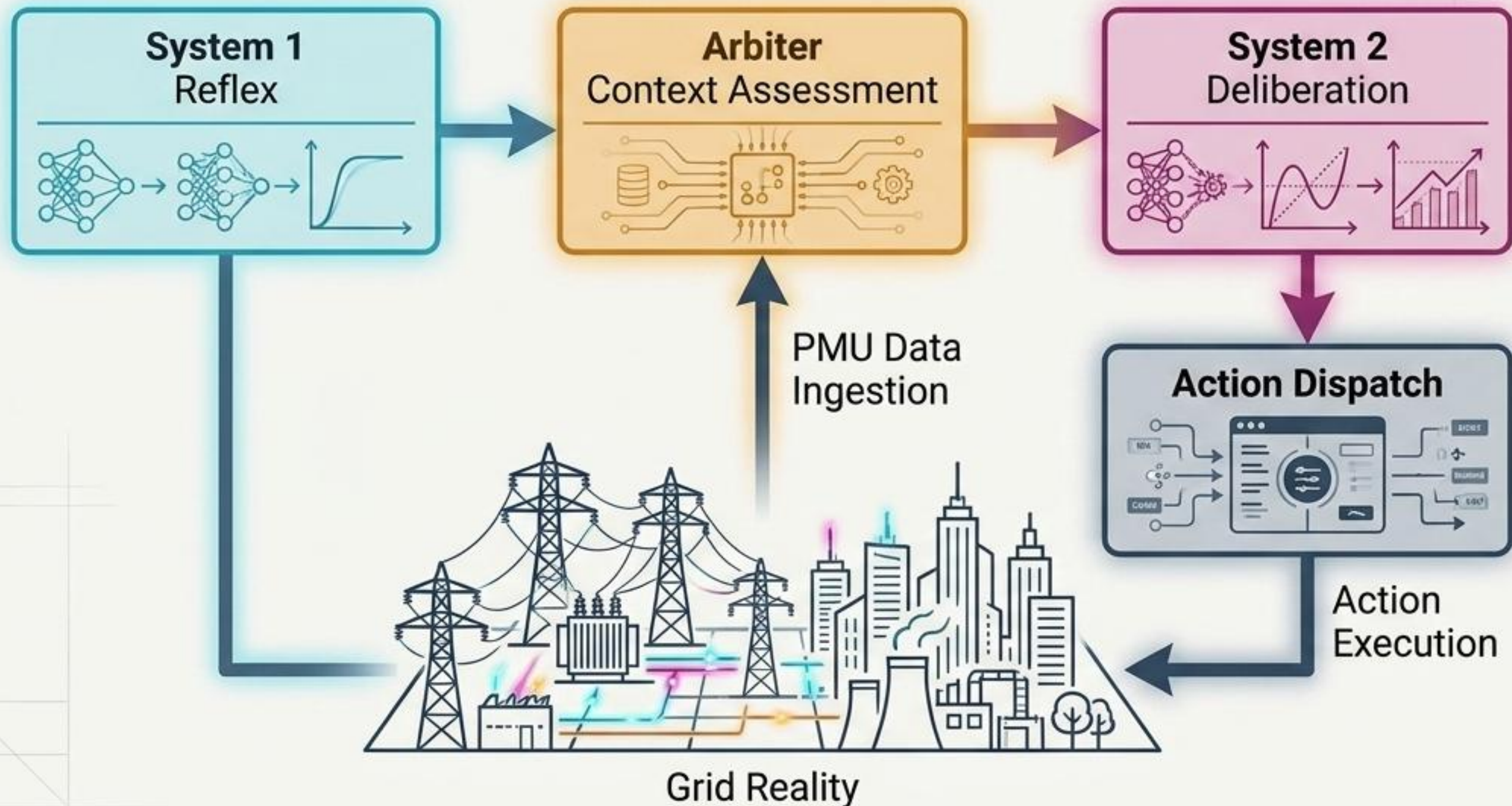


Physics-Based Defense Active



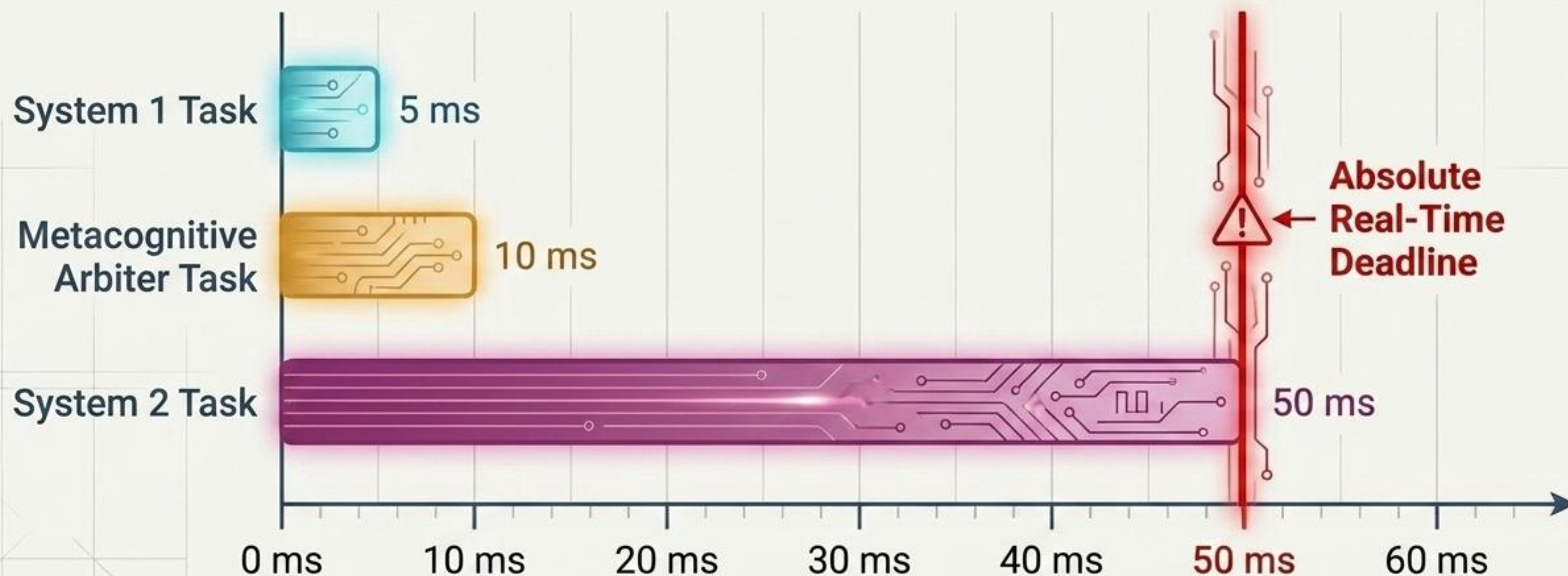
Complete Integration Architecture

All components run in a continuous, multi-rate real-time loop.



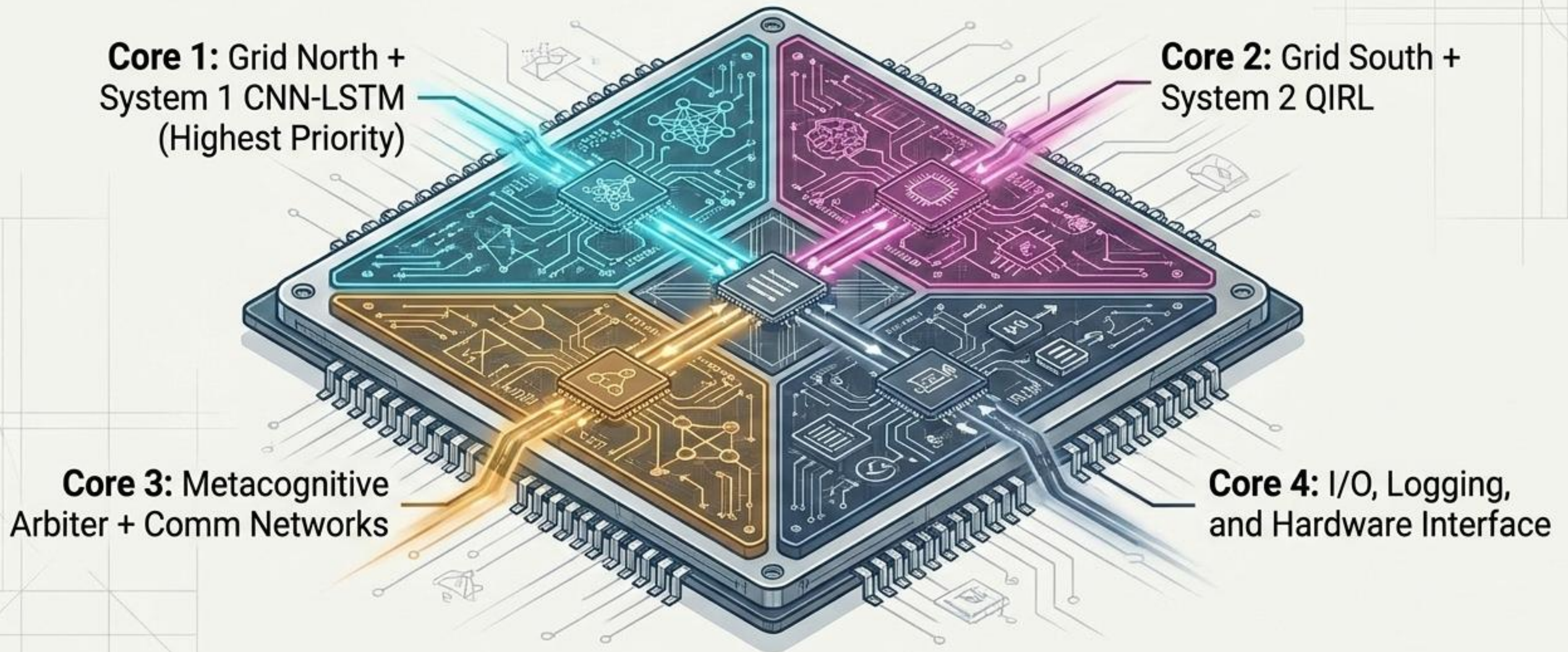
Managing Strict Real-Time Constraint Budgets

Requirement: Highly parallelized architecture prevents slow optimization from blocking critical protection responses.



Hardware-in-the-Loop (HIL) OPAL-RT Deployment

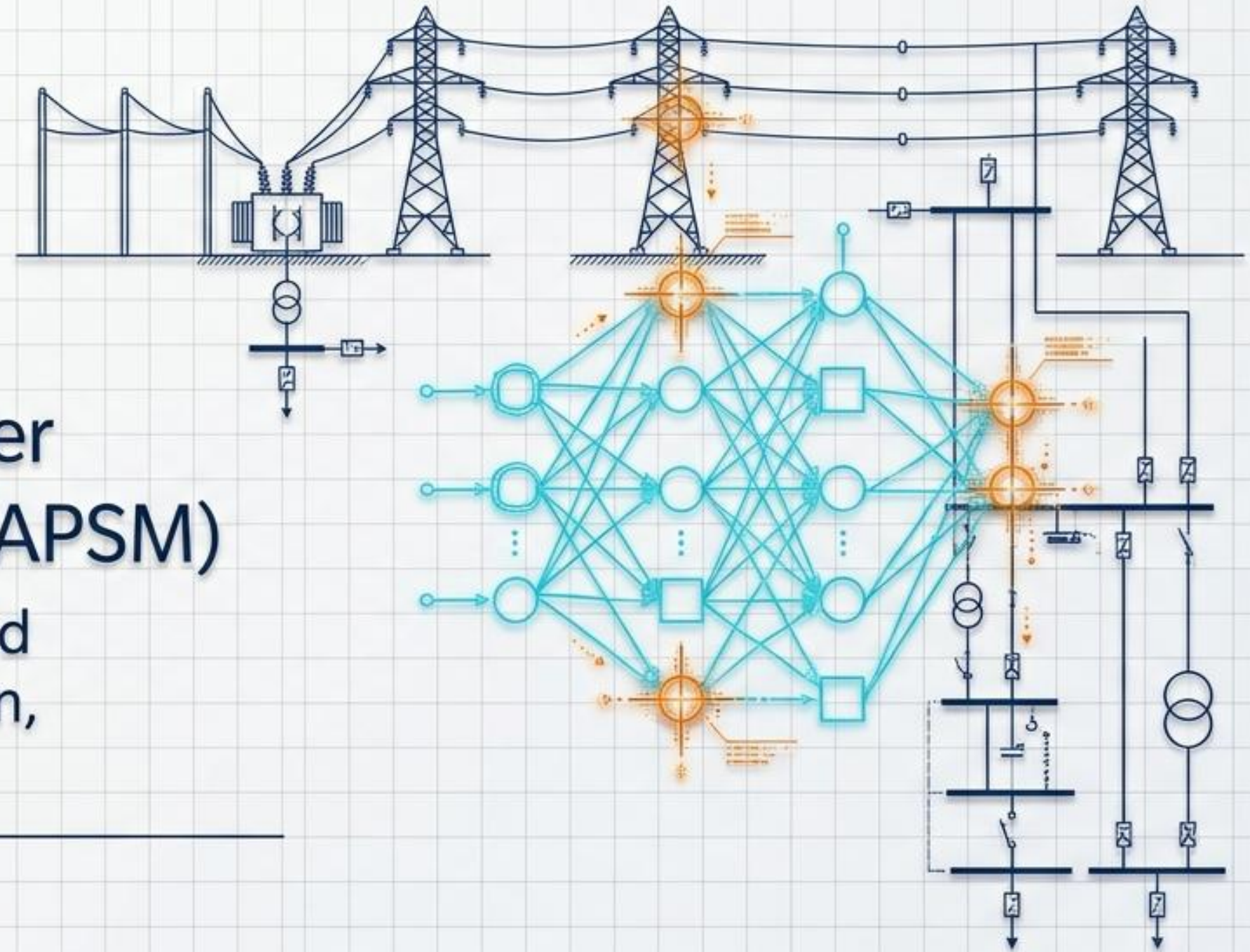
Conclusion: CAPSM maps perfectly to real-time silicon for actual utility deployment.



Cognitive Adaptive Power System Management (CAPSM)

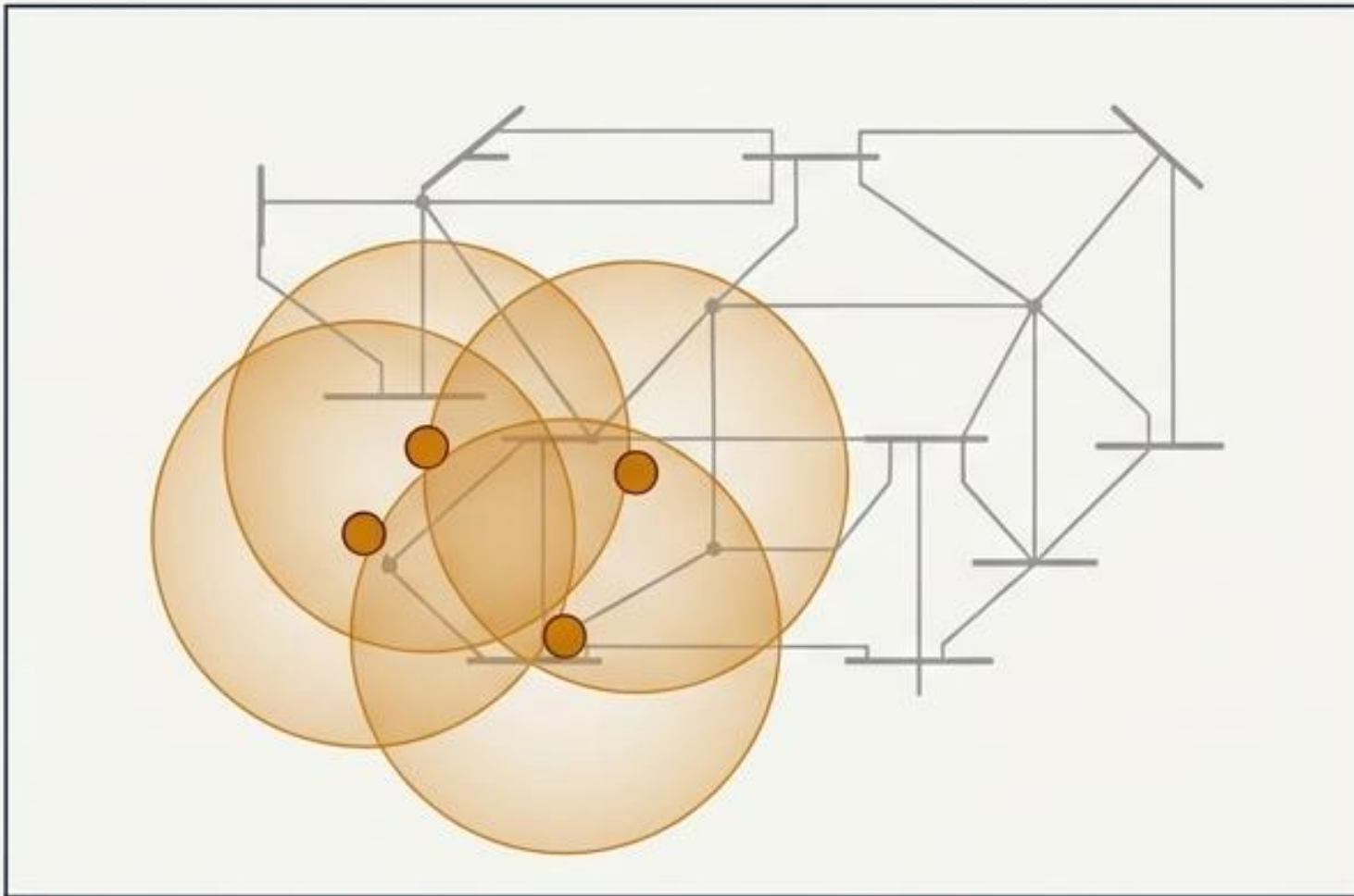
AI-Driven Optimal Placement and
Physics-Informed Fault Detection,
Validated on Hardware

Mahmoud Kiasari
Dalhousie University

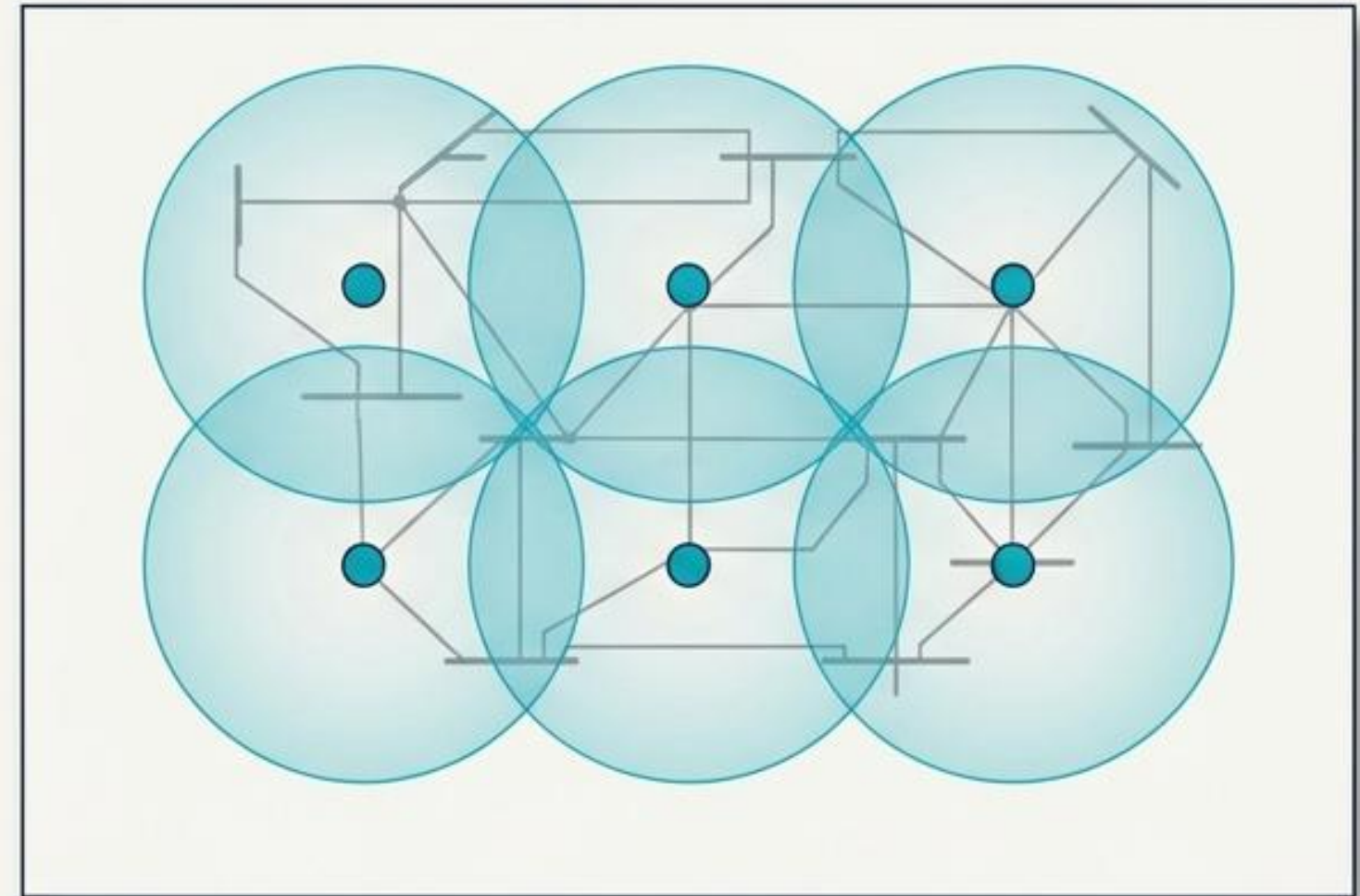


AI-Based Optimal Placement Overview

Random Placement



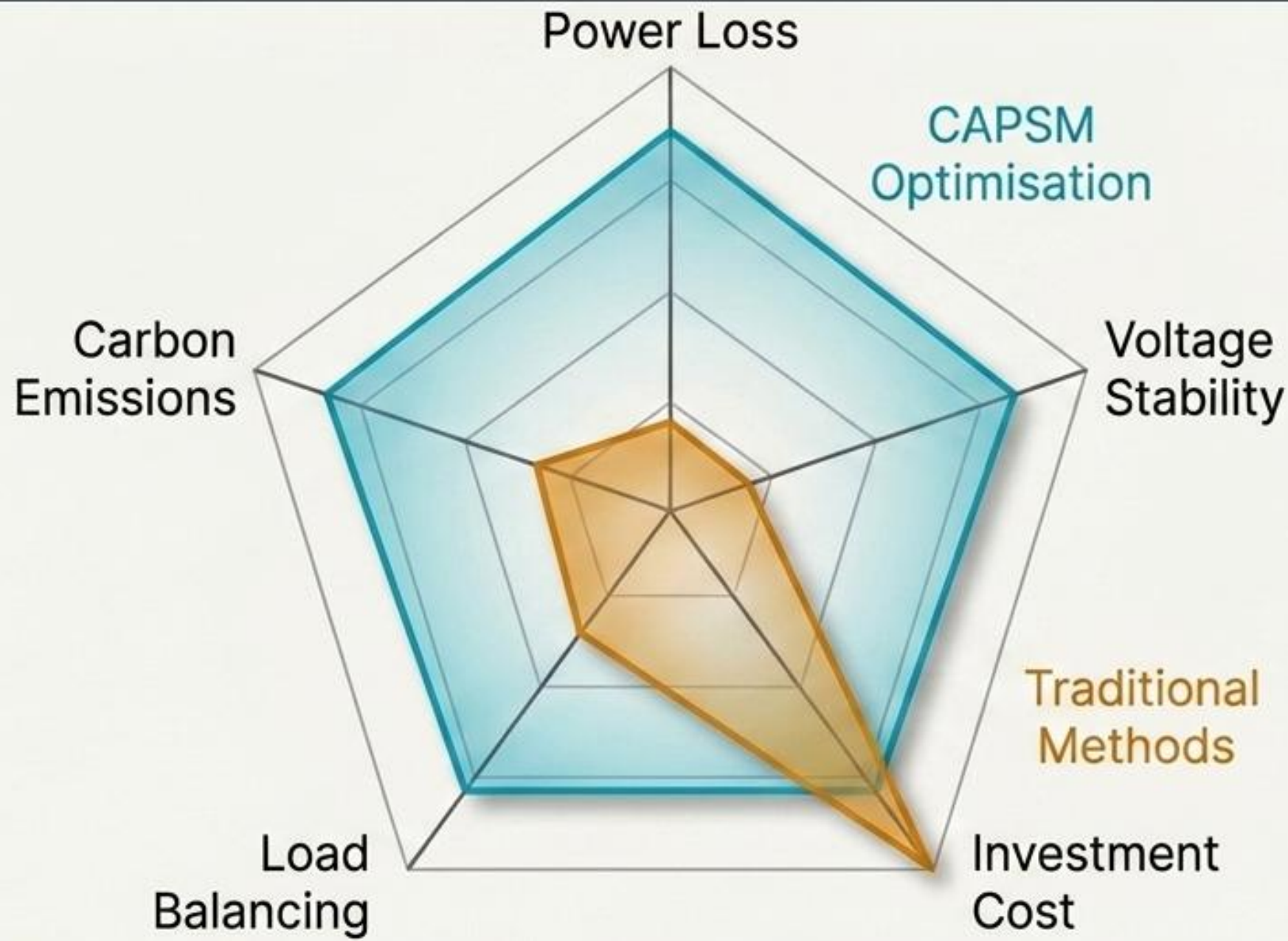
CAPSM AI-Optimised Placement



Control Authority

AI-optimised placement transforms physical grid actuators (FACTS devices) from isolated components into a unified, grid-wide control fabric, maximizing system influence with the same number of devices.

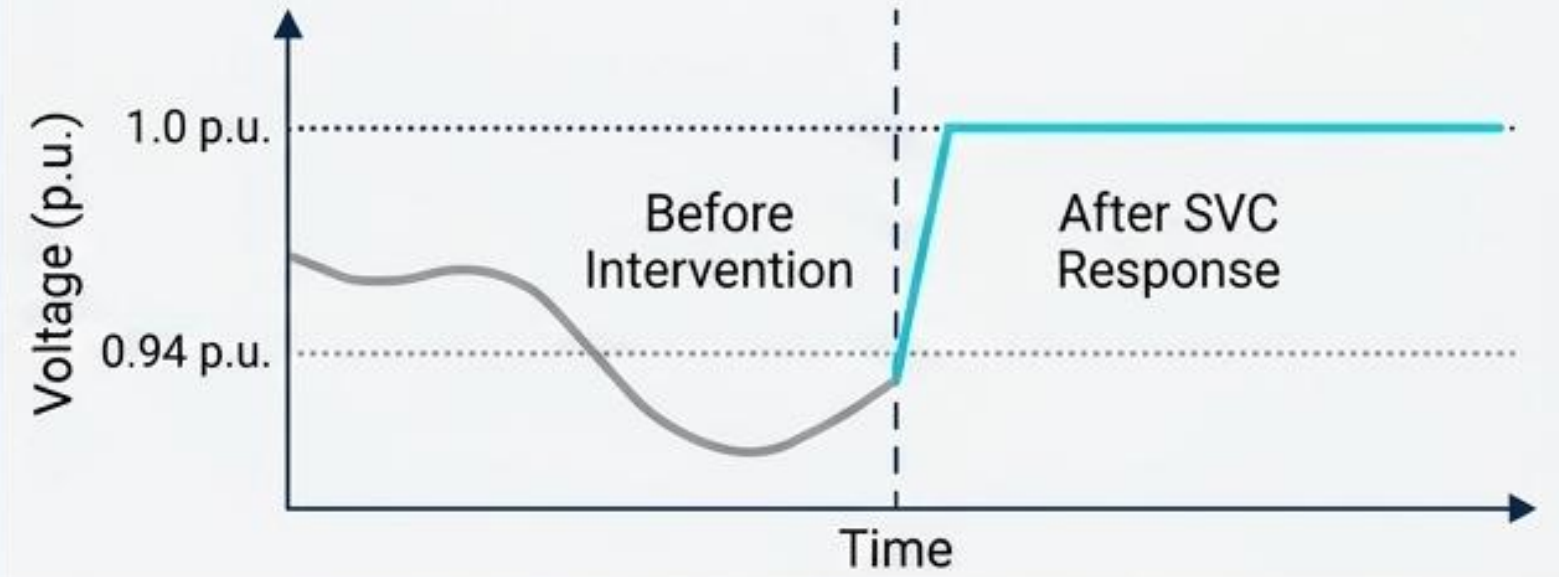
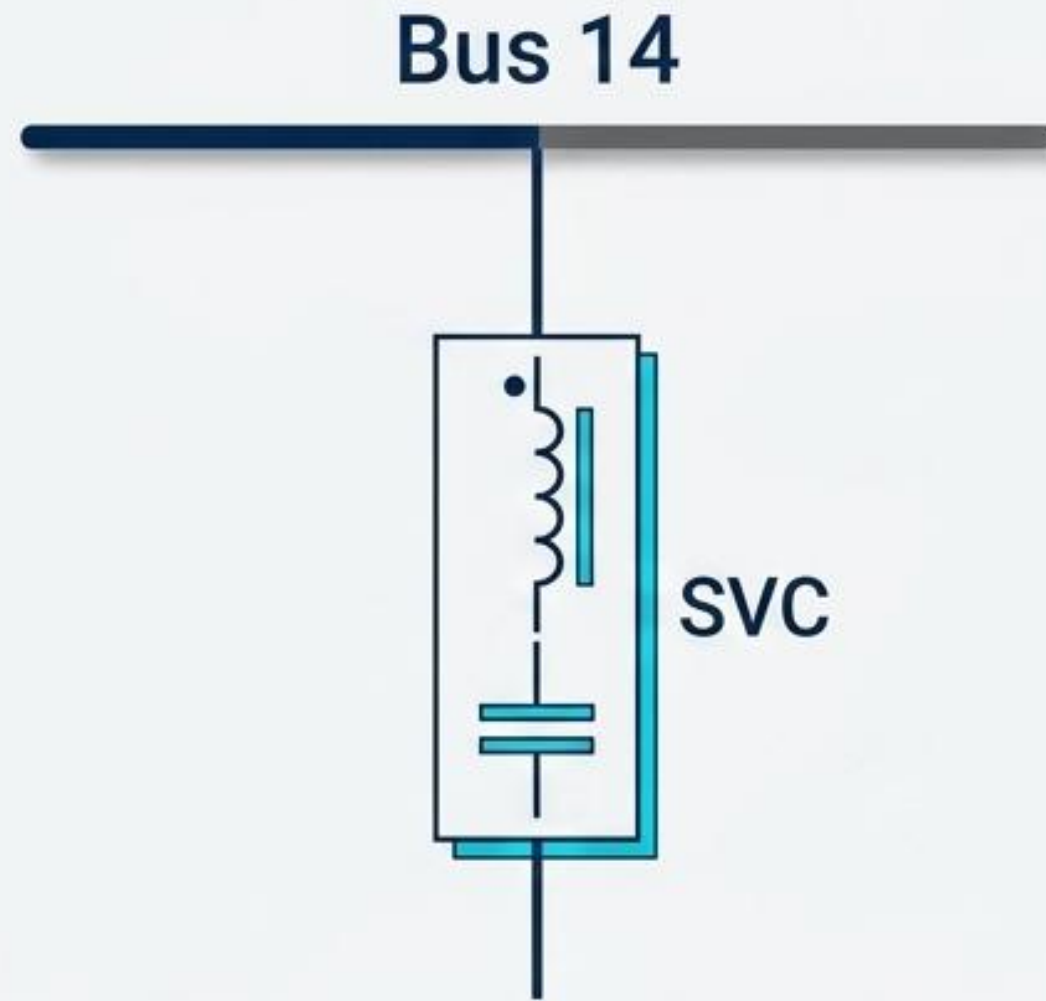
Problem Formulation: Multi-Objective Optimization



$$\min F(x,y,z) = [f_1, \dots, f_m]$$

CAPSM resolves the tension between competing physical grid requirements by balancing five distinct operational objectives simultaneously.

Actuator 1: SVC at Bus 14

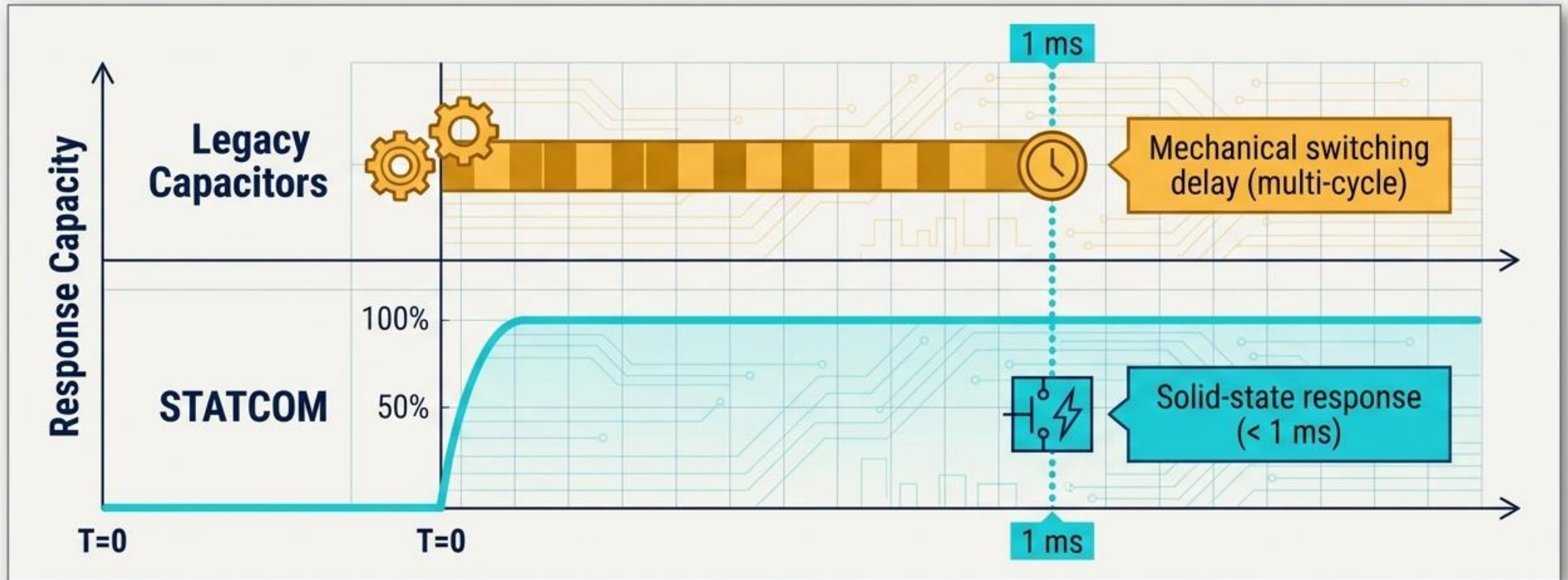


Function: Shunt Voltage Support

Mechanism: Dynamic Reactive Power Injection

Target: Bus 14

Actuator 2: STATCOM at Bus 39

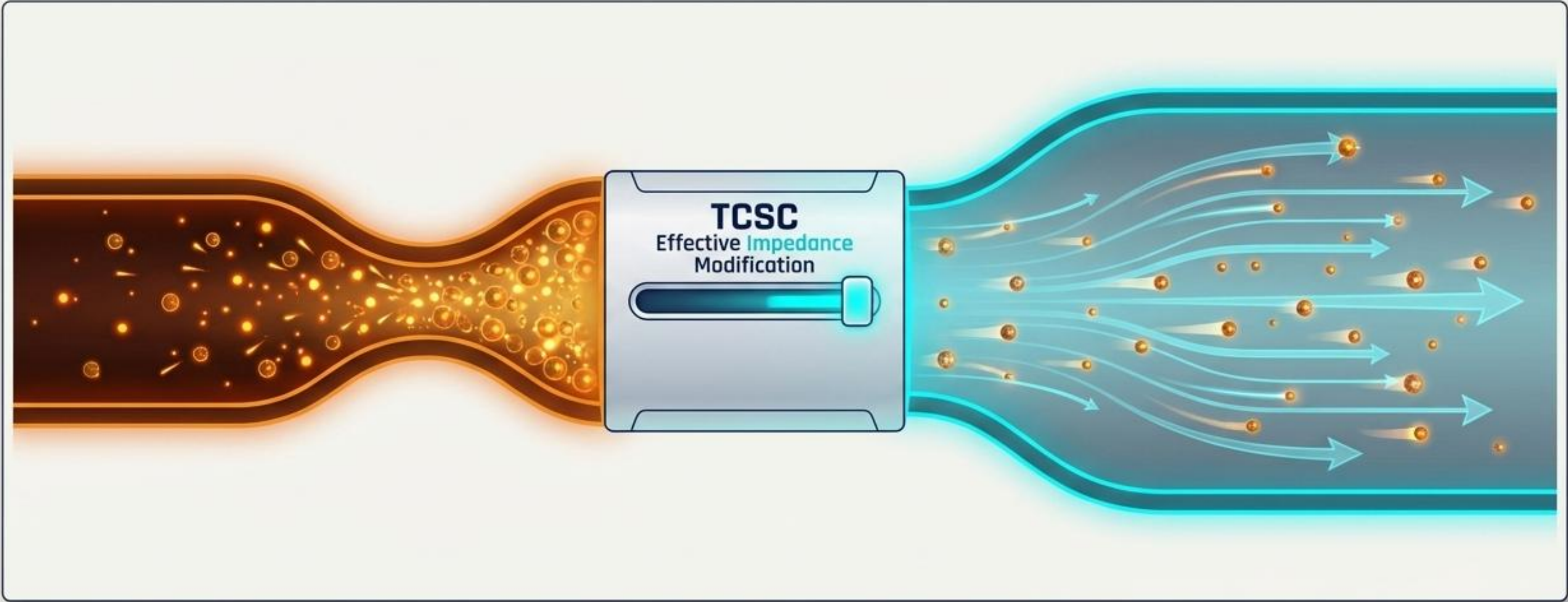


Function: Fast Reactive Power Compensation

Advantage: Sub-cycle solid-state response vs mechanical switching delay

Target: Bus 39

Actuator 3: TCSC on Line 16-17

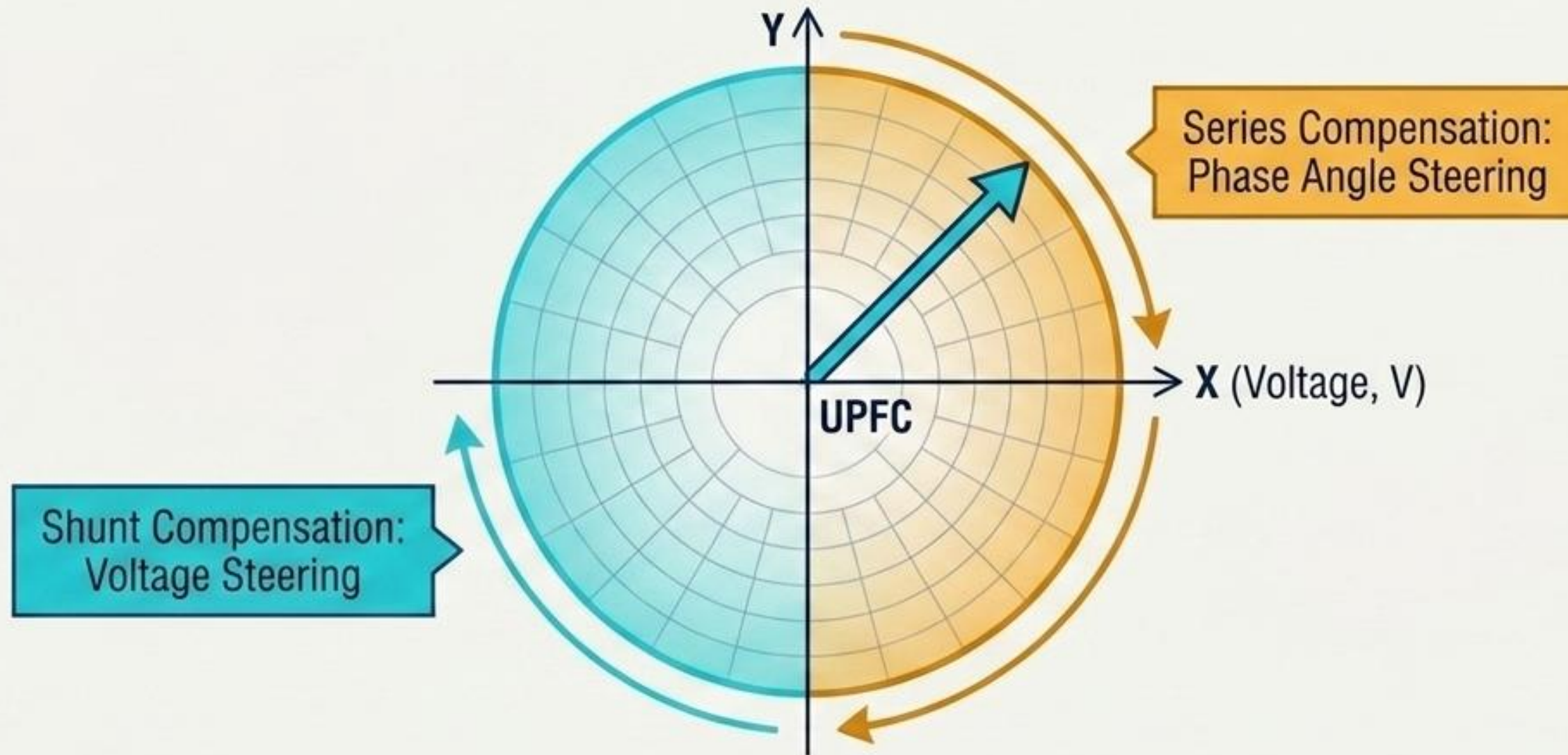


Function: Transmission Capacity Enhancement

Mechanism: Series effective impedance modification

Target: Line 16-17

Actuator 4: UPFC at Bus 26



Function: Unified Power Flow Control

Mechanism: Simultaneous Series + Shunt compensation

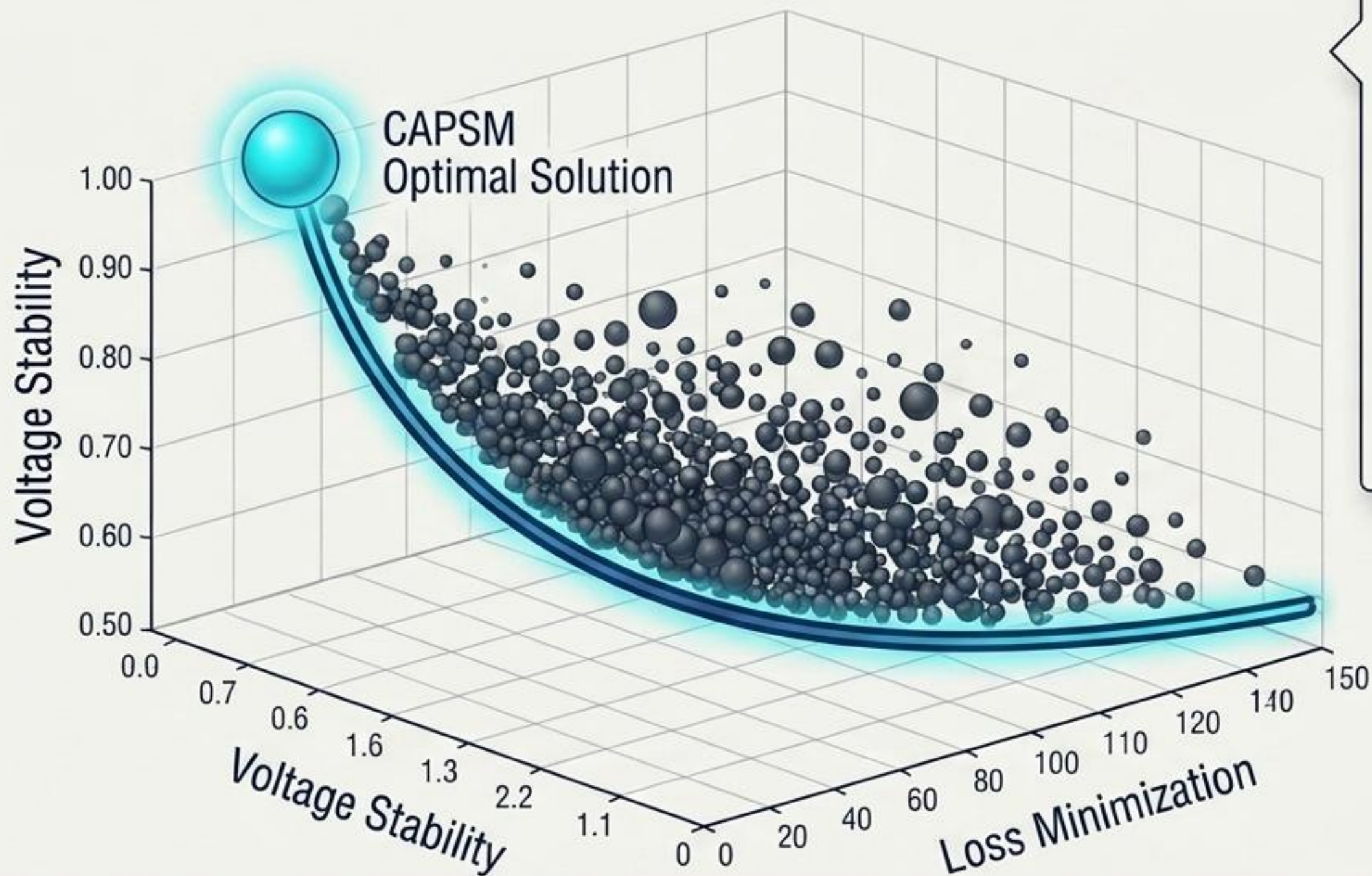
Target: Bus 26

The Multi-Objective Trade-off

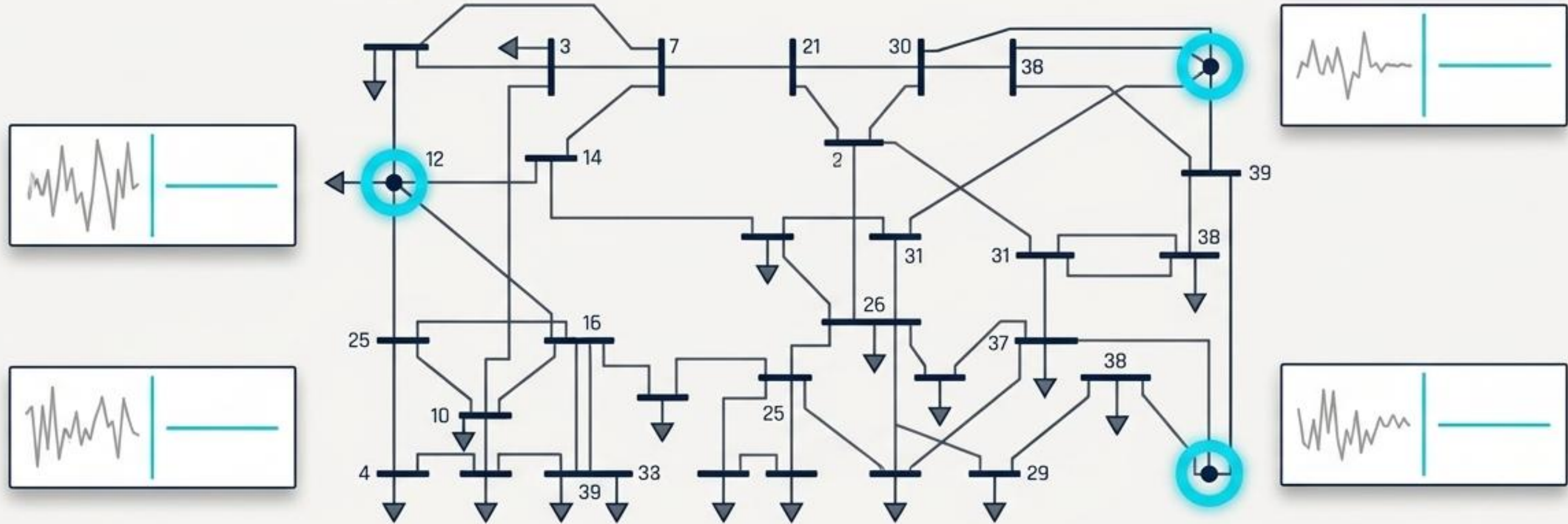
Quantum-Inspired Logic

Quantum-Inspired Reinforcement Learning (QIRL) converged 58% to 59% faster than classical DQN.

Tunneling algorithms escaped local optima to locate the absolute best configuration for the physical grid assets.



Placement Results on IEEE 39-Bus System



+ 37.2%

Improvement in voltage stability

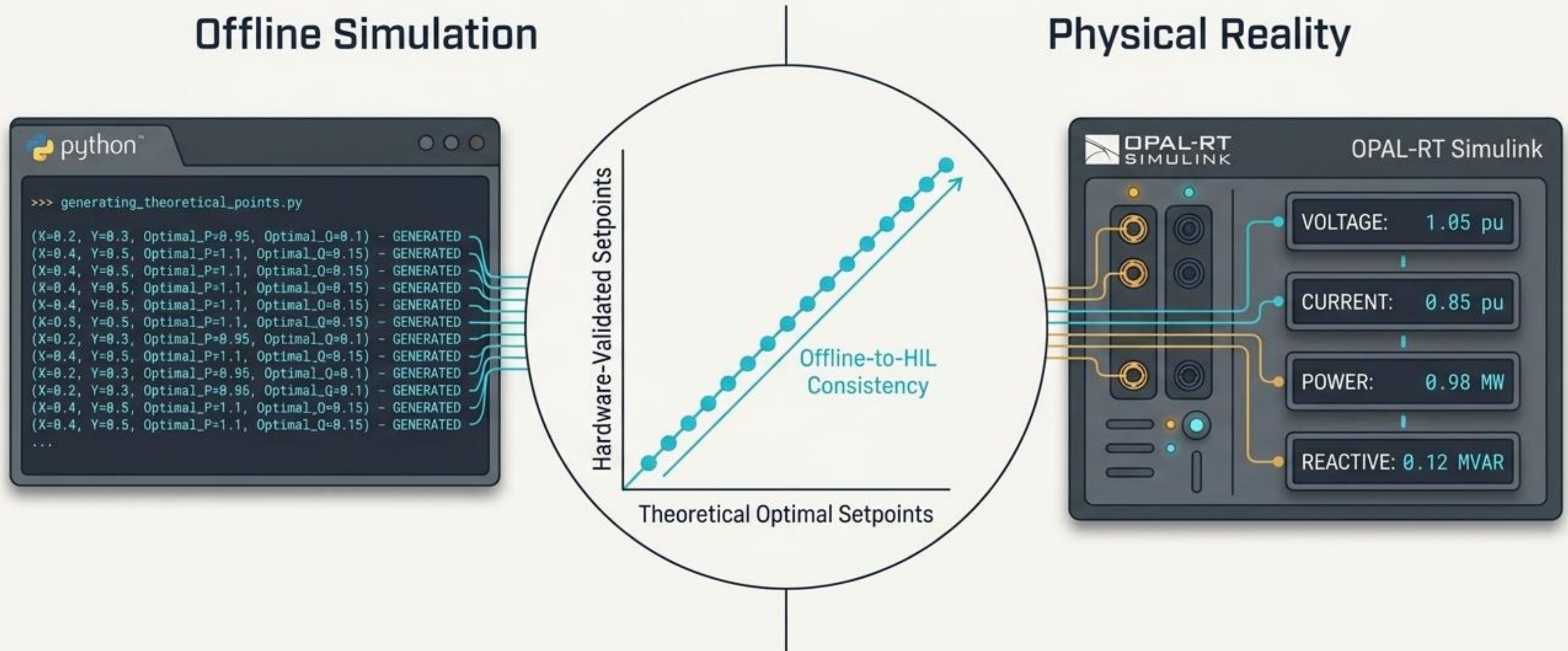
- 24.3%

Decrease in system losses

+ 28.5%

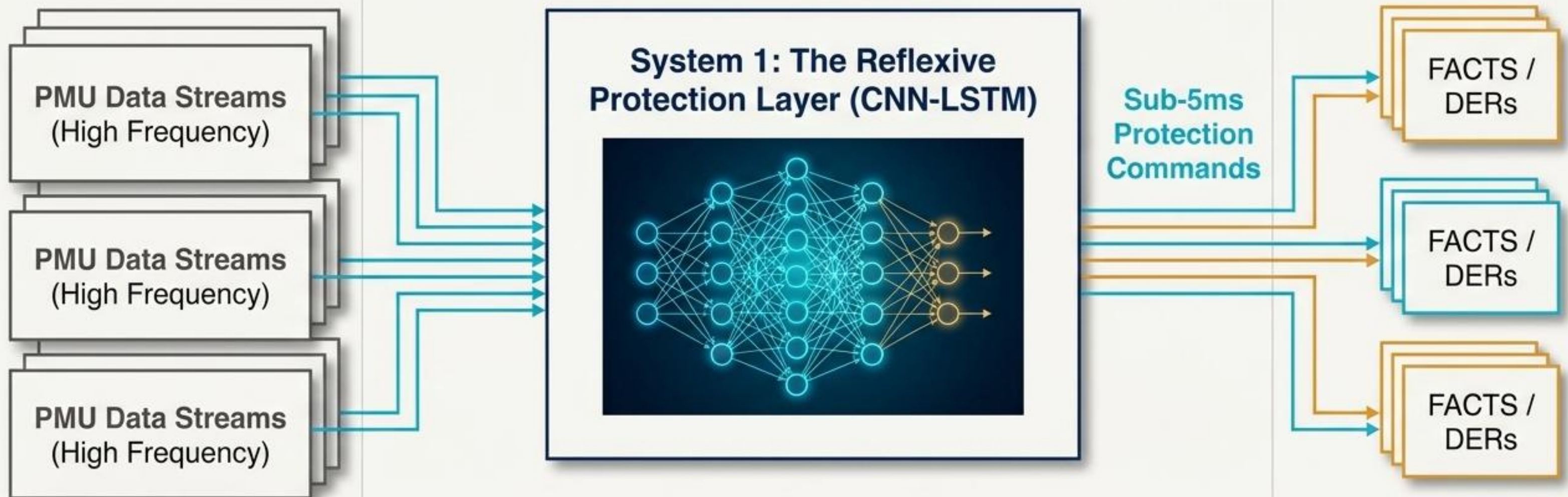
Increase in loading margin

HIL-Validated Placement Methodology



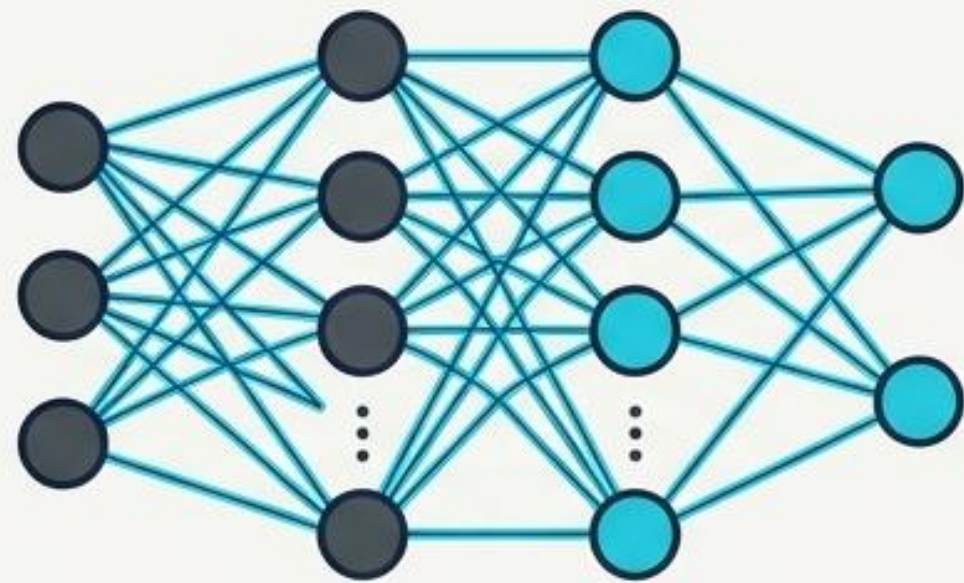
Hardware-in-the-Loop (HIL) validation catches modelling errors invisible in offline simulation, successfully bridging the gap between theoretical AI models and practical grid deployment.

Fault Detection Overview (CAPSM System 1)



This system maps human cognitive theory to grid control, acting as an involuntary, hyper-fast flinch mechanism to protect physical assets before severe damage occurs.

PINN Architecture & Physics Integration



Physics Constraint Bounding Box
 $0.94 < V < 1.06$ p.u.

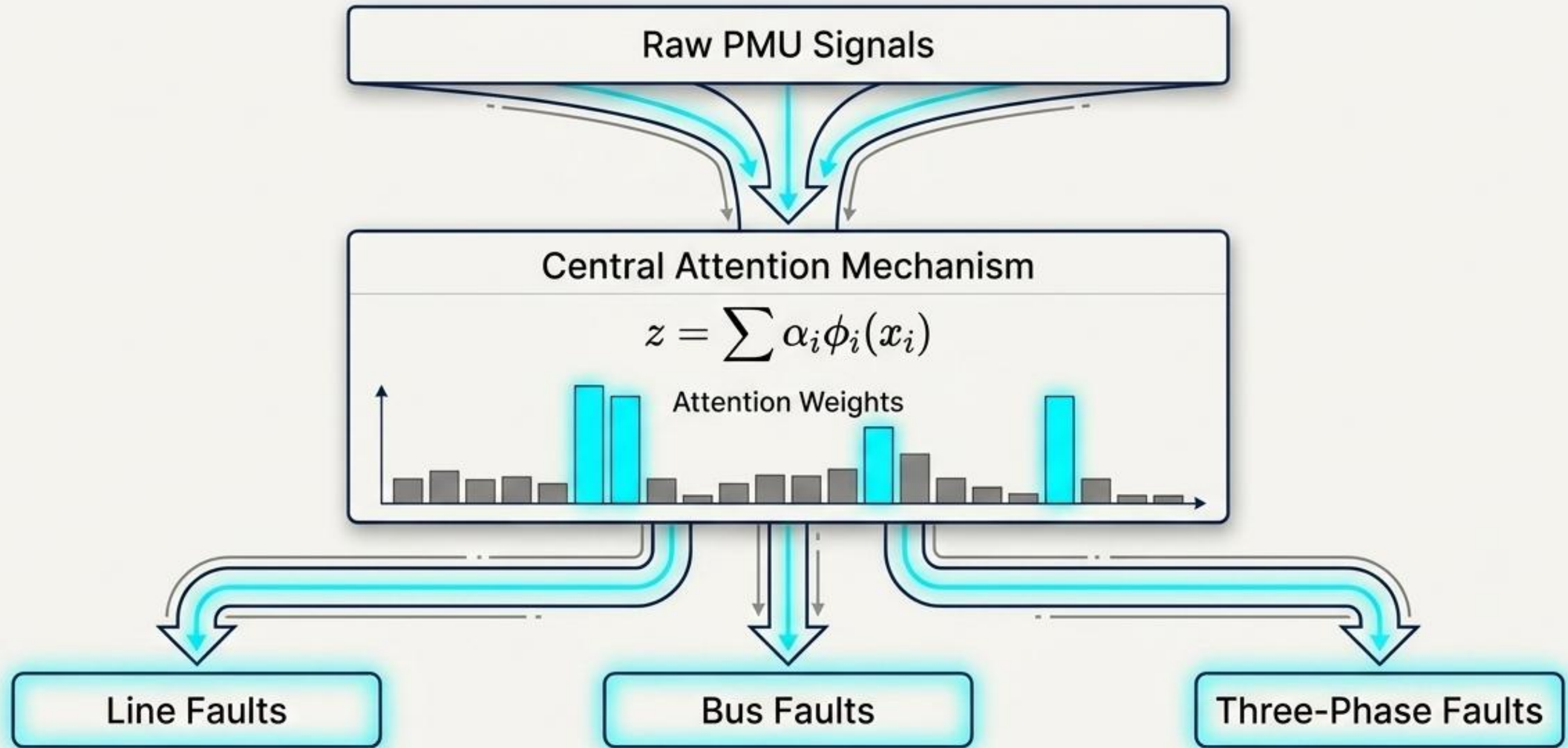
$$\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{data}} + \lambda \mathbf{L}_{\text{physics}}$$

Impossible outputs are mathematically penalized.

99.97%
Constraint
Compliance

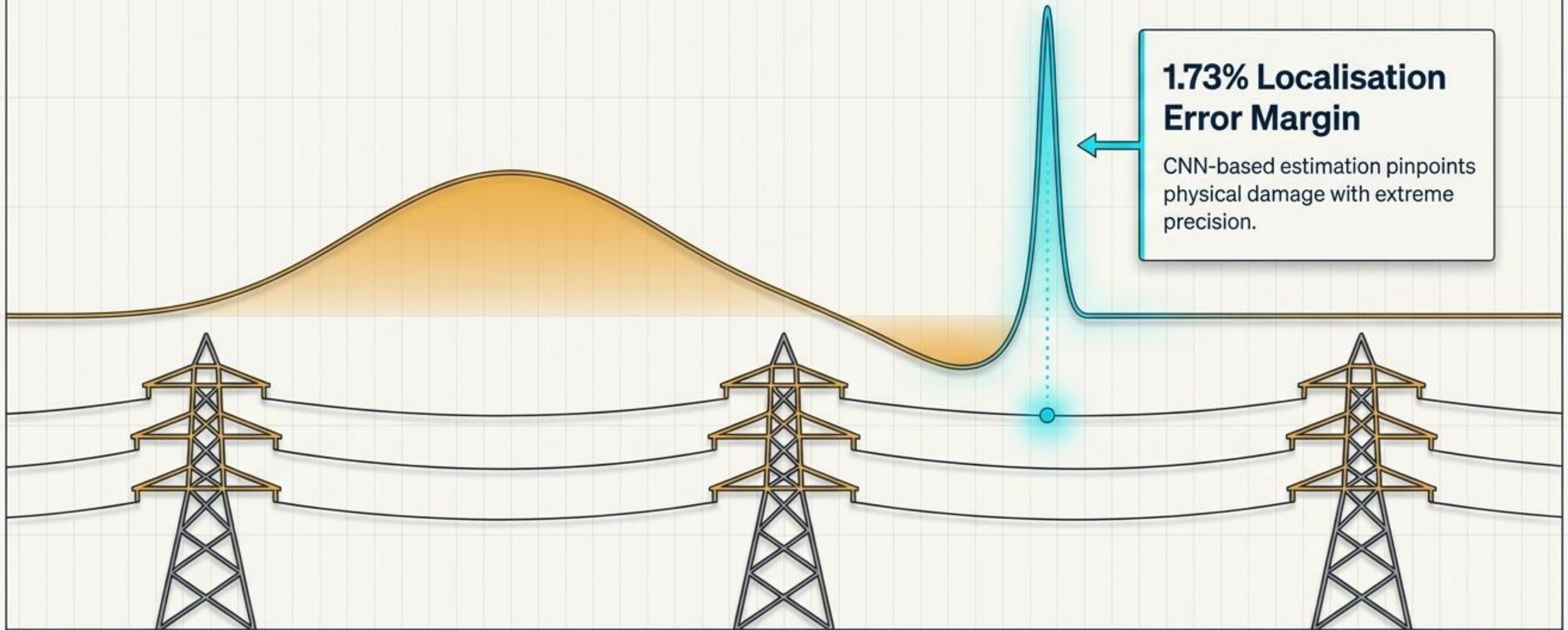
Reduced from **3130** violations
to **11** in 10,000 cycles

Fault Classification Methodology

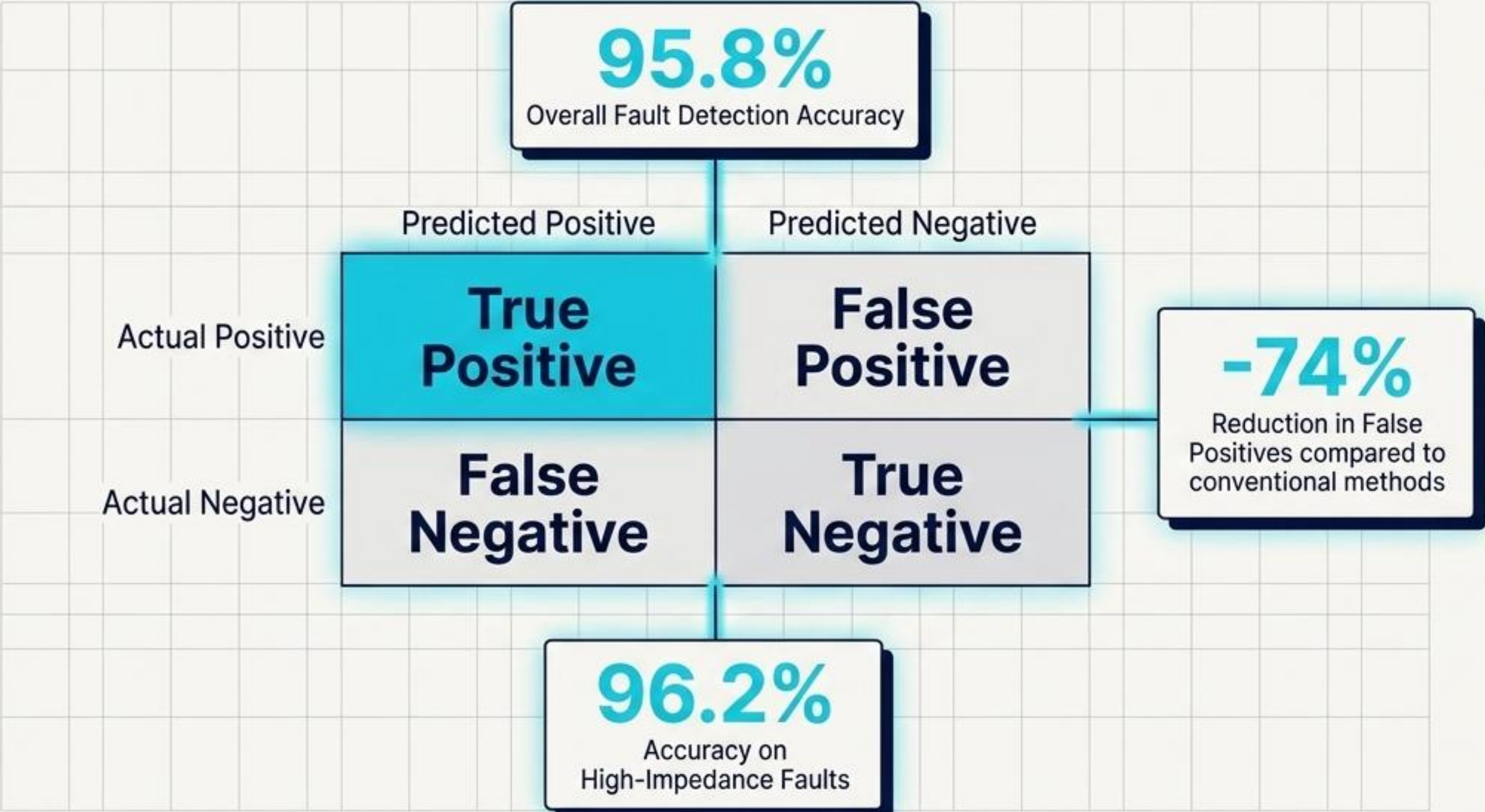


Attention weights filter out noise and prioritize specific time steps in the multi-modal data fusion to instantly identify the true fault signature.

Fault Localization (NN-Based Estimation)

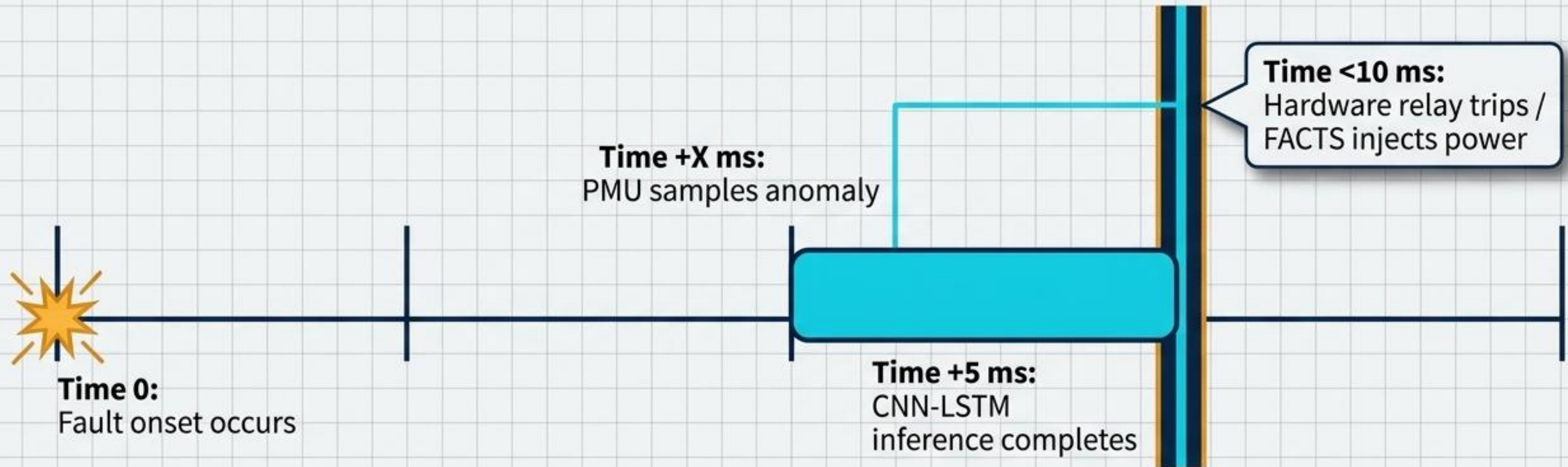


Fault Detection Accuracy Results



Extensive offline performance data confirms the algorithmic accuracy, establishing a flawless baseline before transitioning the system to physical hardware stress-testing.

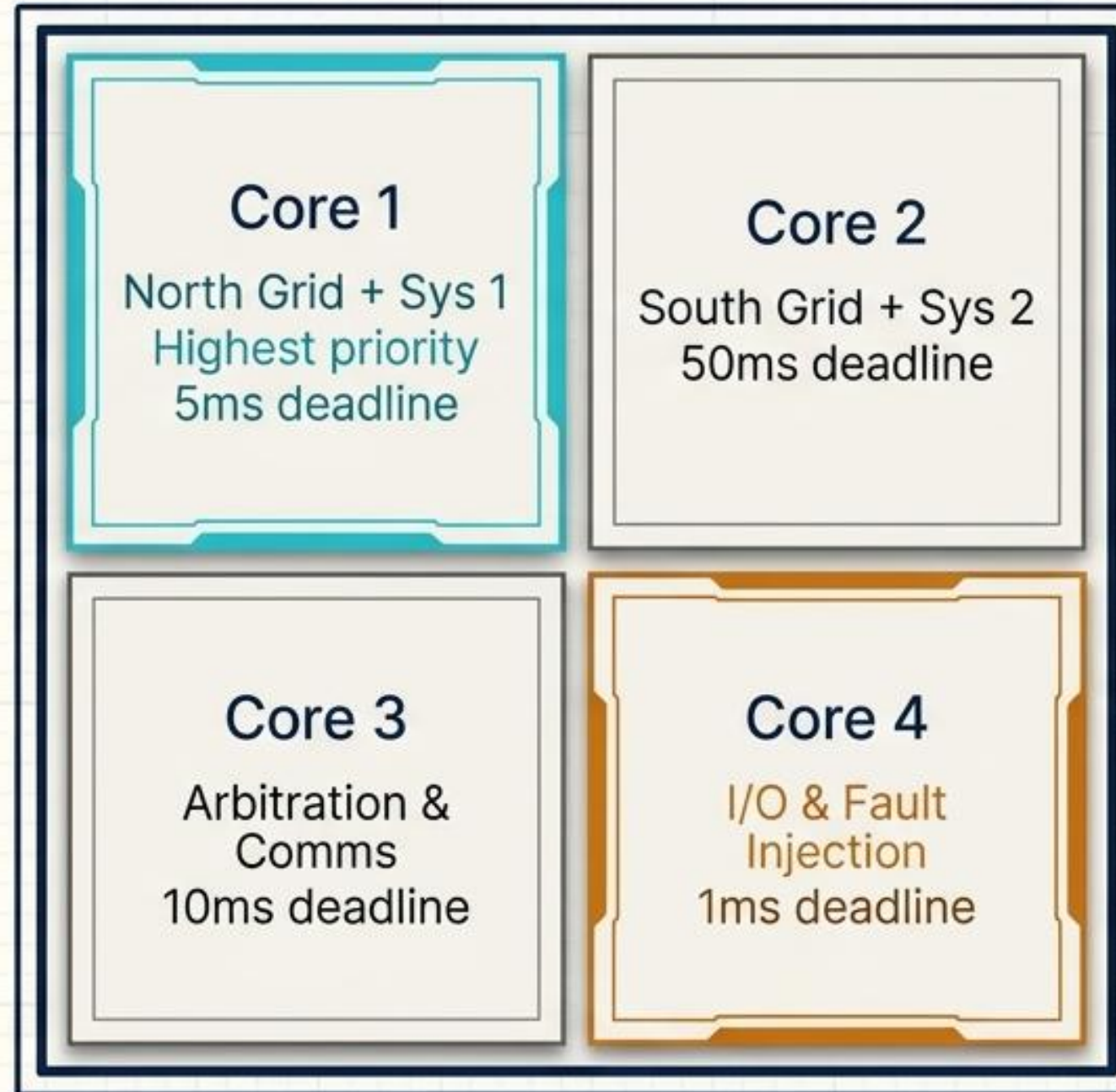
Real-Time Fault Response (<10ms on HIL)



61.4% faster response versus the best traditional rule-based methods.

HIL Fault Validation Setup

Deploying CAPSM on Real-Time Hardware: The OPAL-RT Controller-HIL Setup

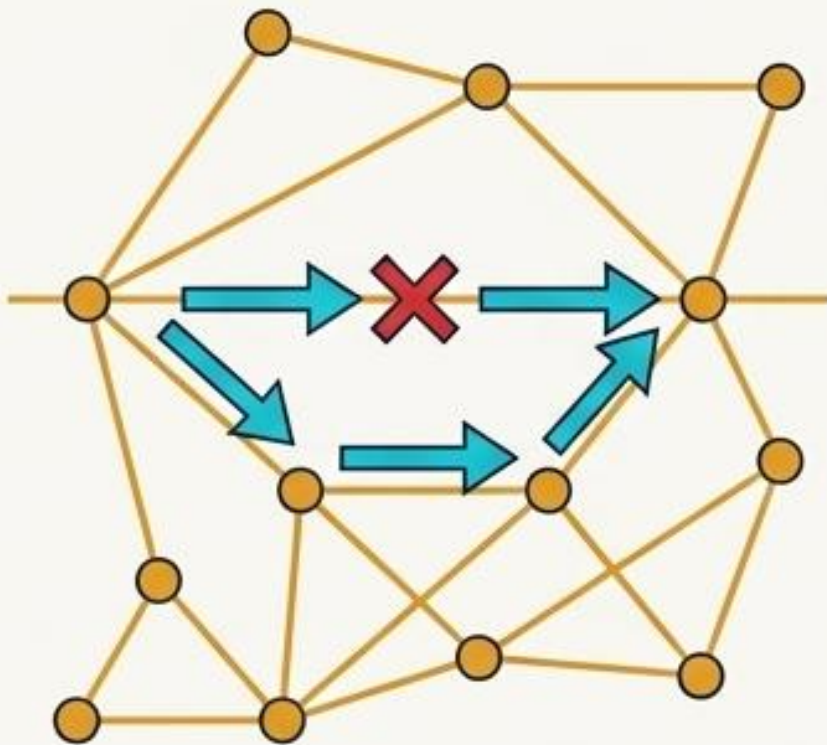


Solver Configuration

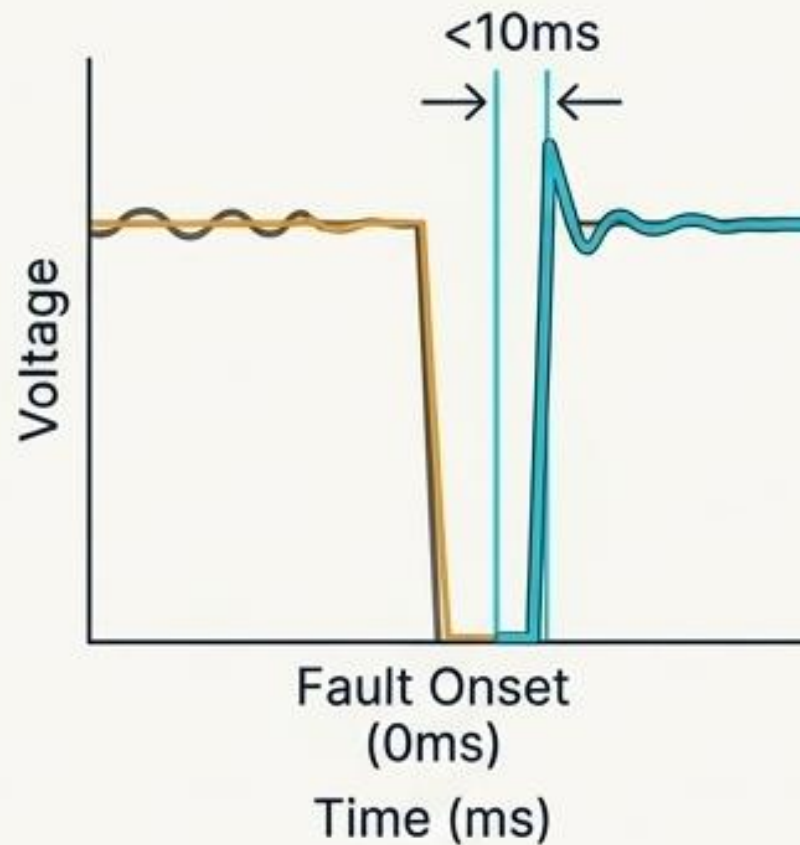
ARTEMIS solver running at a $50\mu\text{s}$ Electromagnetic Transient (EMT) time step.

Fault Scenario Testing

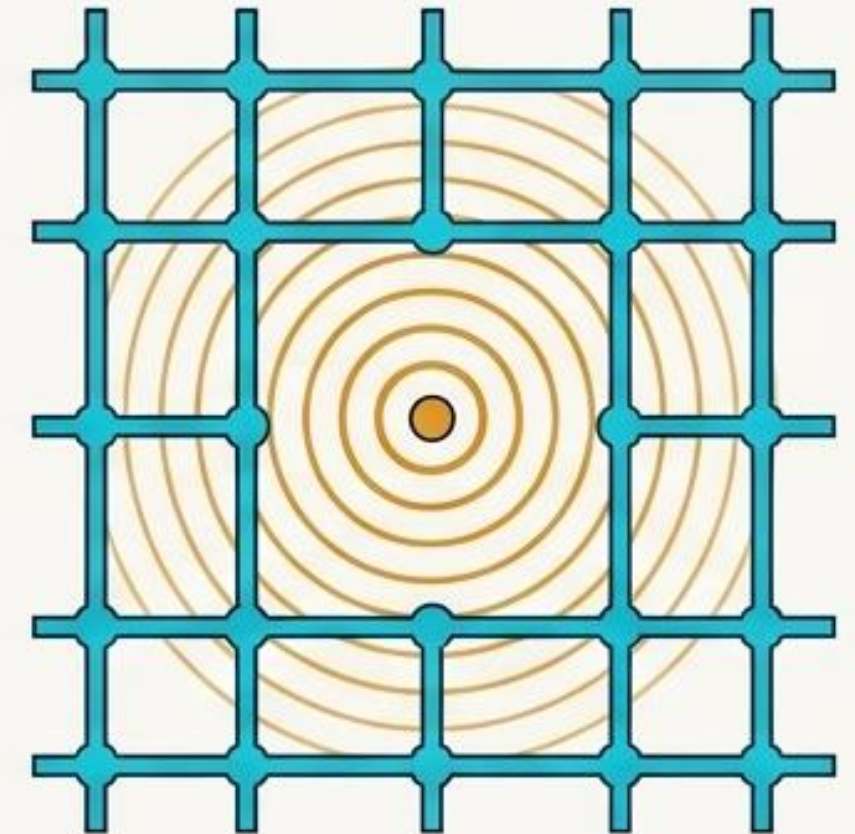
Scenario 1:
Line Tripping



Scenario 2:
Three-Phase Bus Fault



Scenario 3:
Cascading Failures



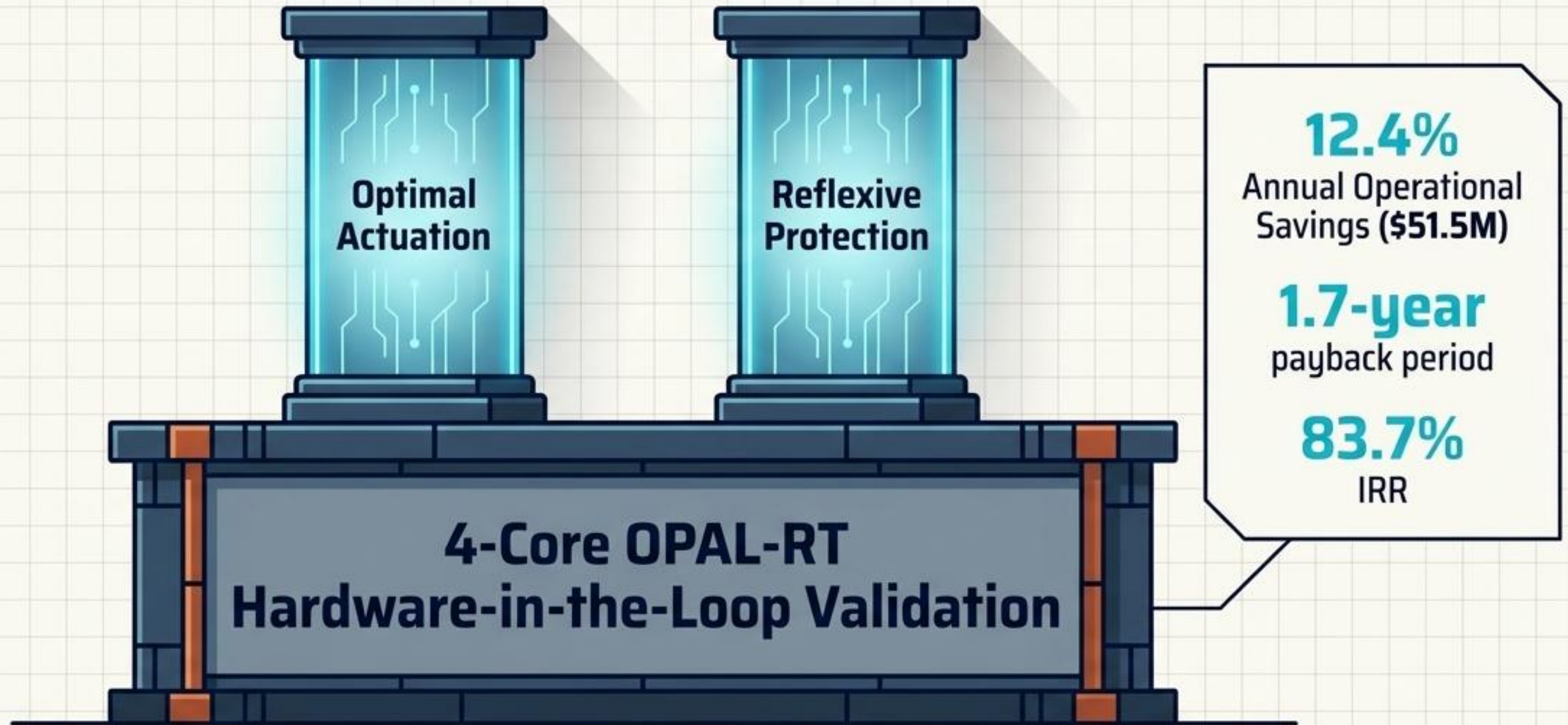
Hardware-in-the-Loop stress-testing proves that the system responds correctly and autonomously to extreme physical damage.

Performance Metrics Comparison

Engineering Dimension	CAPSM PINN	Conventional Impedance Relays
Latency	Sub-10ms	20-40ms
Localisation Error	1.73%	5-10%
False Positives	-74% Reduction	Baseline
High-Impedance Faults	96.2% Sensitivity	Low / Variable
Bidirectional Flow	Native Handling	Requires Complex Elements

AI-driven protection vastly outperforms conventional physics-only relays across all critical engineering dimensions in DER-rich environments.

Summary of Contributions



CAPSM successfully bridges theoretical AI to practical power engineering, delivering a field-ready blueprint for the modern grid.

Cognitive Adaptive Power System Management (CAPSM)

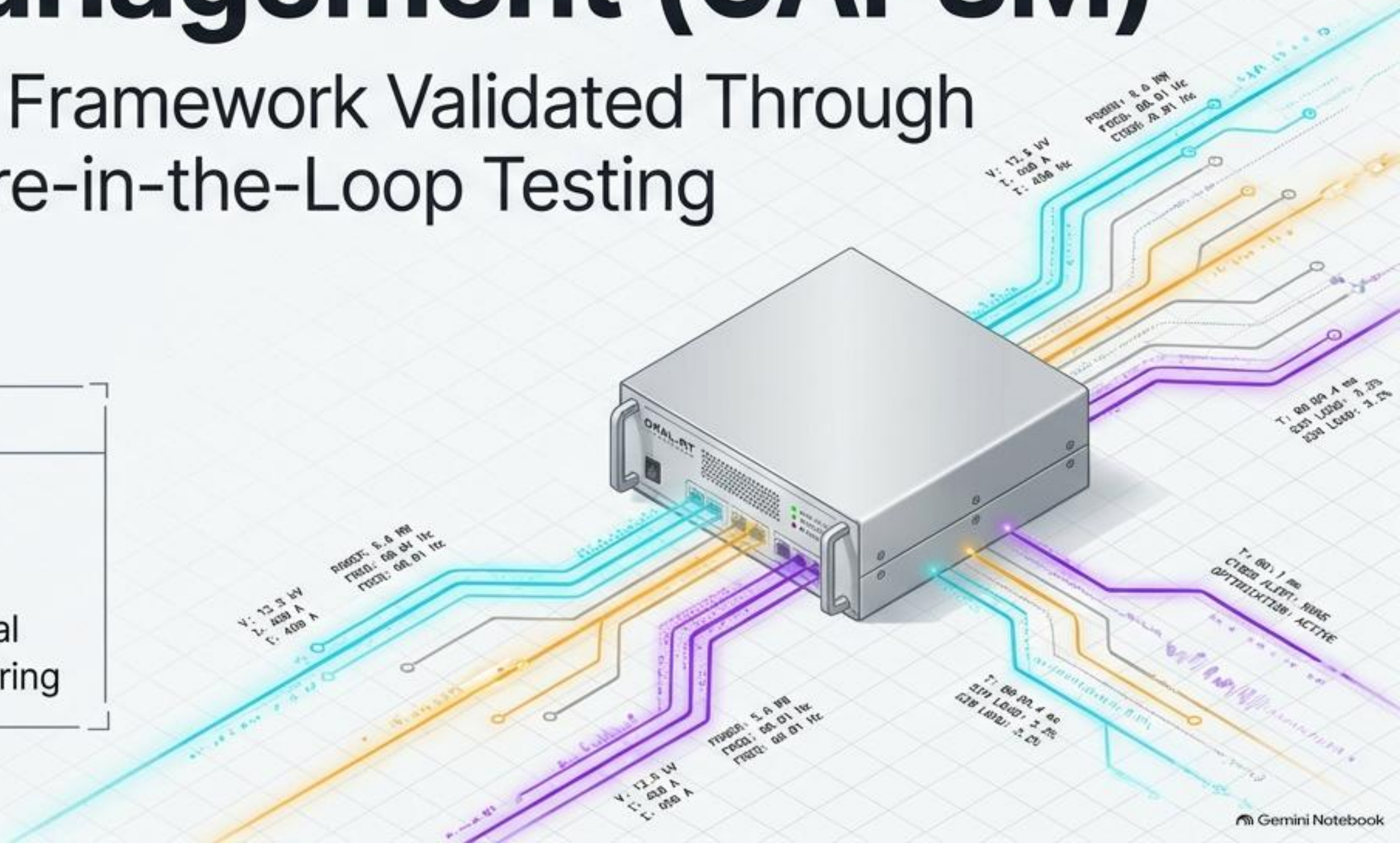
A Brain-Inspired AI Framework Validated Through Sub-10ms Hardware-in-the-Loop Testing

MODULAR TELEMETRY

Author: Mahmoud Kiasari

Institution: Dalhousie University

Department of Electrical
and Computer Engineering

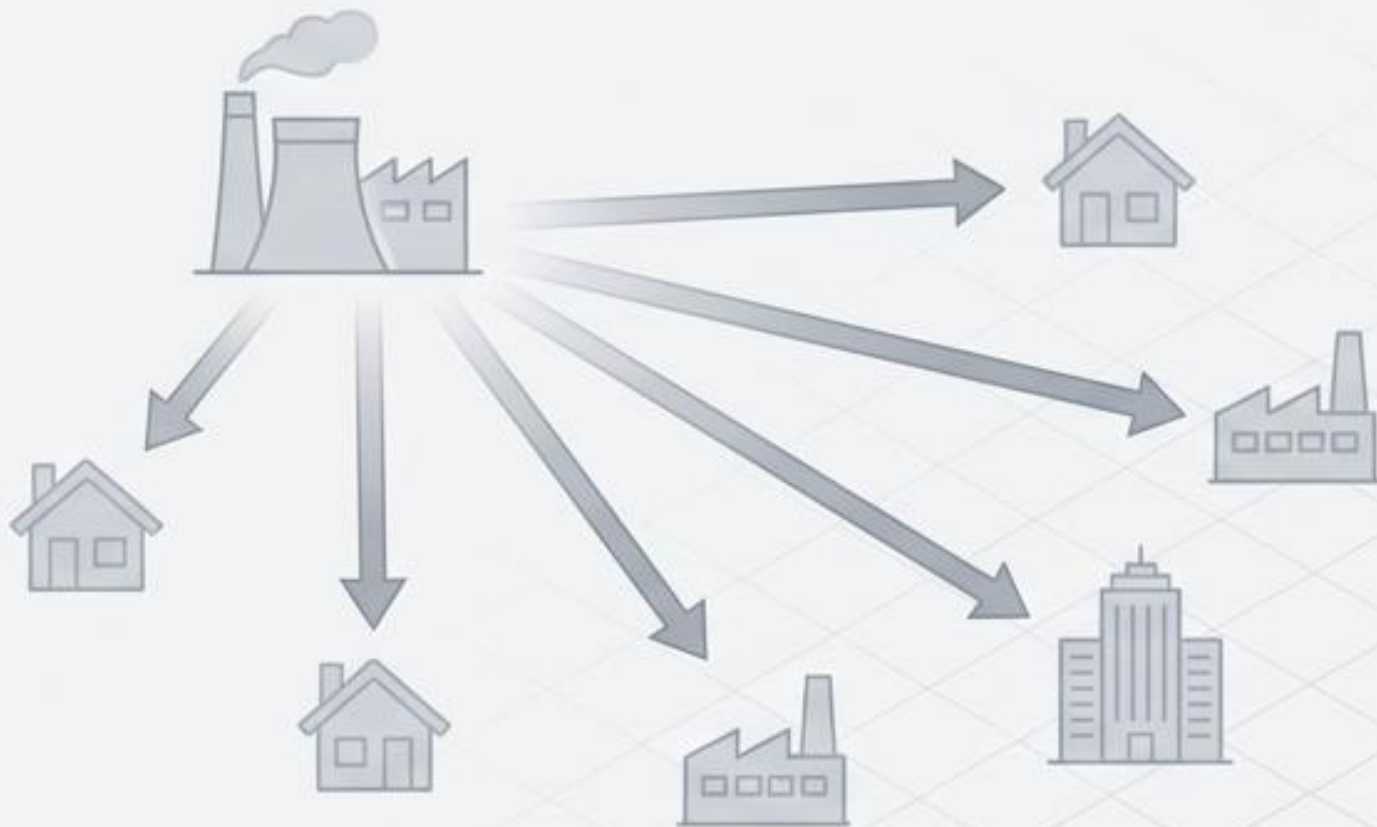


The Intermittency and Complexity Crisis

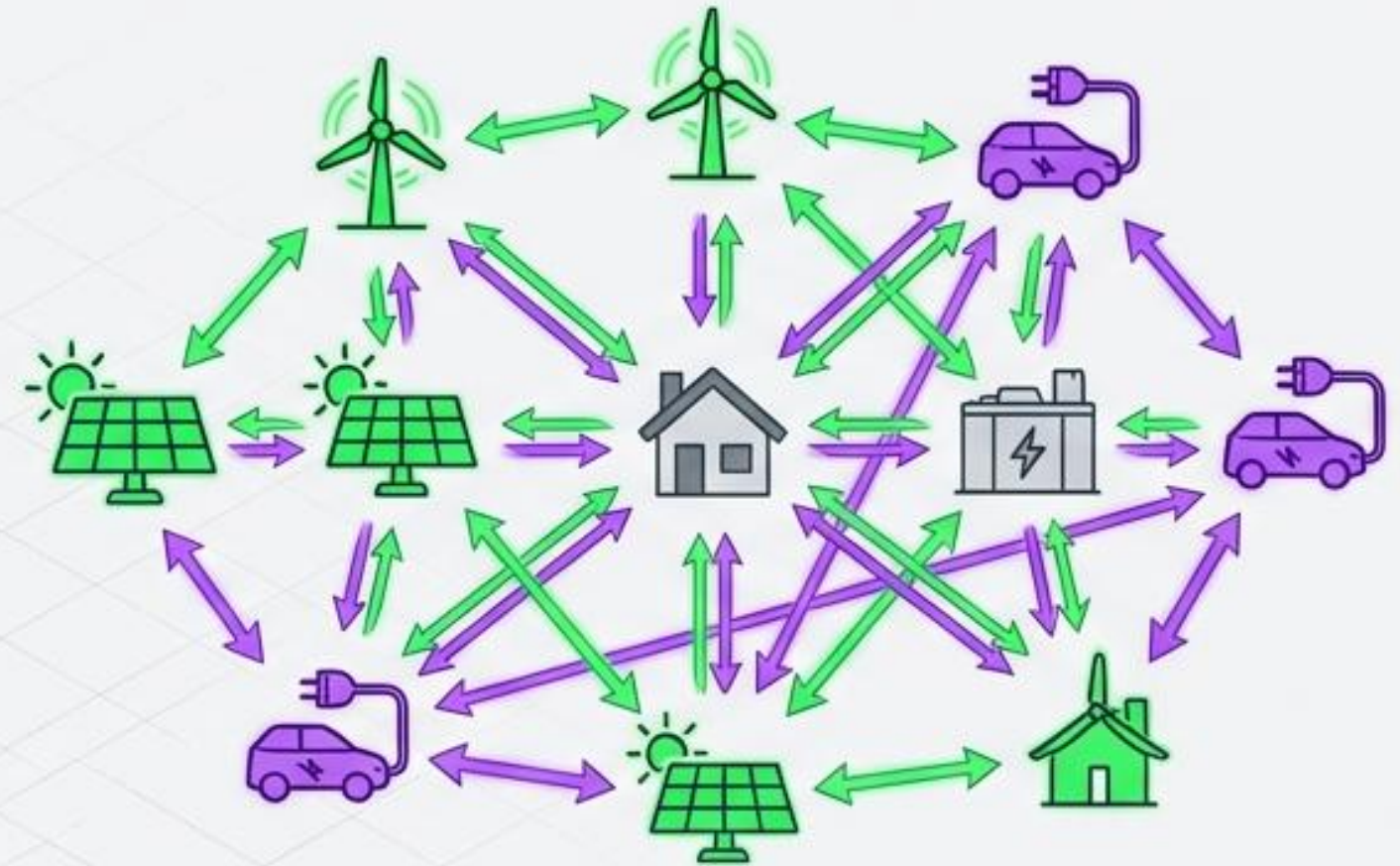
Global Renewable Capacity: +60% (Reaching 4,800 GW)

DER Growth: 132.4 GW to 528.4 GW (16.7% CAGR)

Traditional Architecture



Distributed Architecture



Reduced System Inertia
(Sync generators replaced by inverters)



Bidirectional Power Flows
(Distribution networks behaving like transmission lines)

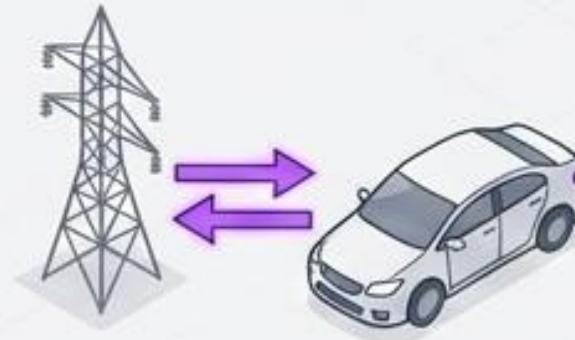


Cybersecurity Vulnerabilities
(Digitalisation of critical infrastructure)

Vehicle-to-Grid (V2G) Architecture Concept



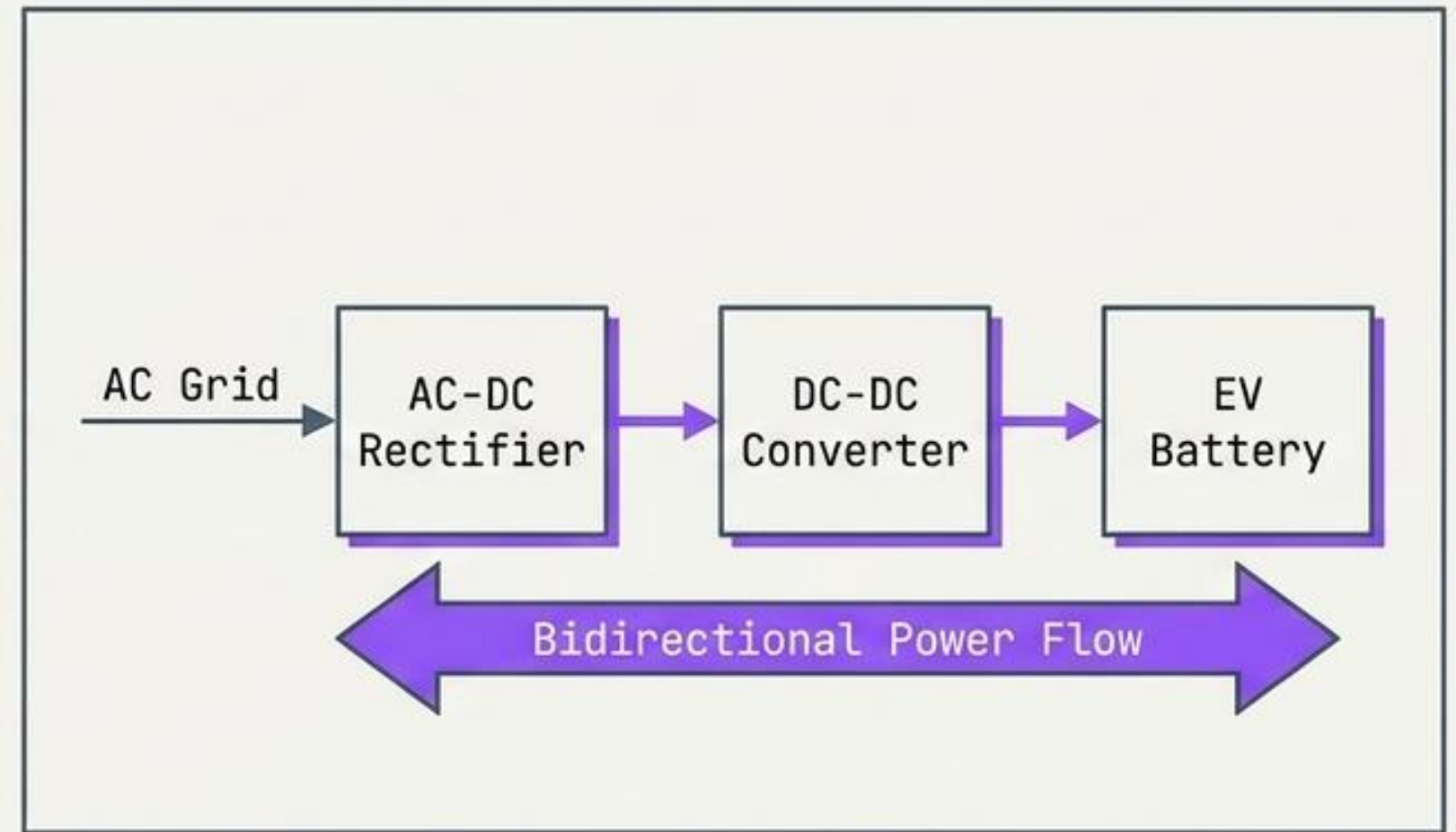
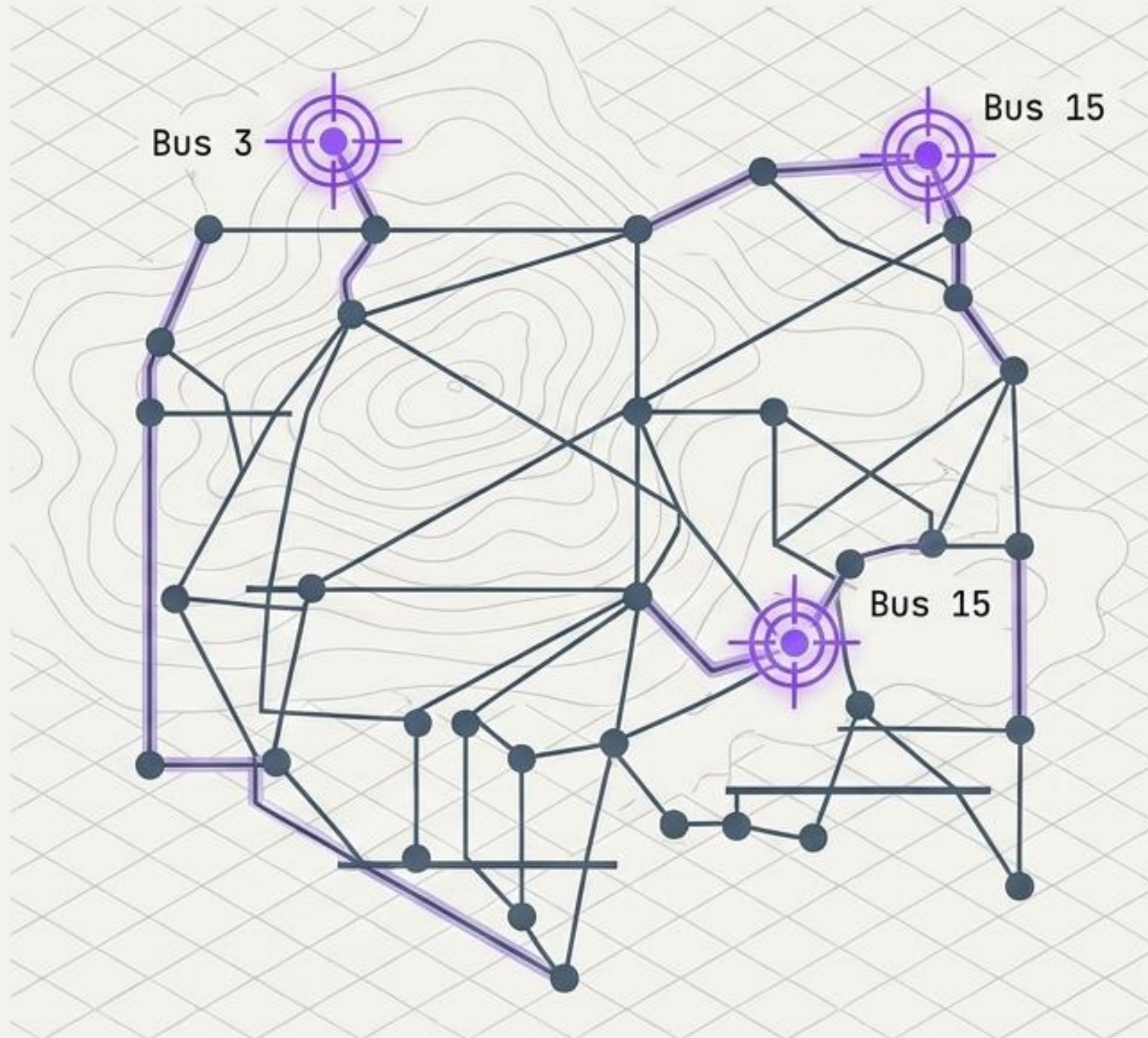
G2V: One-Way Flow



V2G: Two-Way Flow

	Grid-to-Vehicle (G2V)	Vehicle-to-Grid (V2G)
Power Flow	Unidirectional	Bidirectional
Consumer Role	Passive consumer	Active Distributed Energy Resource (DER)
Grid Impact	Acts as grid burden	Provides frequency regulation & storage
Hardware Requirements	Standard rectifiers	Complex bidirectional power electronics and Phase-Locked Loop (PLL) synchronisation

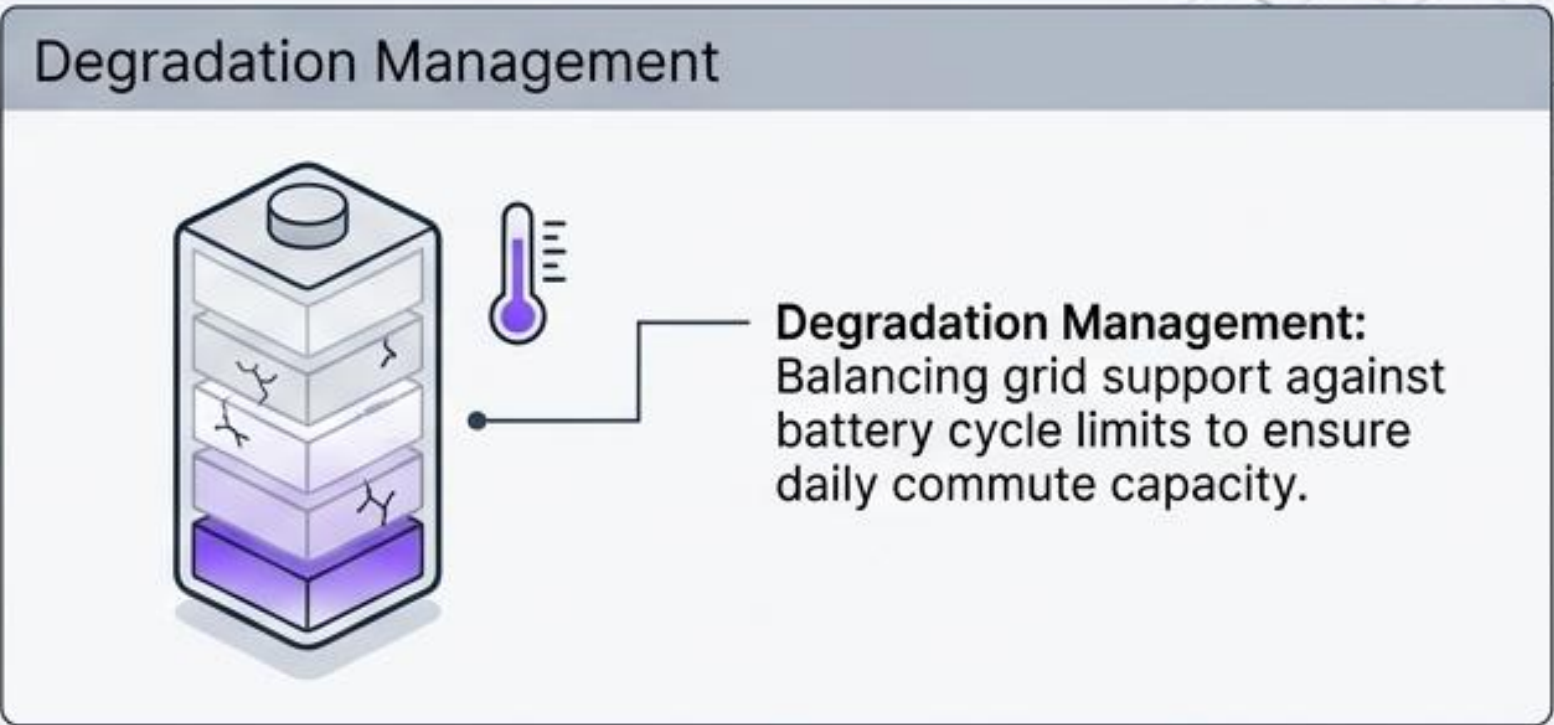
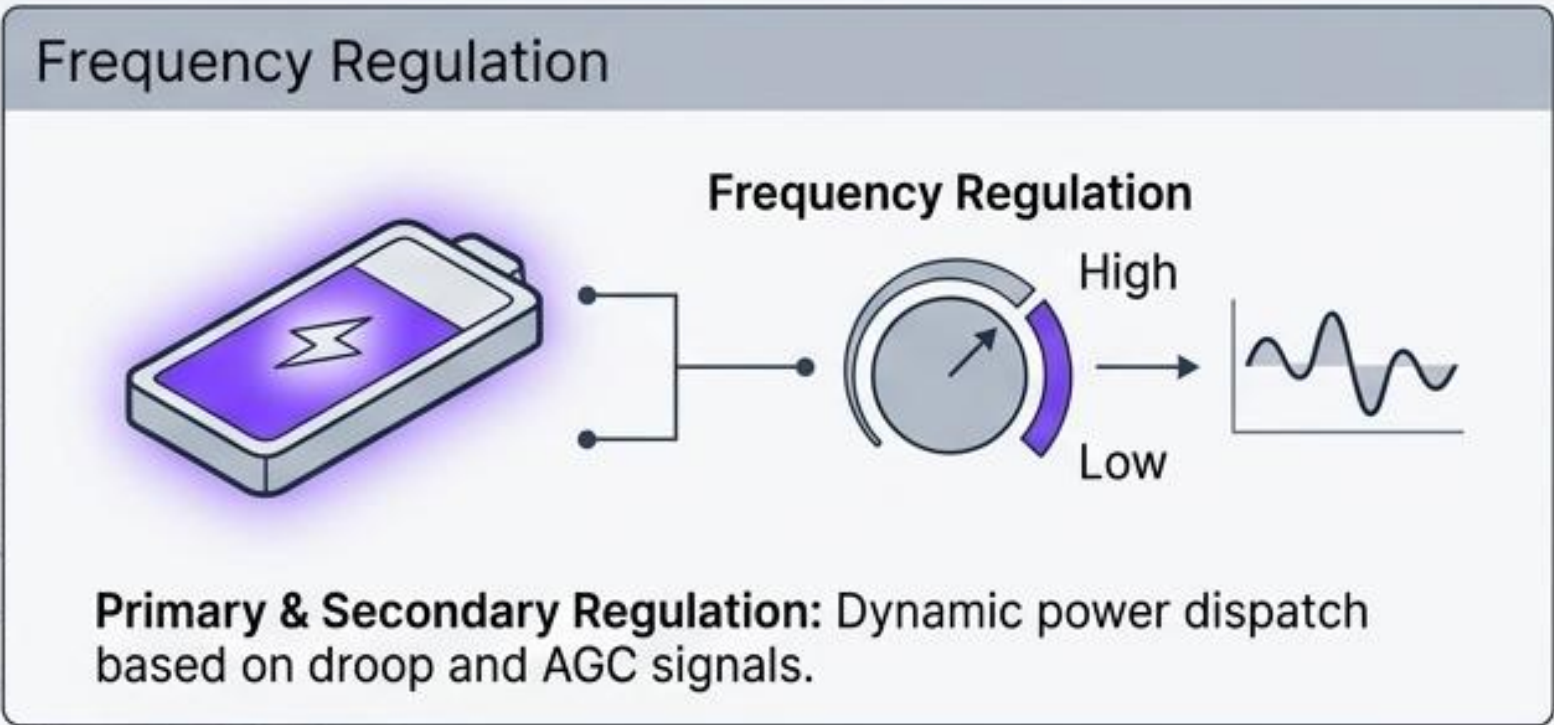
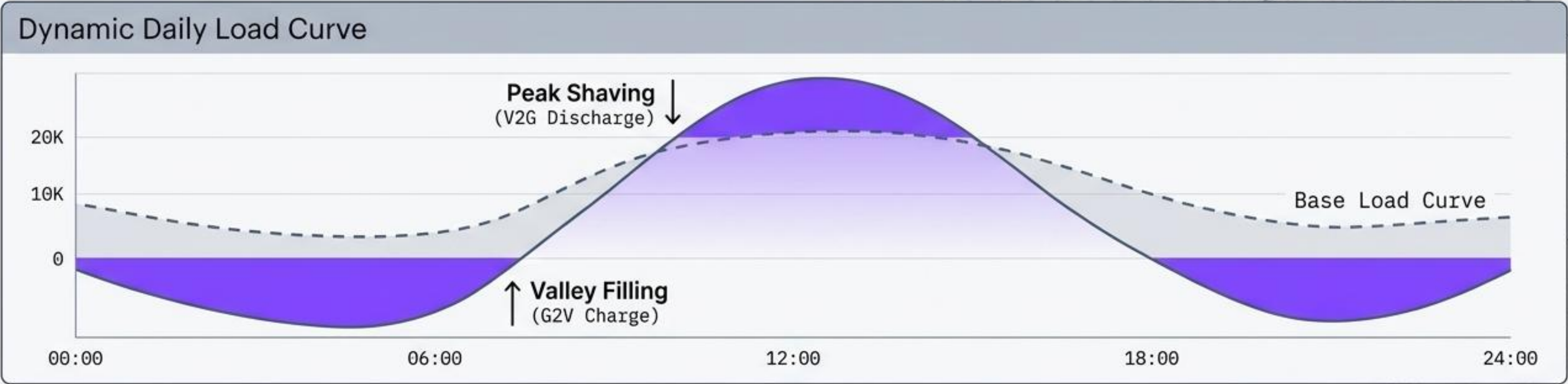
EV Charger Modeling and Placement



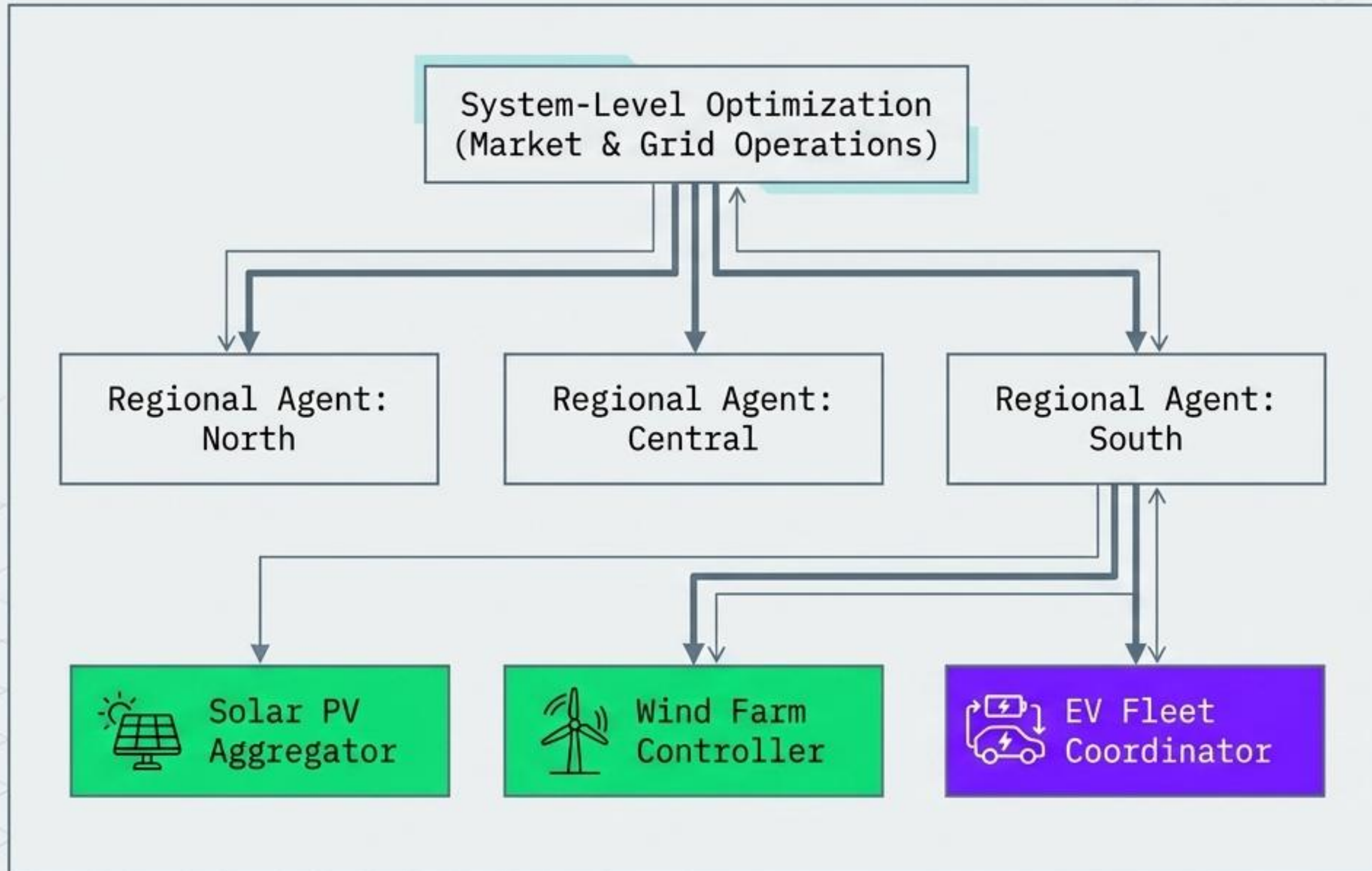
CAPSM EV Fleet Parameters:

- 3 EV fleets modeled
- 19.2 kW power capability per unit
- 40 kWh capacity per unit

Smart Charging Strategies & Grid Support



DER Management & Coordination Framework

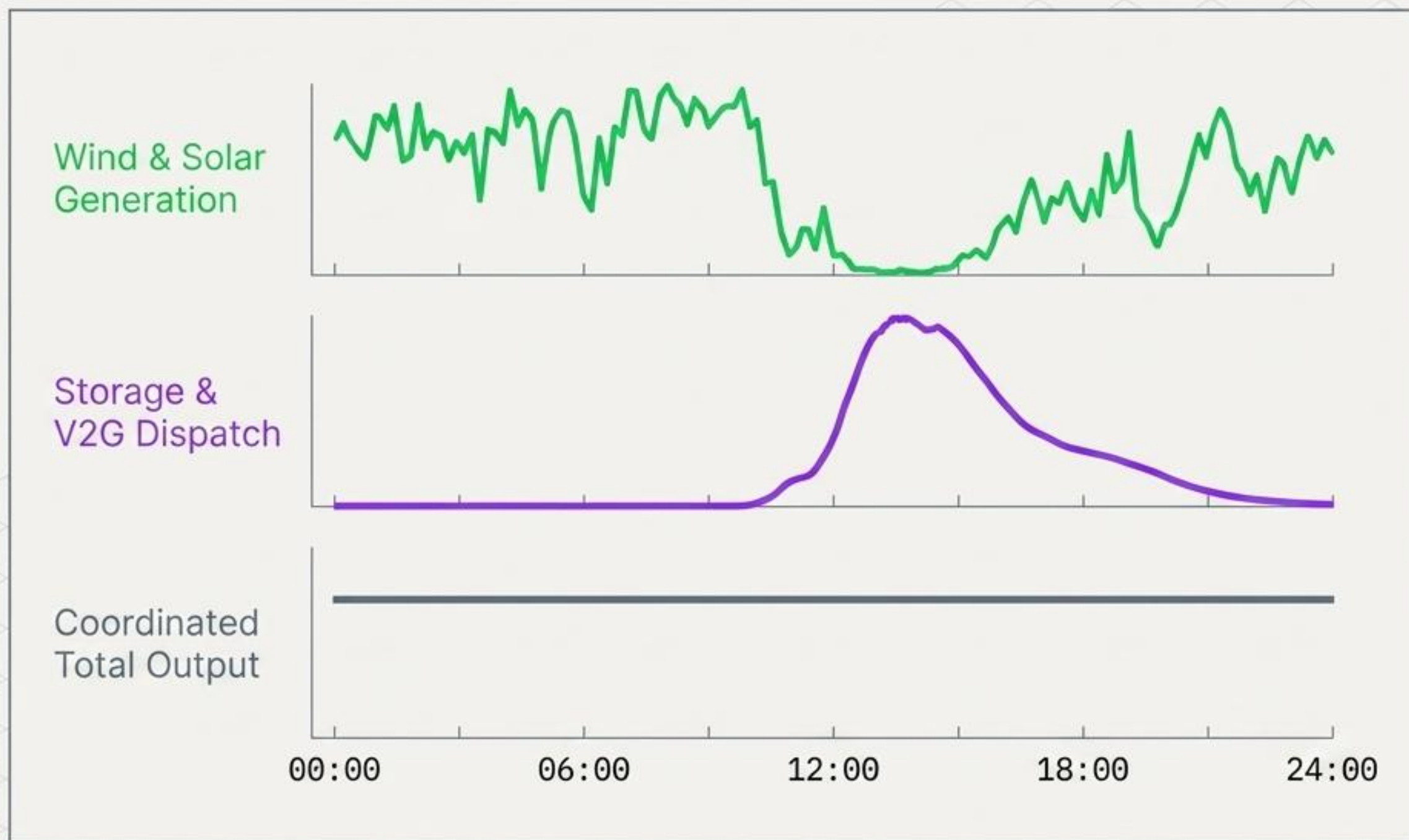


- **Hierarchical Architecture:** Distributes computational load and prevents central bottlenecks.

- **Multi-Agent Coordination:** Employs ADMM consensus algorithms to synchronize independent DERs.

- **Temporal Coordination:** Aligns fast-acting inverters with slow-moving mechanical generation.

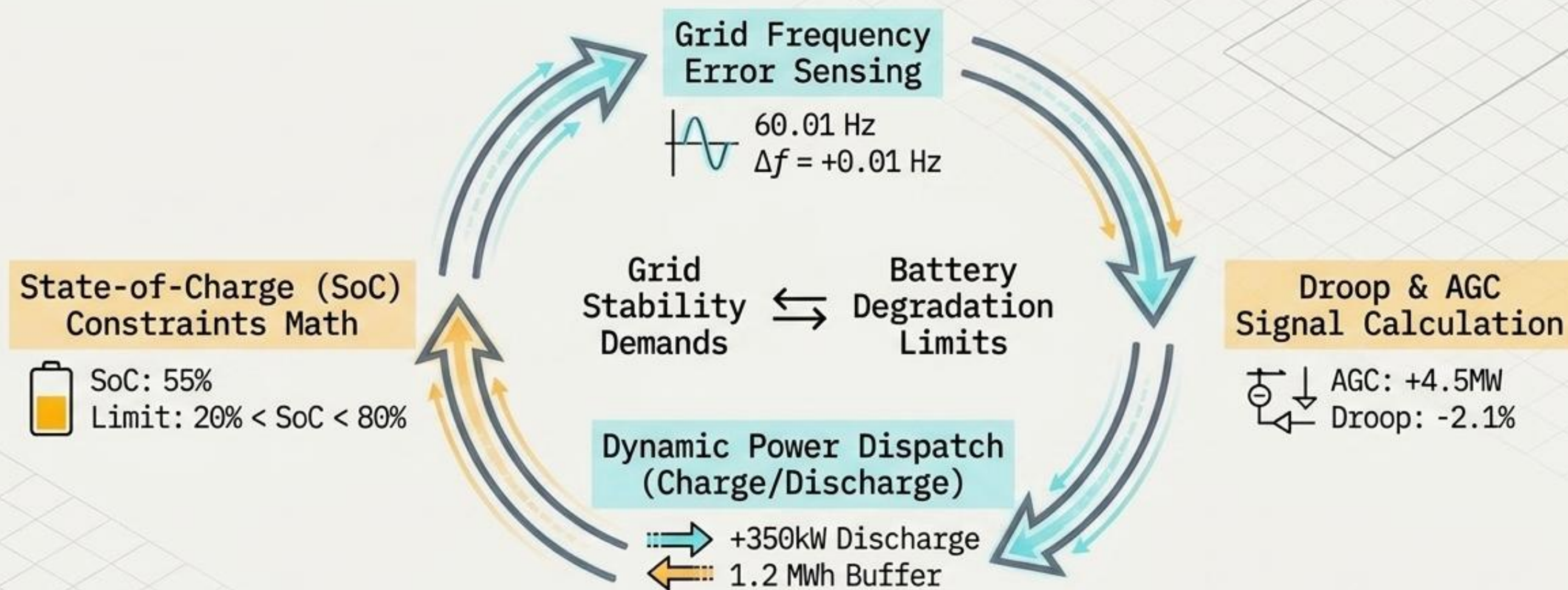
Renewable Energy Coordination



Intermittency Smoothing Profile

1. Senses weather-dependent generation dips via PMU data.
2. Temporally coordinates battery storage buffers.
3. Results in a flat, predictable grid injection.

Battery Coordination & SoC Management



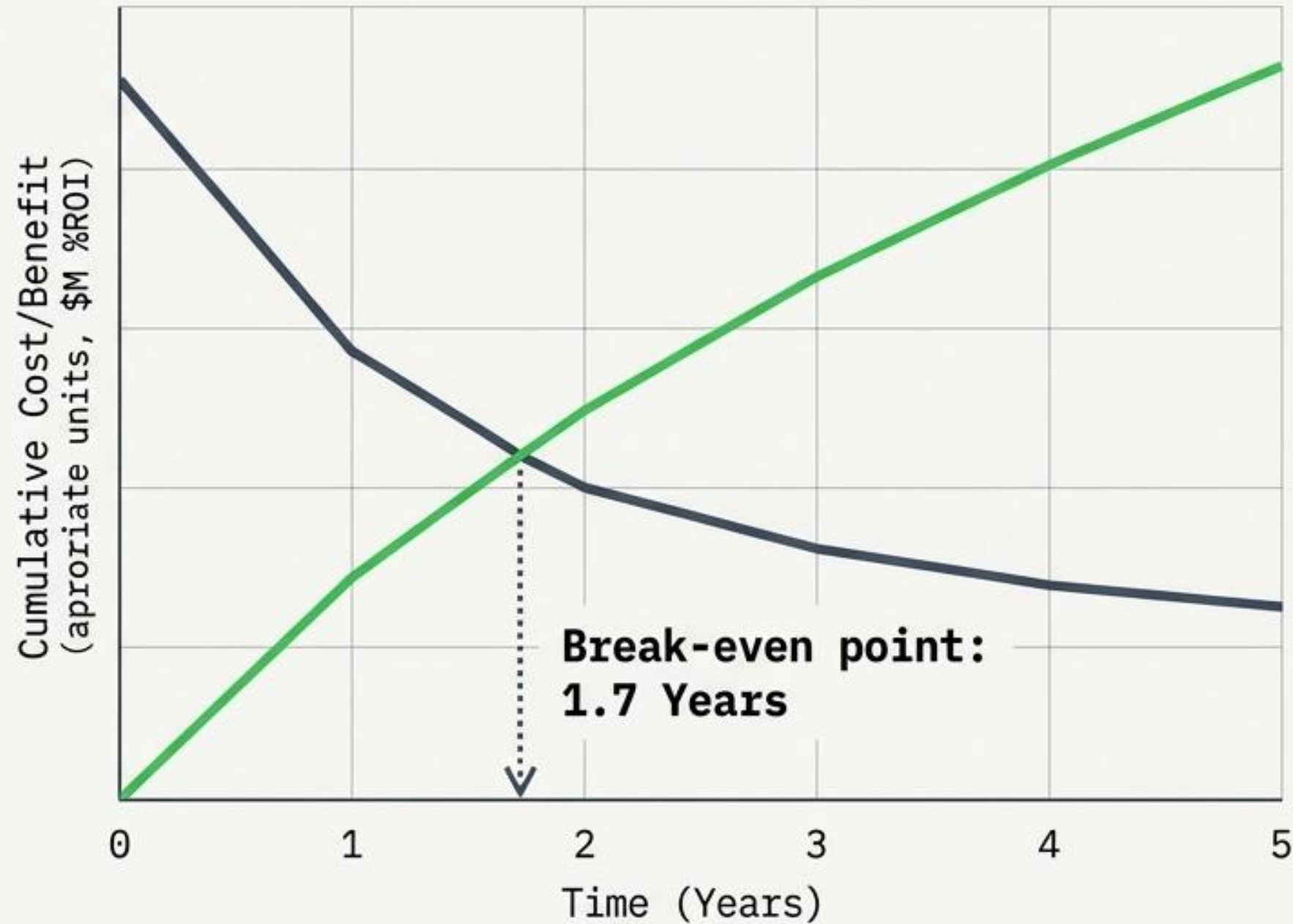
Maintains optimal battery levels for consumer EV driving schedules.



Constrains V2G cycling to prevent accelerated capacity fade and semi-empirical degradation.



Economic Impact & V2G Simulation Results



12.4%



Annual operational cost reduction
(\$51.5M/year on IEEE 118-bus)

42.5%



Reduction in renewable energy
curtailment

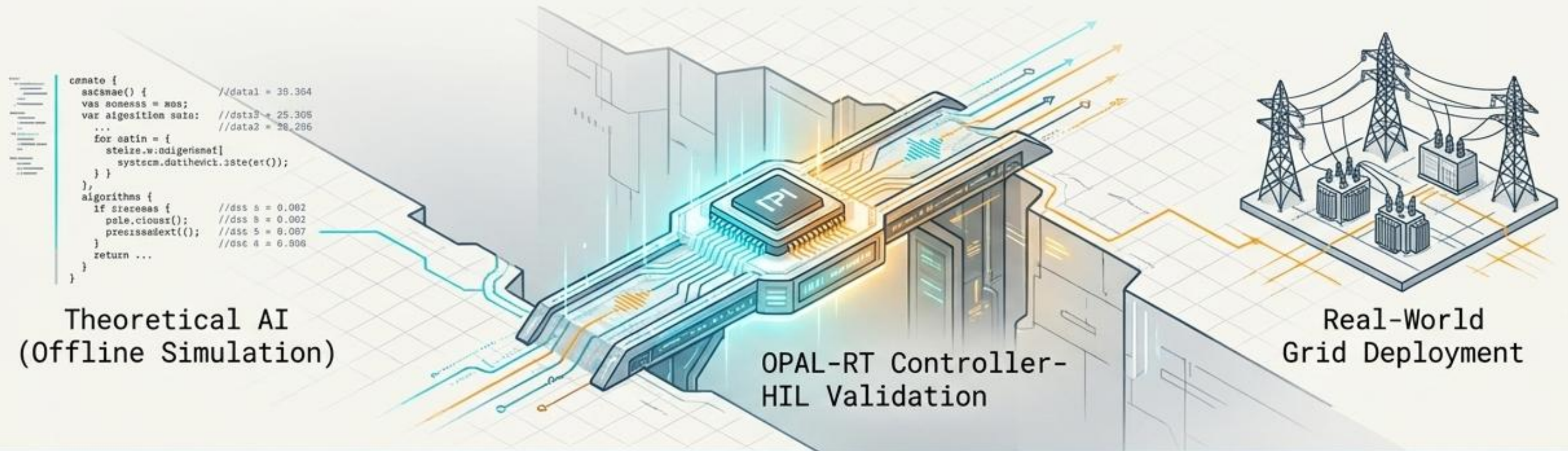
28.0%



Increase in Distributed Energy
Resource (DER) utilisation efficiency

Integrating wind and solar intermittency with V2G dispatch yields an **83.7% Internal Rate of Return**.

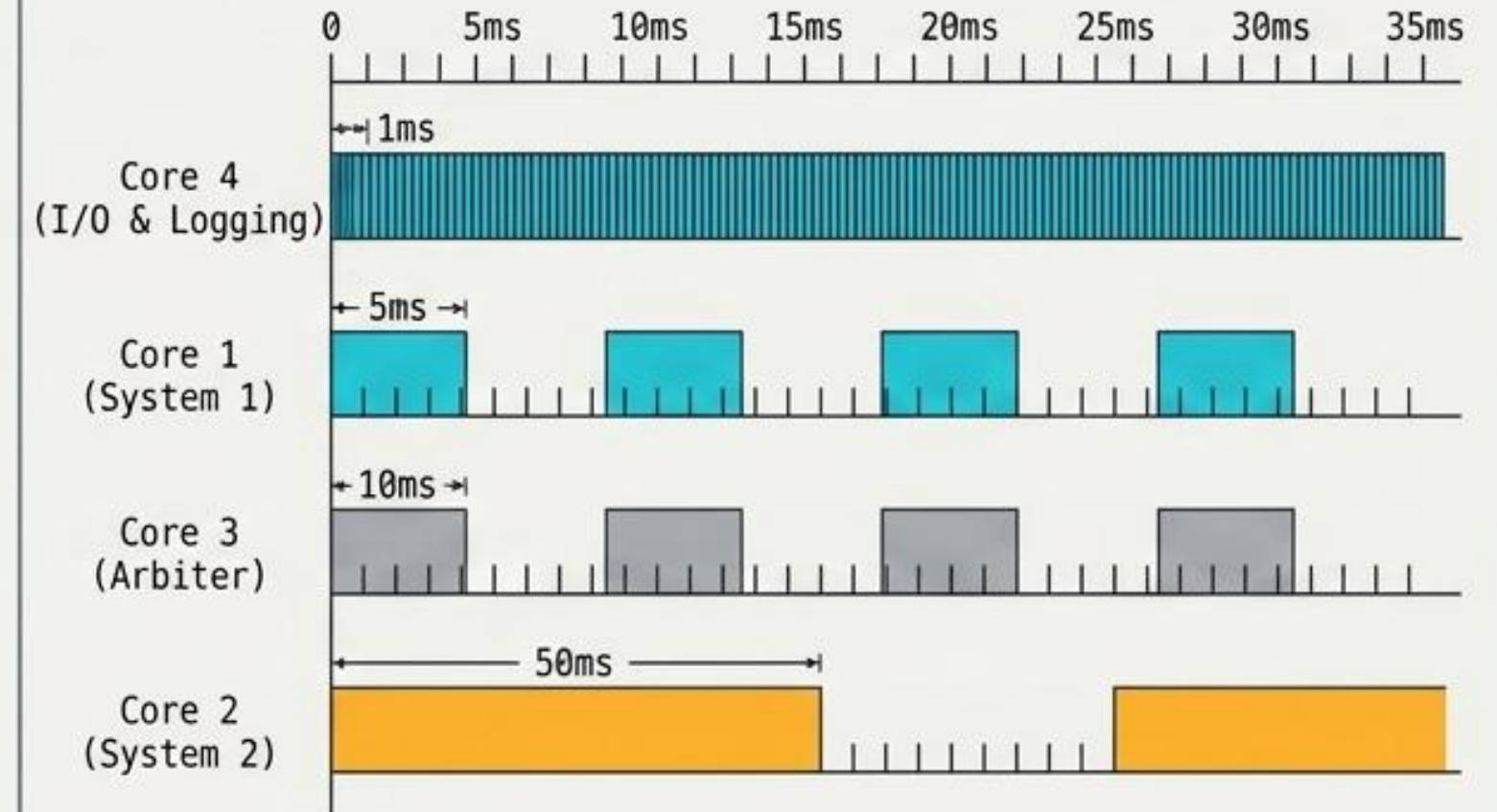
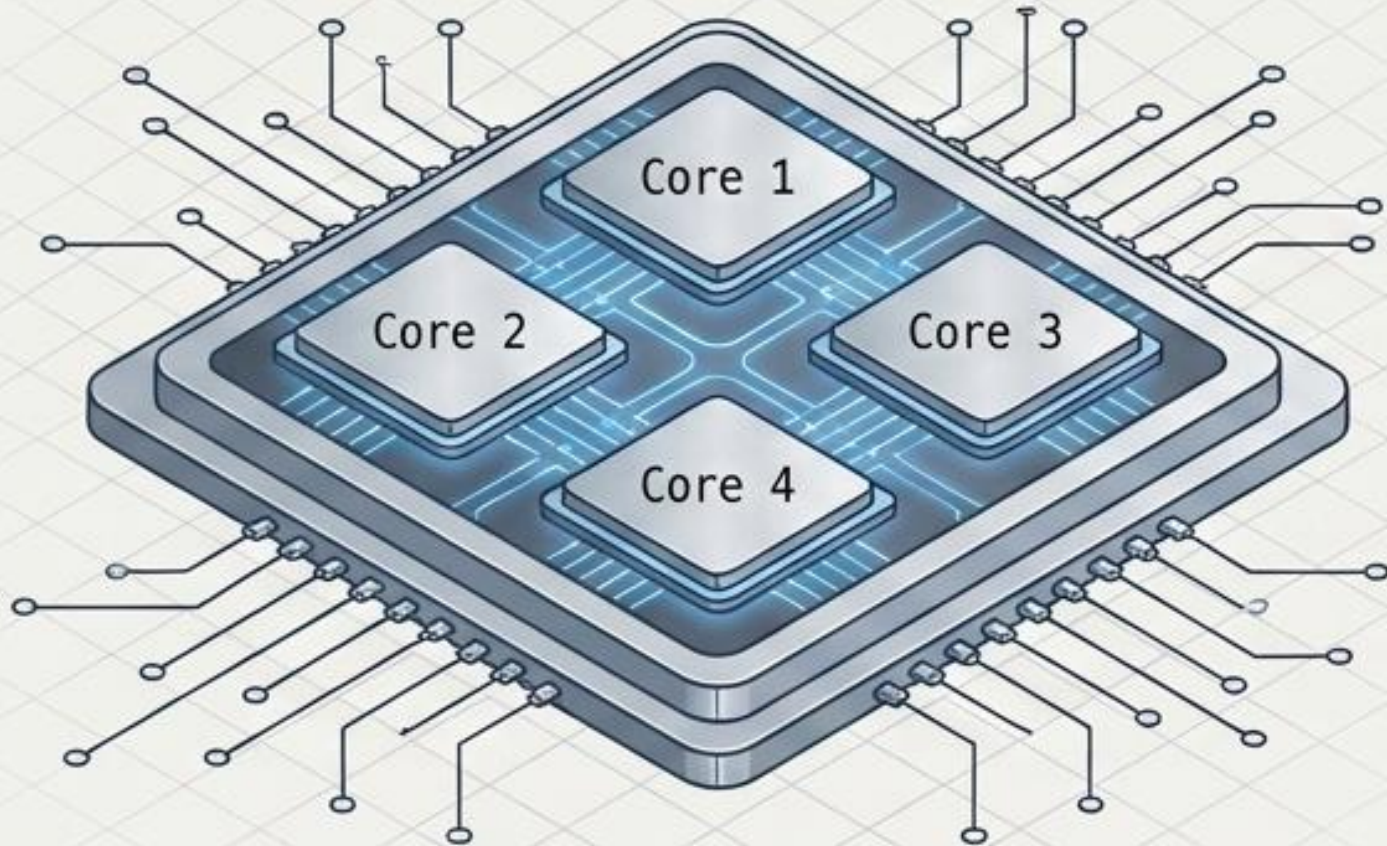
The Validation Gap: Moving AI to Physical Hardware



Constraint	Offline Simulation	Hardware-in-the-Loop (HIL)
Time Management	Pauses for infinite compute	Strict real-time milliseconds
Latency	Idealised zero delay	Actual physical comms delay
Processing Limits	Infinite cloud computing	Hardware-bound (4-Core)
Validity Level	Theoretical Math Proof	TRL-6 Field Ready

Theoretical AI models often fail in the field due to communication latency and real-time execution limits. HIL validation is the essential bridge.

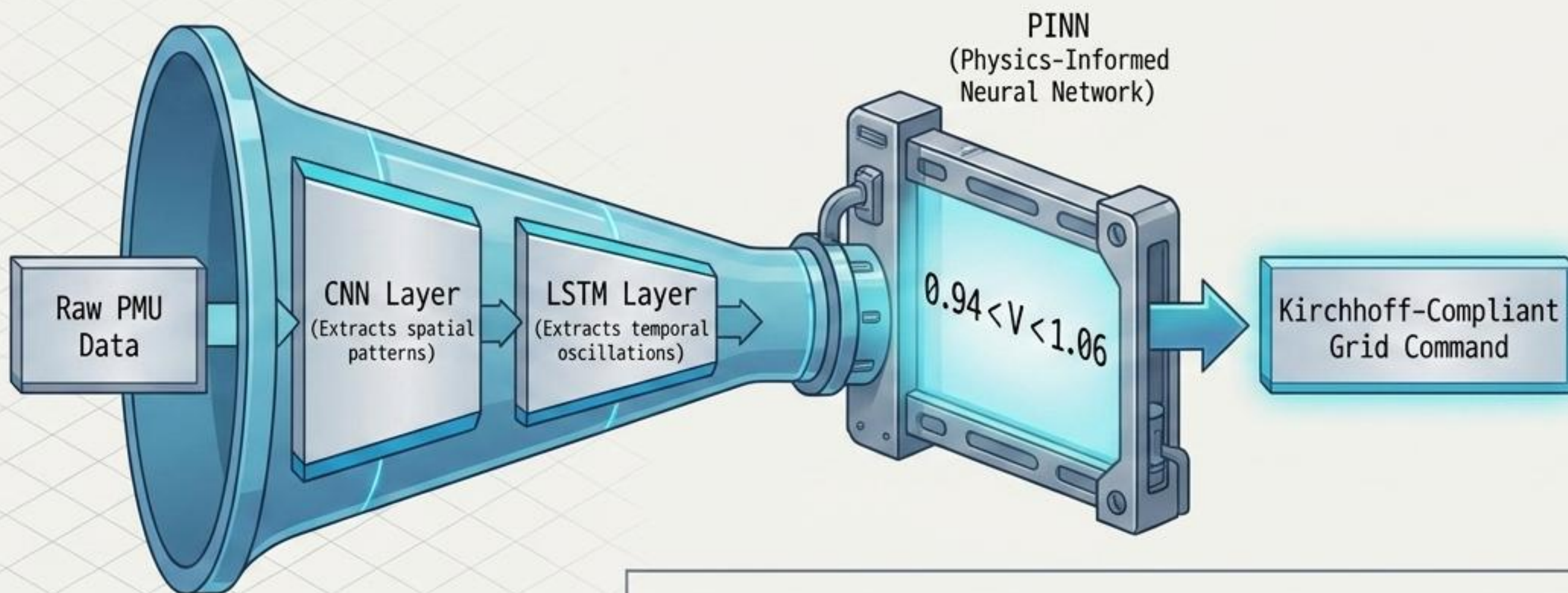
4-Core OPAL-RT Architecture & Task Partitioning



Multi-timescale AI operation requires strict rate-monotonic scheduling. High-frequency I/O fires rapidly while cognitive planning runs on longer, overlapping cycles.



Core 1: System 1 CNN-LSTM Reflex



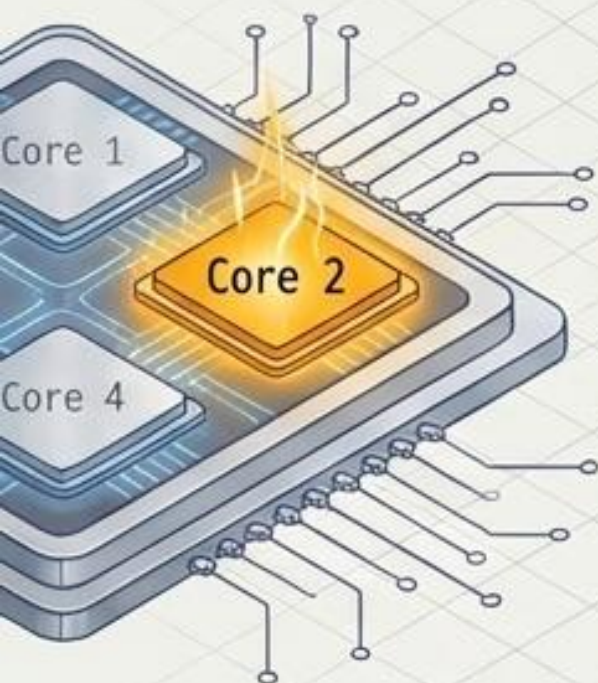
Assignment: North Region (Buses 1-19)

Constraint: ⌚ 5ms timestep (Highest Priority)

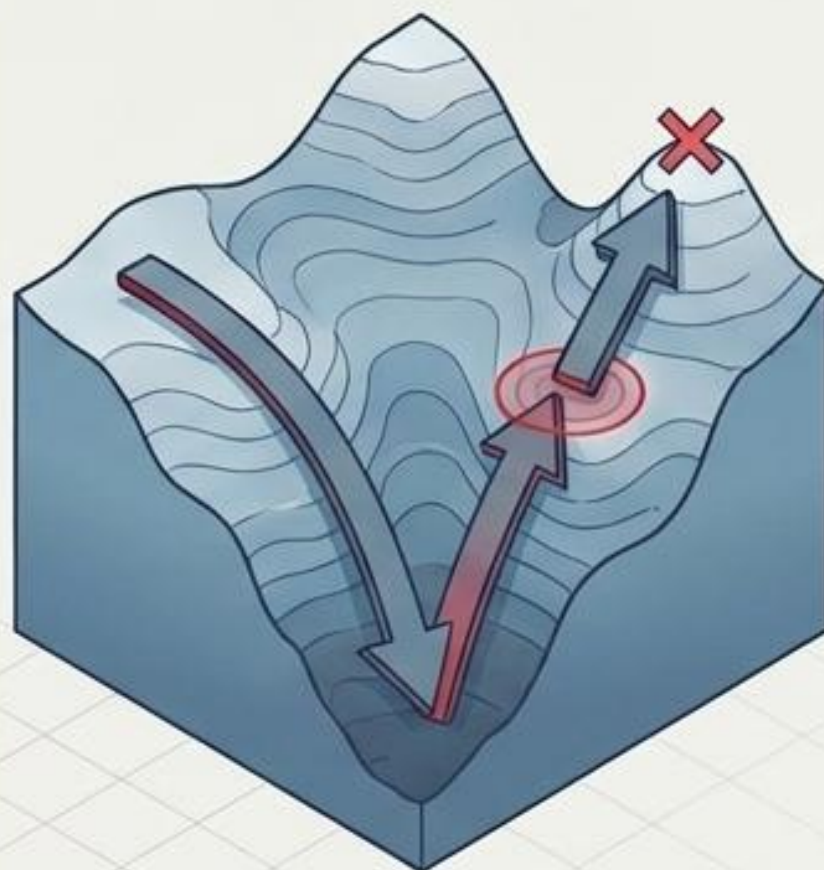
Function: Sub-millisecond inference for immediate grid protection and fault detection.



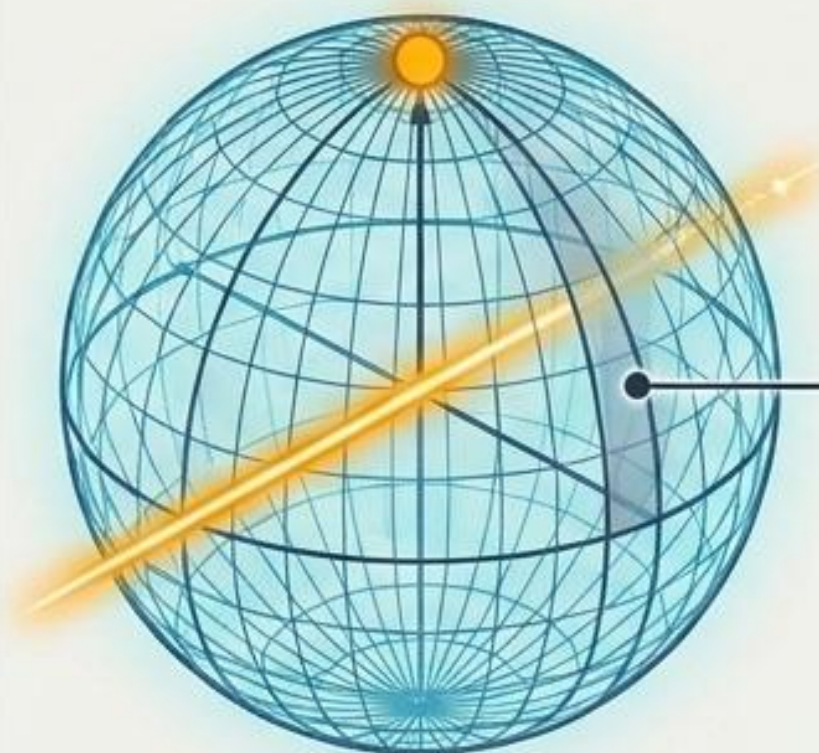
Core 2: System 2 QIRL Optimisation



Standard Algorithm
(Trapped in Local Optimum)



Quantum-Inspired Tunneling
(Finds Global Optimum)

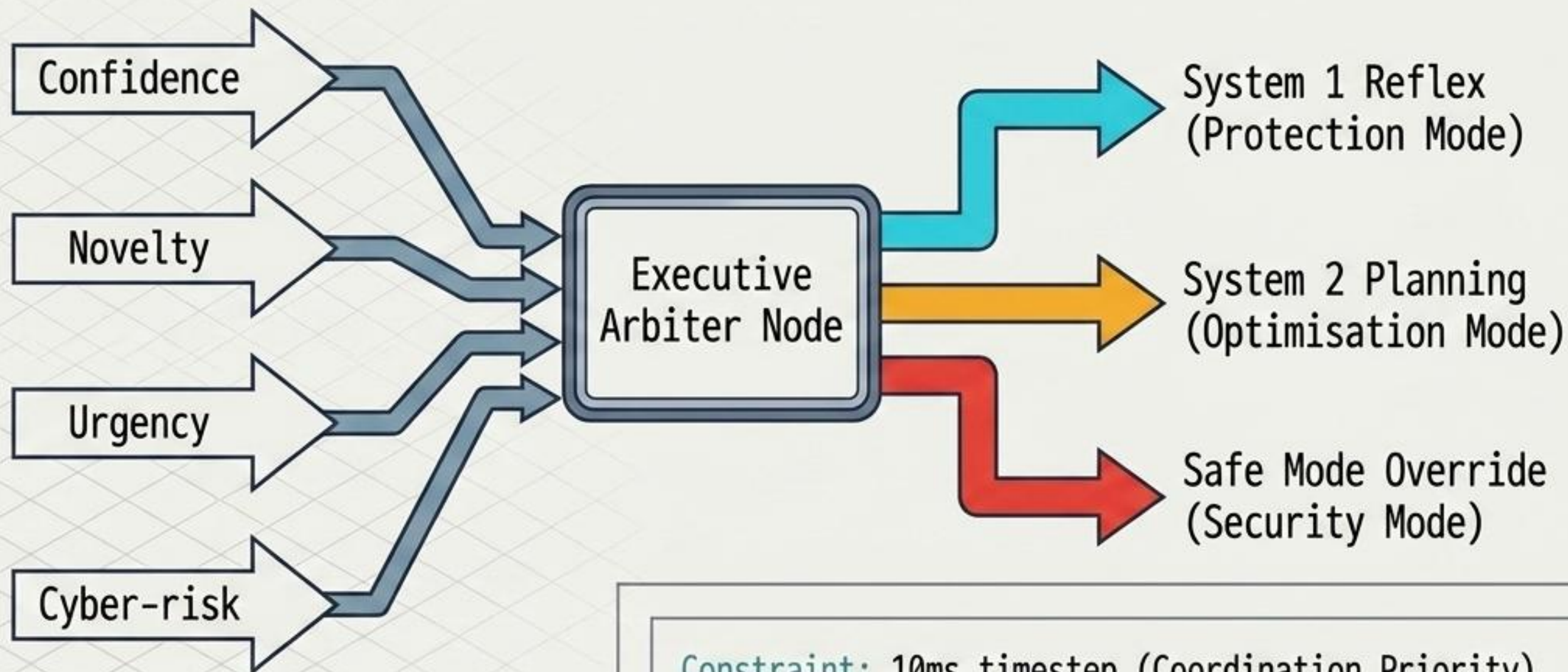
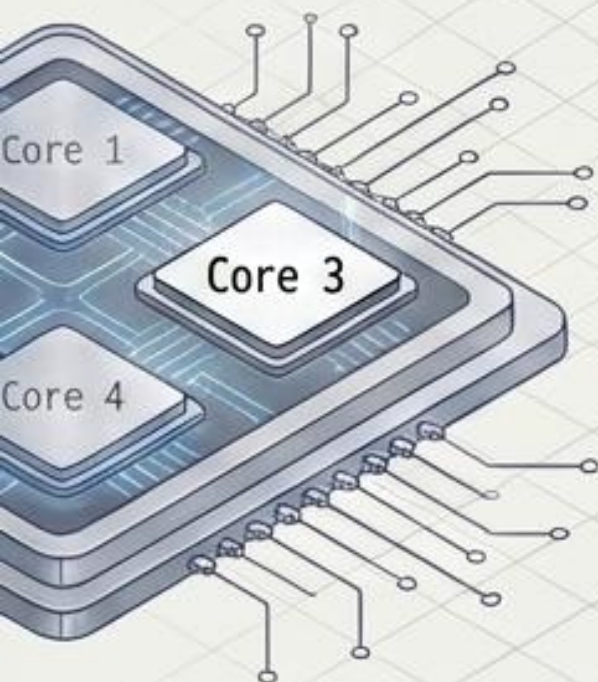


Quantum-Inspired
Tunneling

Assignment: South
Region (Buses 20-39)
Constraint: 50ms
timestep
(Medium Priority)
Performance:
Quantum-Inspired
Reinforcement
Learning (QIRL)
achieves **58% faster**
convergence than
classical Deep
Q-Networks
(DQN) for economic
dispatch and
scheduling.

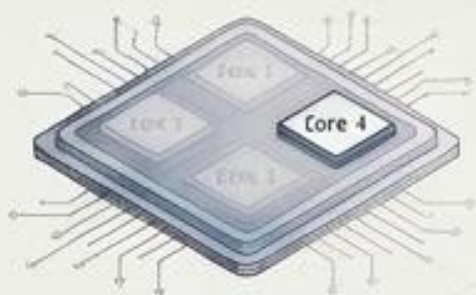


Core 3: Metacognitive Arbitration

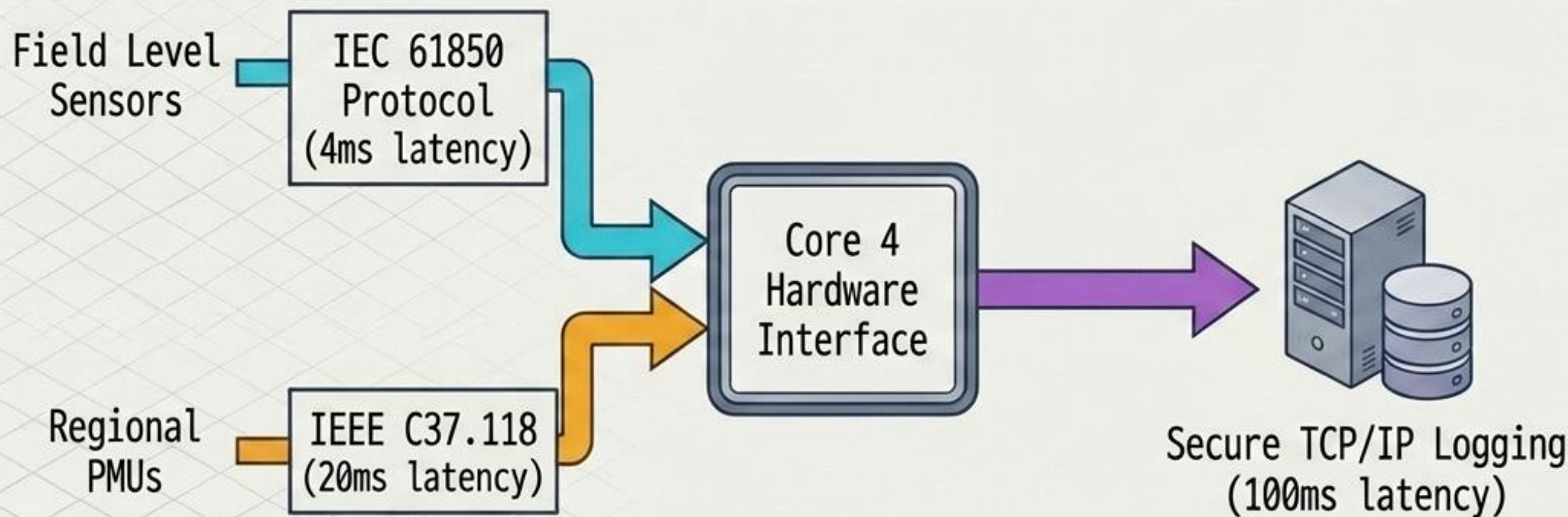
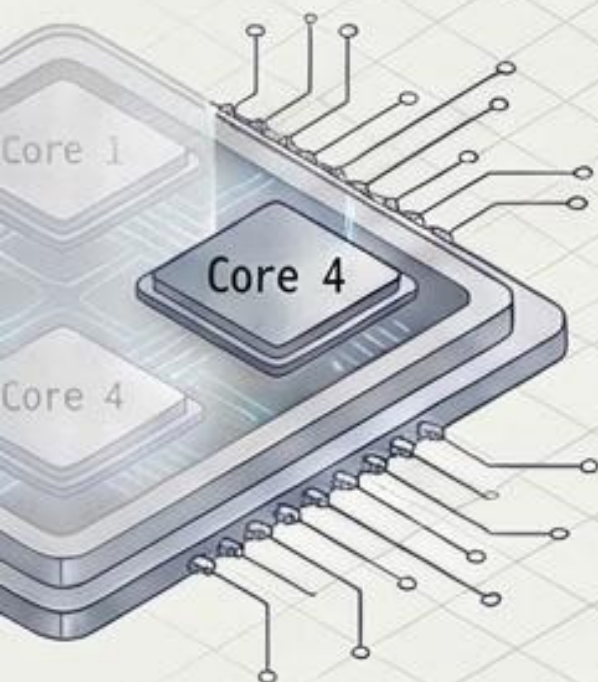


Constraint: 10ms timestep (Coordination Priority)

Function: Manages the tension between fast protection and slow optimisation, instantly reverting to safe physical setpoints if cyber-anomalies are detected.



Core 4: I/O and Logging Interface



Constraint: 1ms timestep (Lowest Computational Priority)

Function: Dedicated exclusively to physical hardware synchronization, data logging, and HIL I/O interfaces to prevent network traffic from interrupting AI inference on Cores 1-3.

RT-LAB Configuration & The 5-Scenario Gauntlet

OPAL-RT RT-LAB Configuration

ARTEMIS Solver locked at strict **50 μ s**
Electromagnetic Transient (EMT) timestep.

Scenario 1

Normal
Operation

(Baseline
tracking)

Scenario 2

Fault
Response

(3-phase short
circuit)

Scenario 3

Intermittency

(Wind generation
drop)

Scenario 4

Cyber
Attack

(False Data
Injection)

Scenario 5

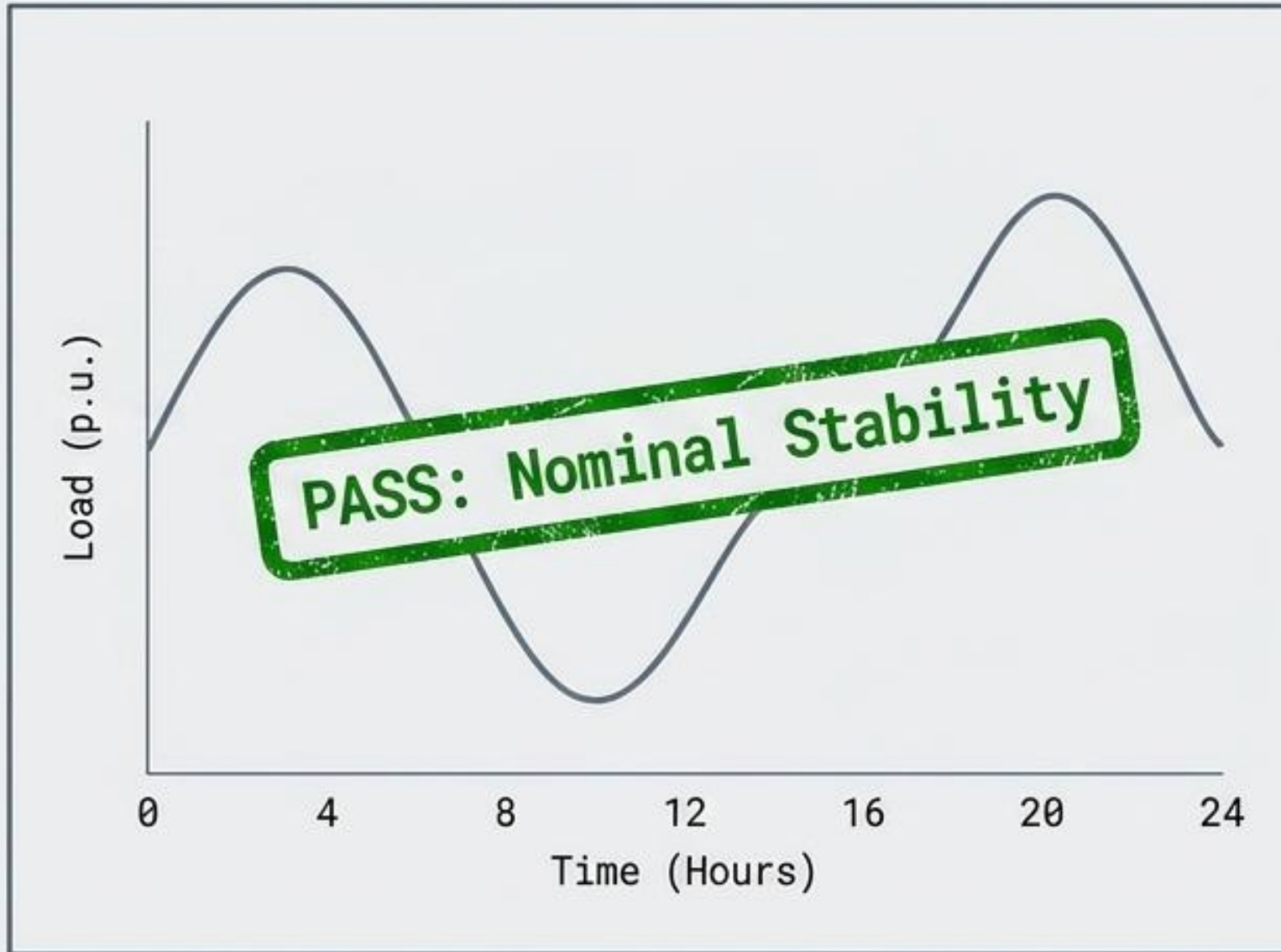
EV Fleet

(V2G
Coordination)

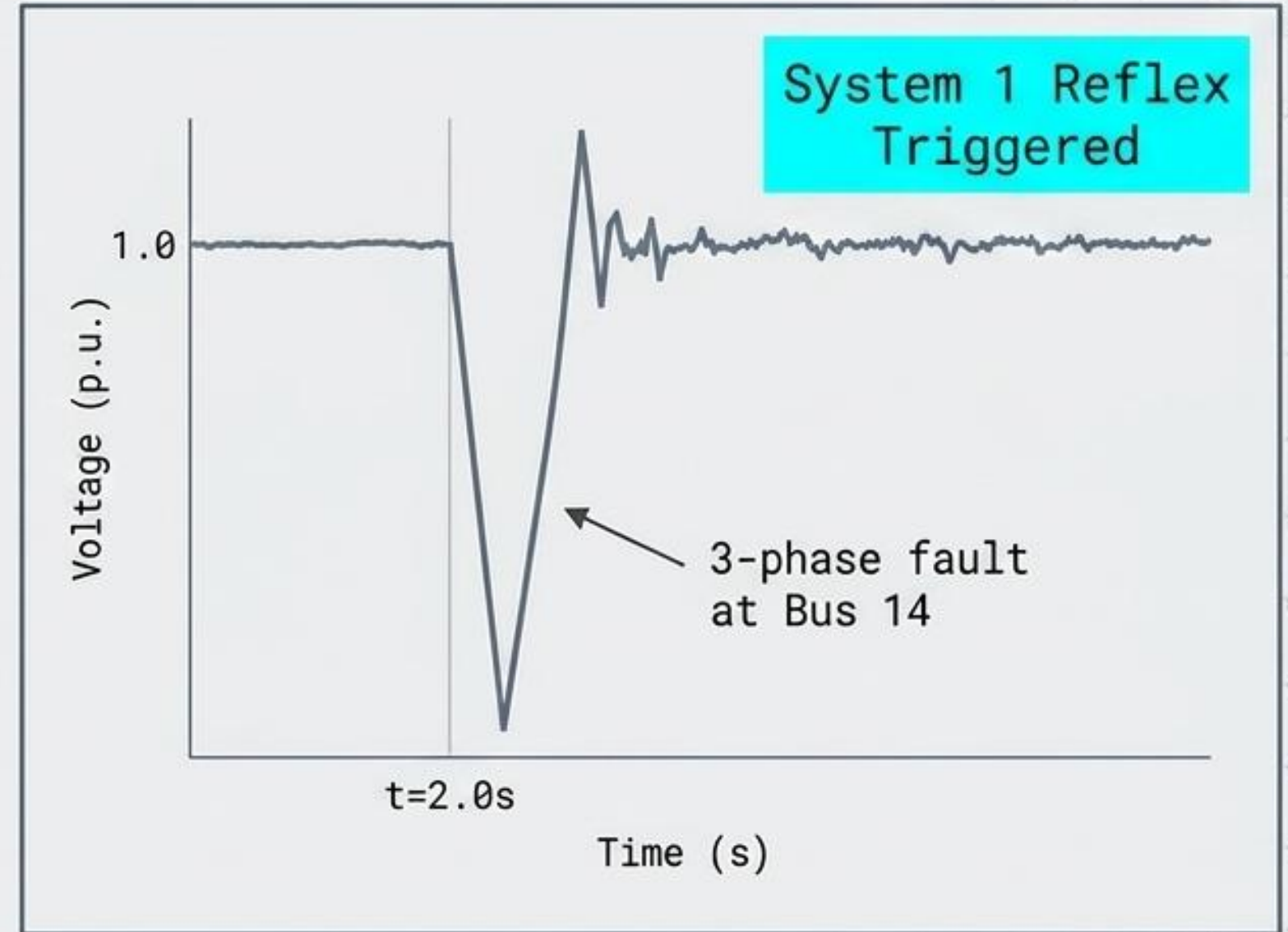
Note: The following telemetry results are recorded live from physical OPAL-RT hardware, not simulated offline outputs.

Telemetry: Normal Operation & Extreme Faults

Scenario 1: Daily Operations



Scenario 2: Physical Grid Fault



Fault Detection Metrics: Achieved with 95.8% accuracy and 1.73% localisation error, preventing cascading grid failure.

Telemetry: Intermittency & Cyber-Physical Resilience

Scenario 3: Renewable Drop

WIND GENERATION (MW)



SYSTEM 2 BATTERY DISPATCH (MW)



SYSTEM FREQUENCY (Hz)

60.00 Hz — STEADY

Scenario 4: Cyber Attack (FDI) ⚠



CRITICAL ALERT: False Data Injection detected at Buses 5 and 10

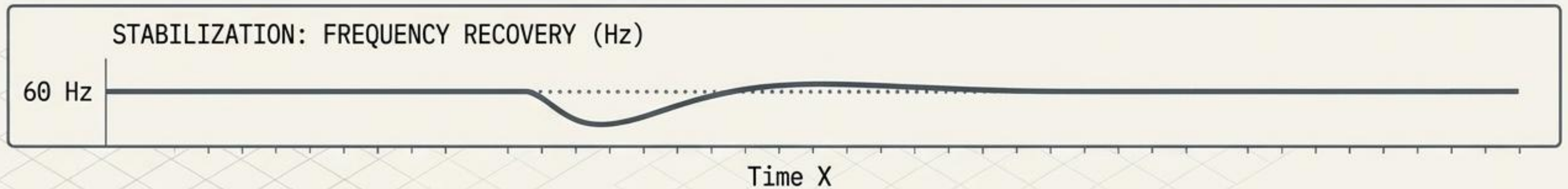
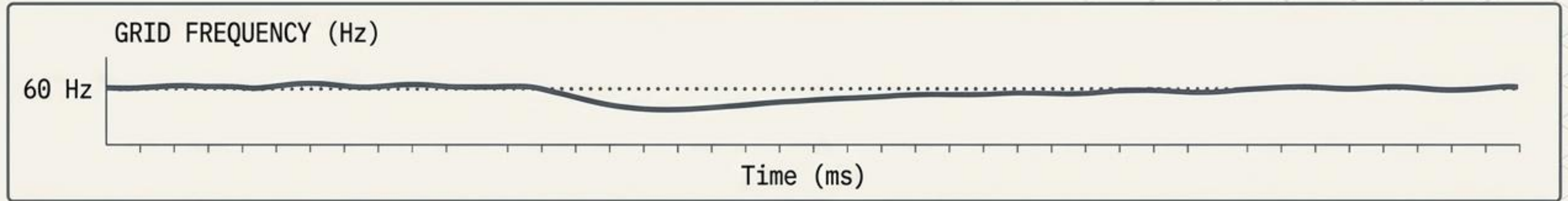
SCADA INPUTS

DATA VALIDATION
(FAILED)

SAFE MODE ACTIVATED
Algorithmic optimisation
overridden to maintain
physical stability.

The Metacognitive Arbiter successfully manages both environmental variability and malicious external data poisoning.

Scenario 5: Real-Time EV V2G Coordination



Hardware Validation Confirmed: Bidirectional EV chargers successfully synchronized via Phase-Locked Loops and executed smart charging strategies within strict 50 μ s solver constraints.

The Reality of CAPSM: Hardware Validated, Economically Proven



Real-Time Response Confirmed on physical OPAL-RT silicon.

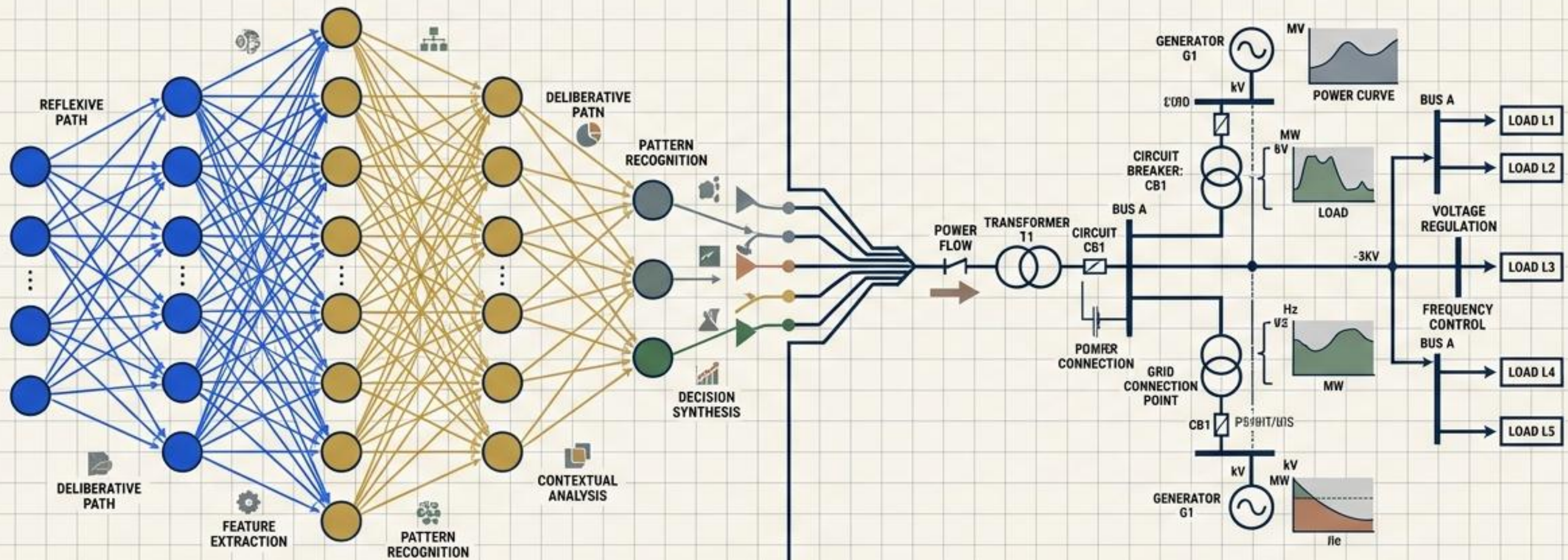
12.4%
Cost Savings



The 12.4% economic savings and 37% stability enhancements are no longer theoretical. The brain-inspired dual-process AI successfully controlled a 39-bus power system in sub-10ms conditions and is ready for industry integration.

Cognitive Adaptive Power System Management

A Brain-Inspired Dual-Process AI Framework Validated
Through Controller-Hardware-in-the-Loop Testing



PhD Defense Presentation | Mahmoud Kiasari | Dalhousie University

Comprehensive Validation Yields Unprecedented Grid Performance

System 1 (Reflexive)

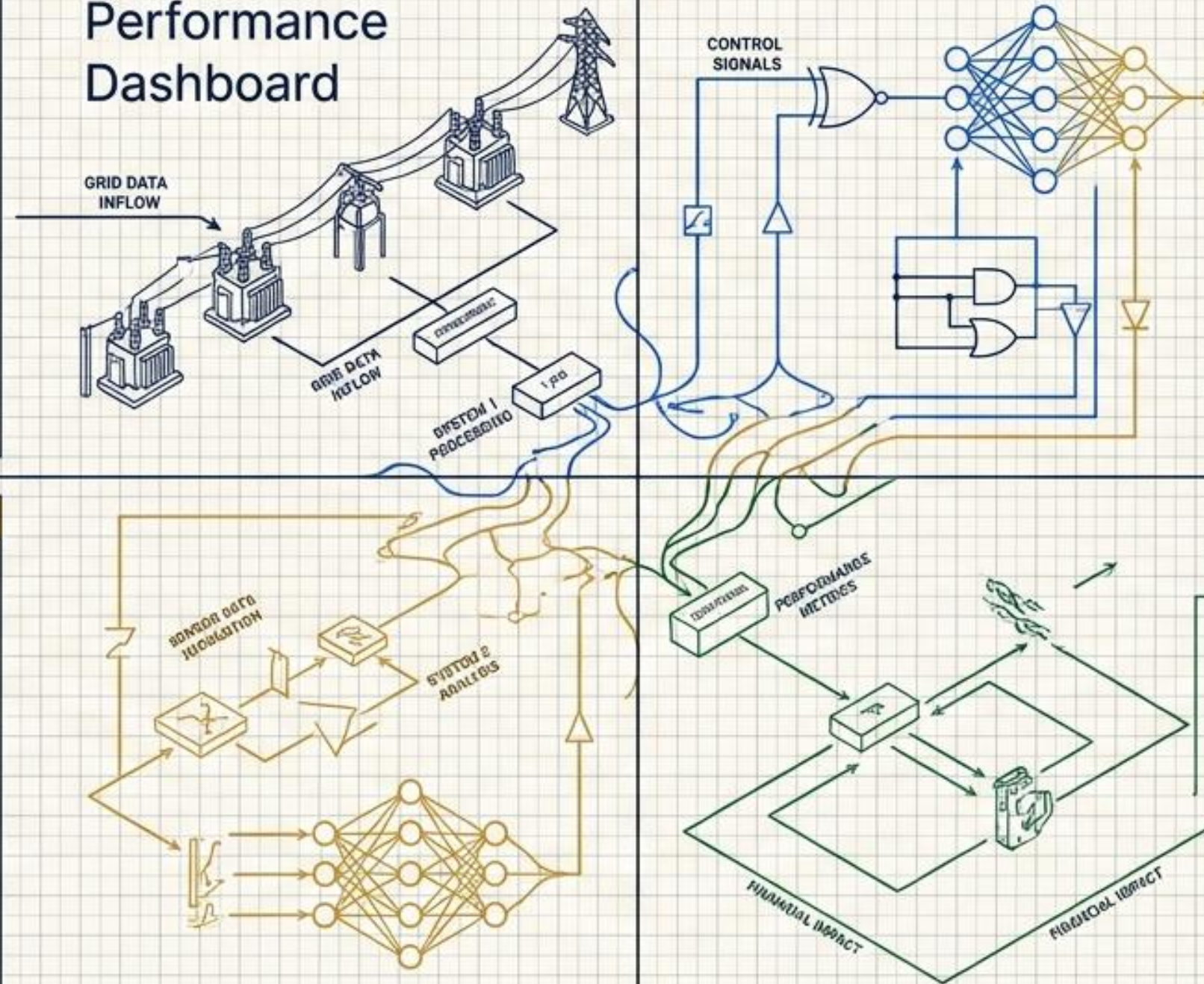
37%



Faster Response

Sub-10ms response times to critical events achieved on physical HIL.

Performance Dashboard



System 1

System 2

42%



Greater Stability

Overall voltage stability margin increased during extreme grid events.

System 2 (Deliberative)

95.8%



Detection Accuracy

Fault detection and localization accuracy via multi-modal fusion.

12.4%



Cost Savings

Annual operational cost reduction, yielding \$51.5M/year in savings.

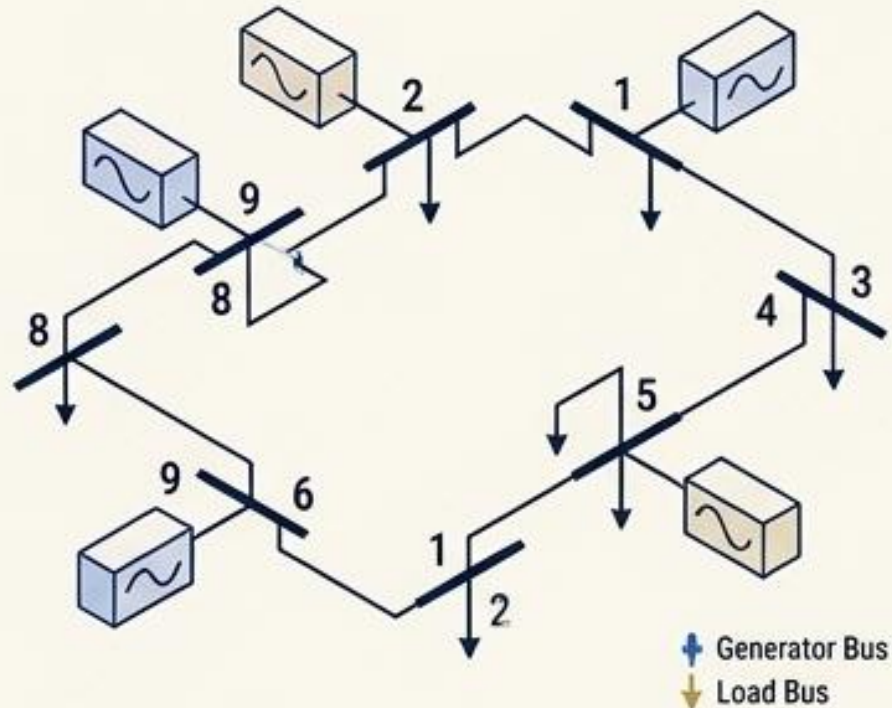
Performance validated consistently from Python simulations to real-time hardware.

Progressive Scaling From Debug to Dense Grid Topologies

Validation progression ensures robustness from simulation to real-time hardware environments.

Stage 1: Offline Python Validation & Algorithm Training

OpenAI Gym-like PYPOWER interface; rapid iteration without real-time constraints.

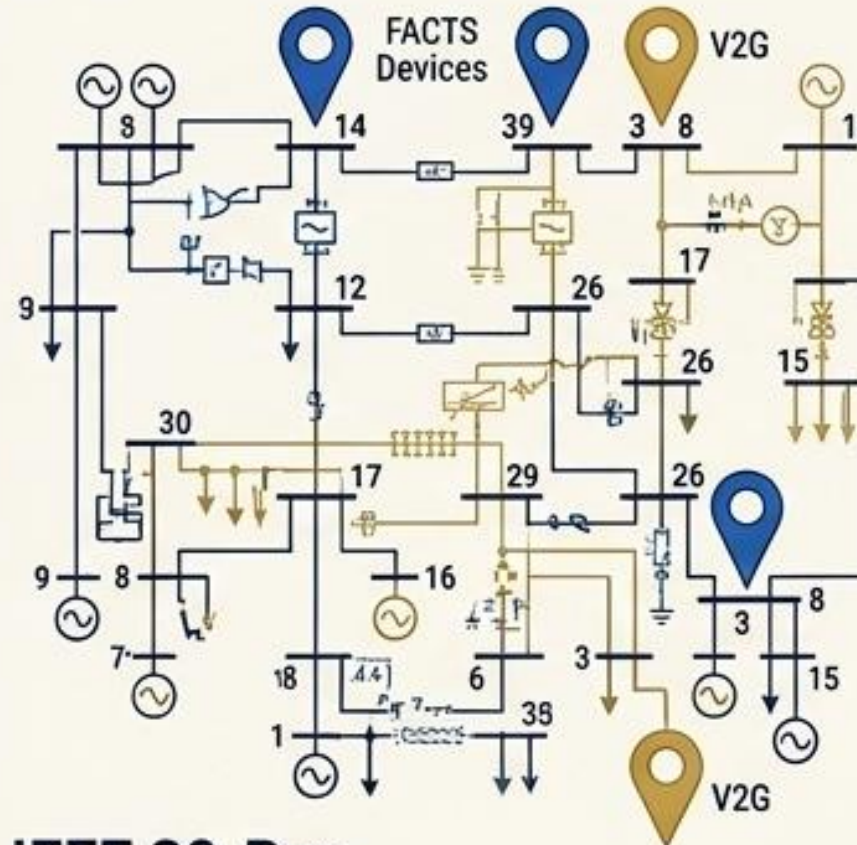


IEEE 9-Bus

OpenAI Gym-like PYPOWER interface; rapid iteration without real-time constraints.

Stage 2: Primary Test System

High-fidelity Simulink environment; handles dynamic 20ms resolution load profiles.

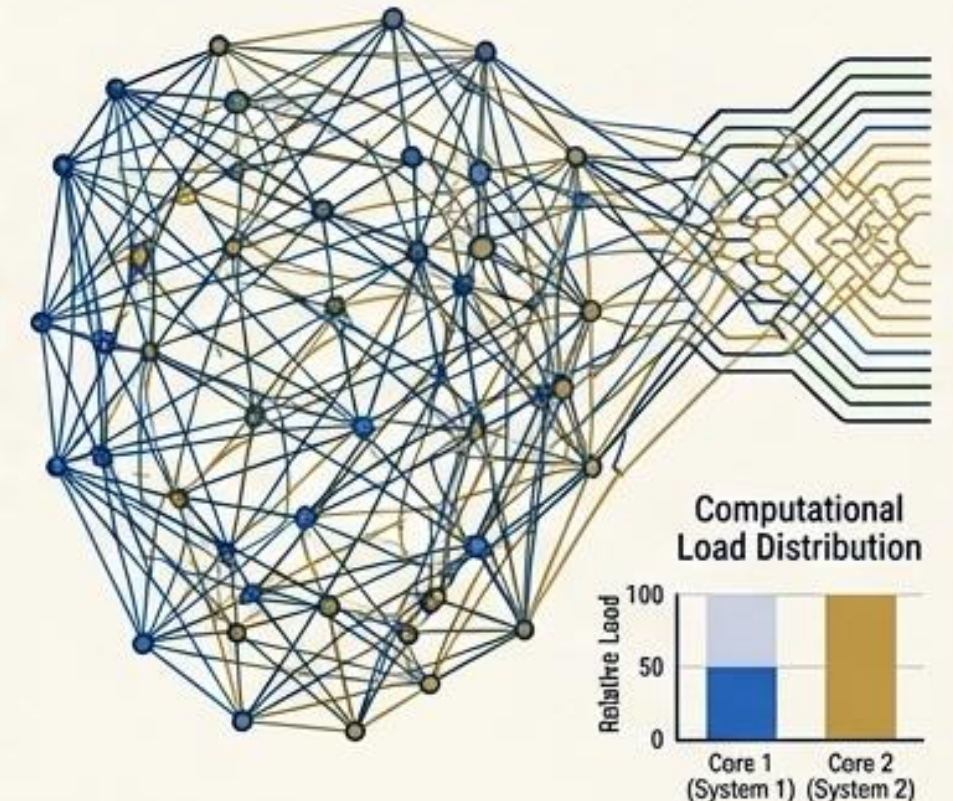


IEEE 39-Bus

High-fidelity Simulink environment; handles dynamic 20ms resolution load profiles.

Stage 3: Scalability Validation

Validates multi-core allocation feasibility; achieves \$51.5M/year operational savings on larger topologies.



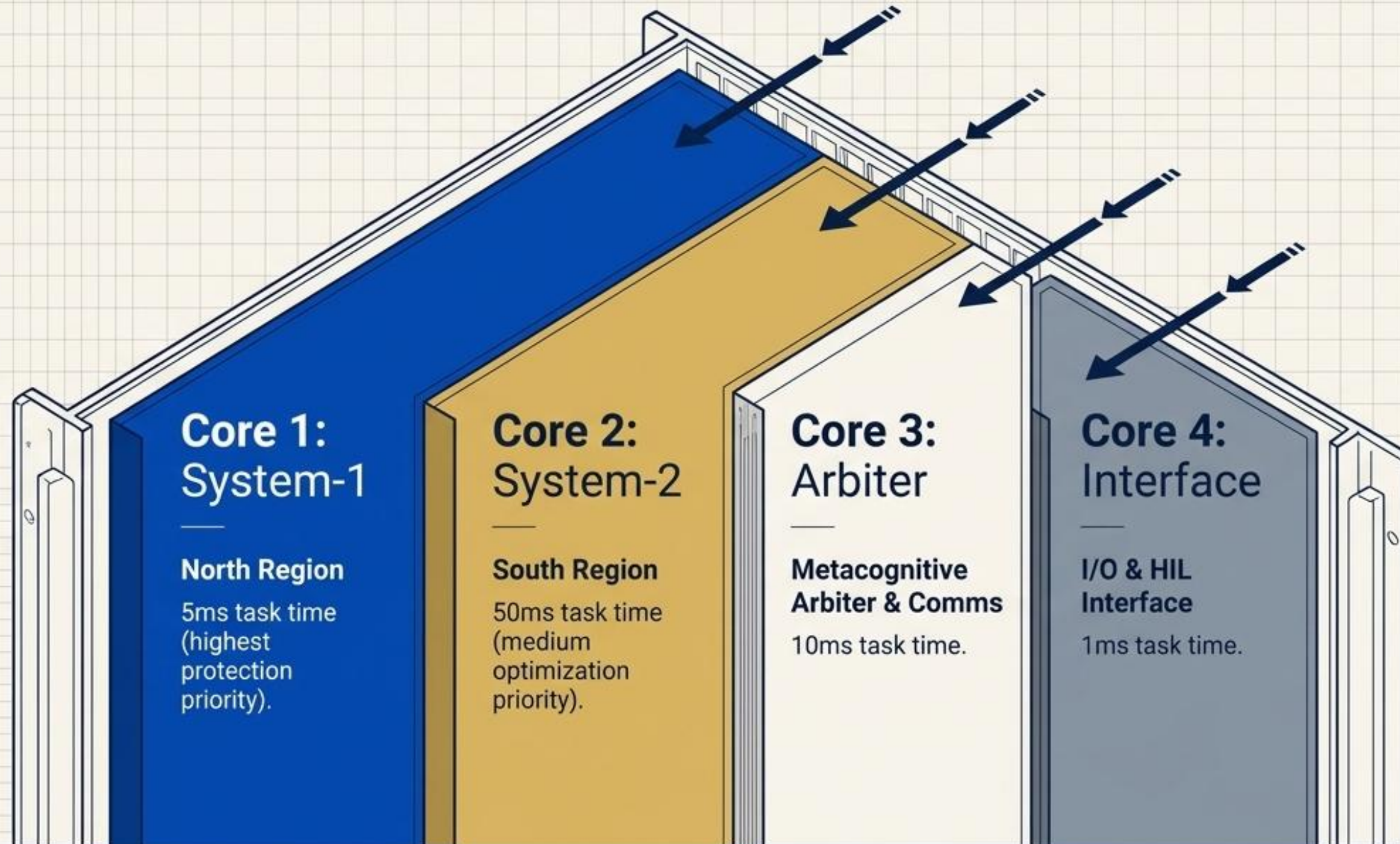
IEEE 118-Bus

Validates multi-core allocation feasibility; achieves \$51.5M/year operational savings on larger topologies.

Savings based on annual operational cost reduction.

Performance and scalability metrics validated through rigorous simulation and analysis across all stages.

Bridging Theory and Practice on OPAL-RT Real-Time Hardware

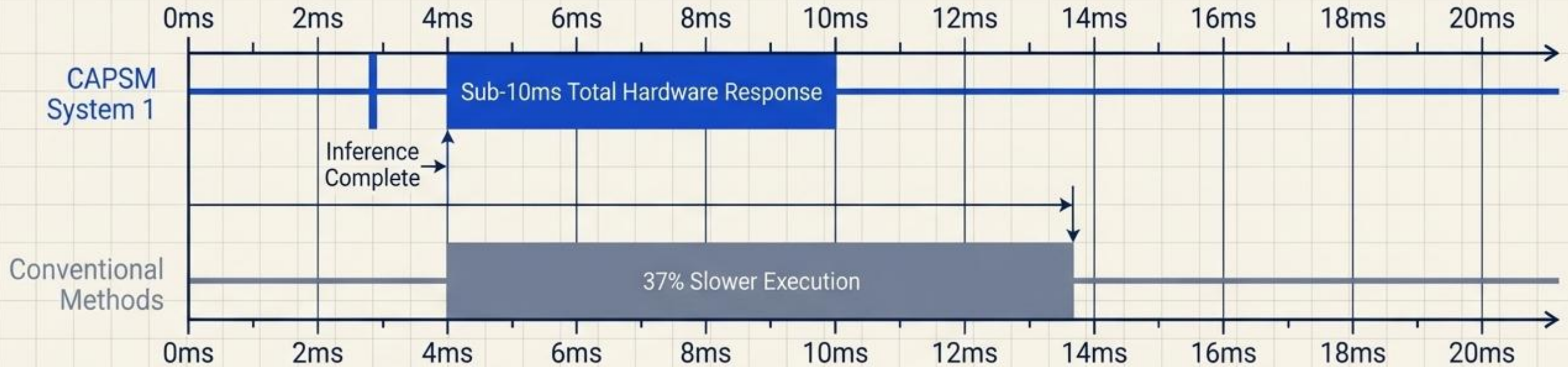


Hardware Reality

Executes at a 50 μ s
Electromagnetic
Transient (EMT) time step.

AI runs directly on
simulator CPUs.

The Reflexive Brain Executes Sub-10ms Protection Responses



Mechanism

CNN extracts spatial patterns from PMU bus data; LSTM tracks temporal oscillation dynamics.

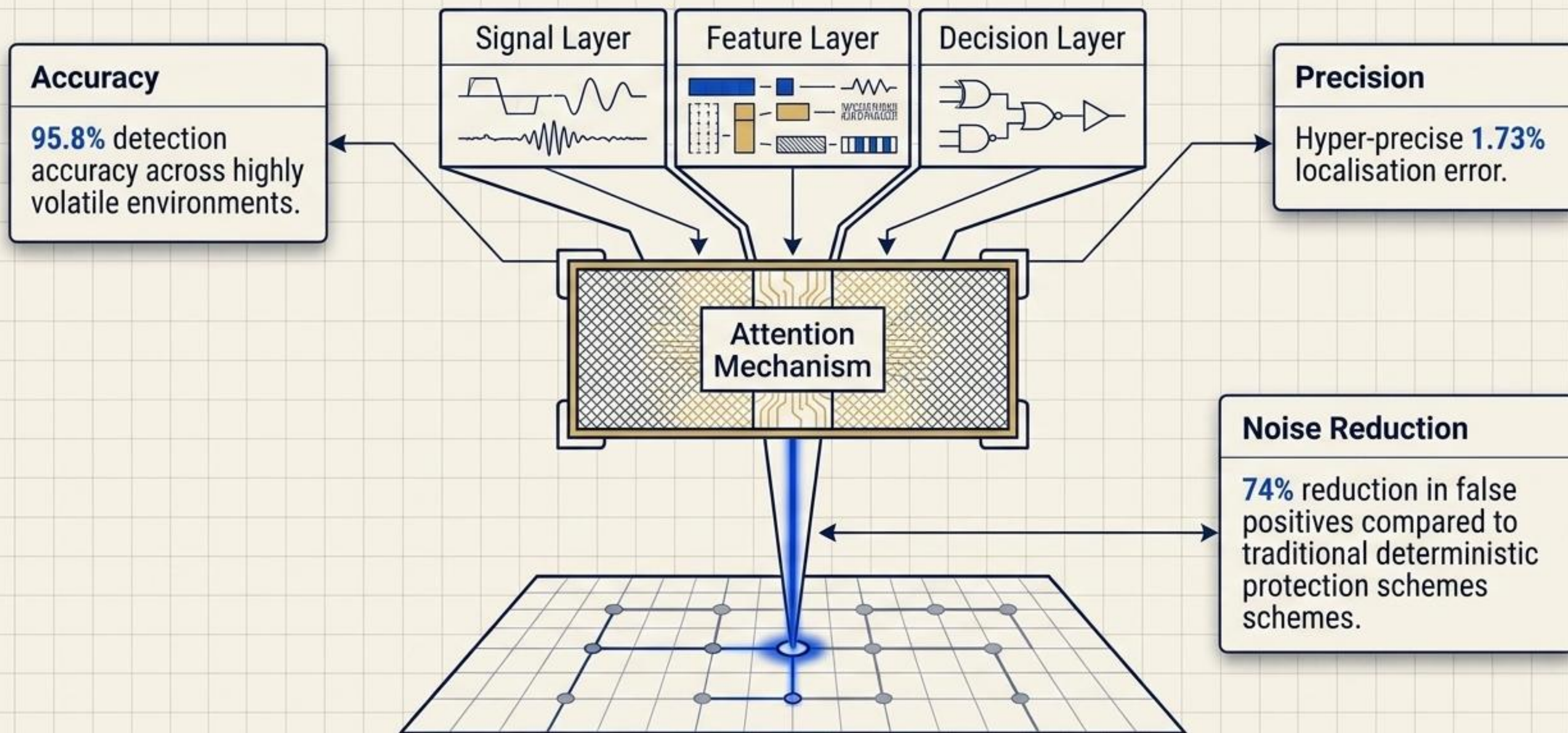
Hardware Reality

Achieves sub-10ms fault detection on physical HIL hardware, not just theoretical simulation.

Metacognitive Trigger

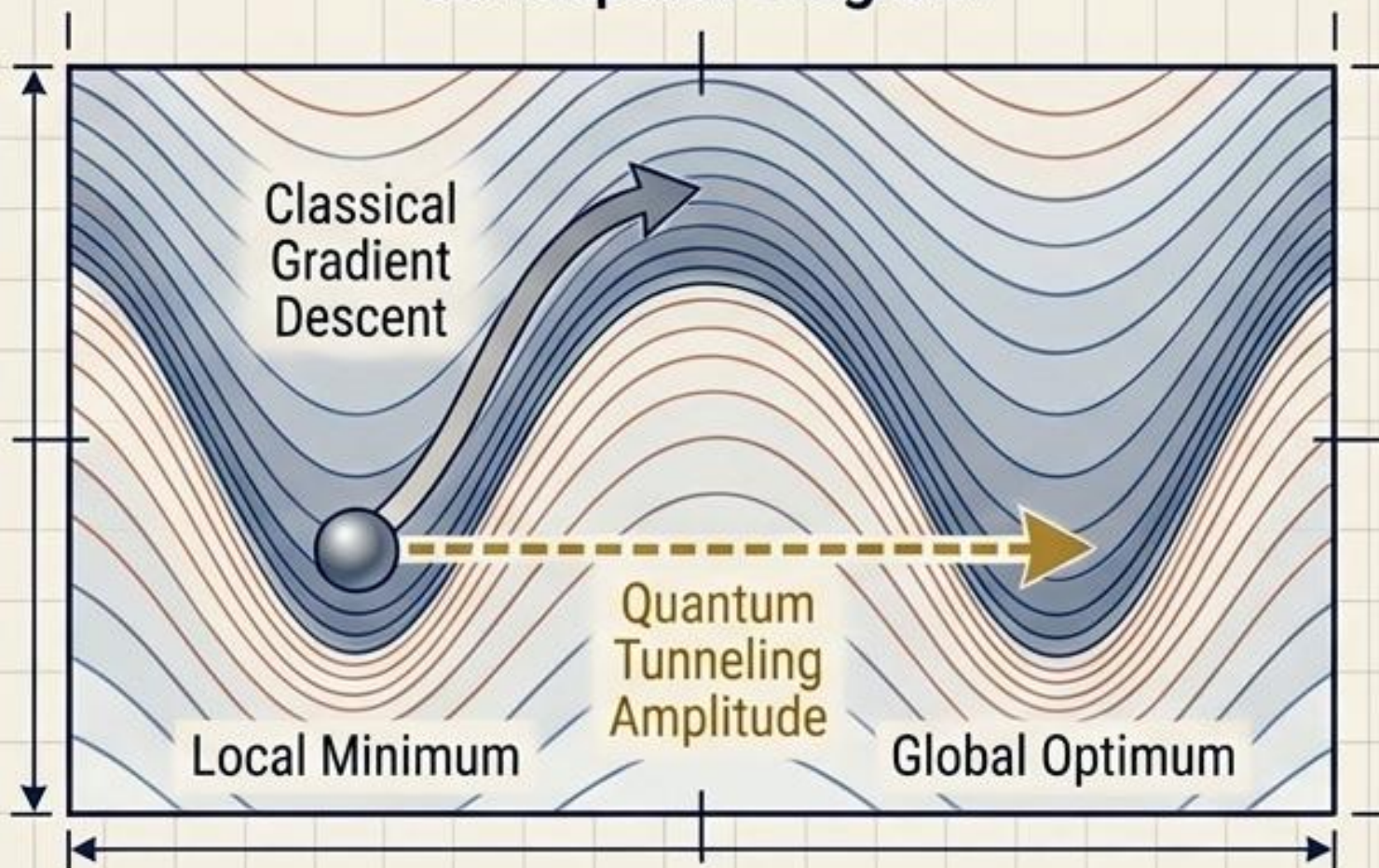
Automatically activated during high-urgency, low-novelty events (e.g., sudden voltage drops).

Multi-Modal Fusion Achieves Near-Perfect Fault Localisation

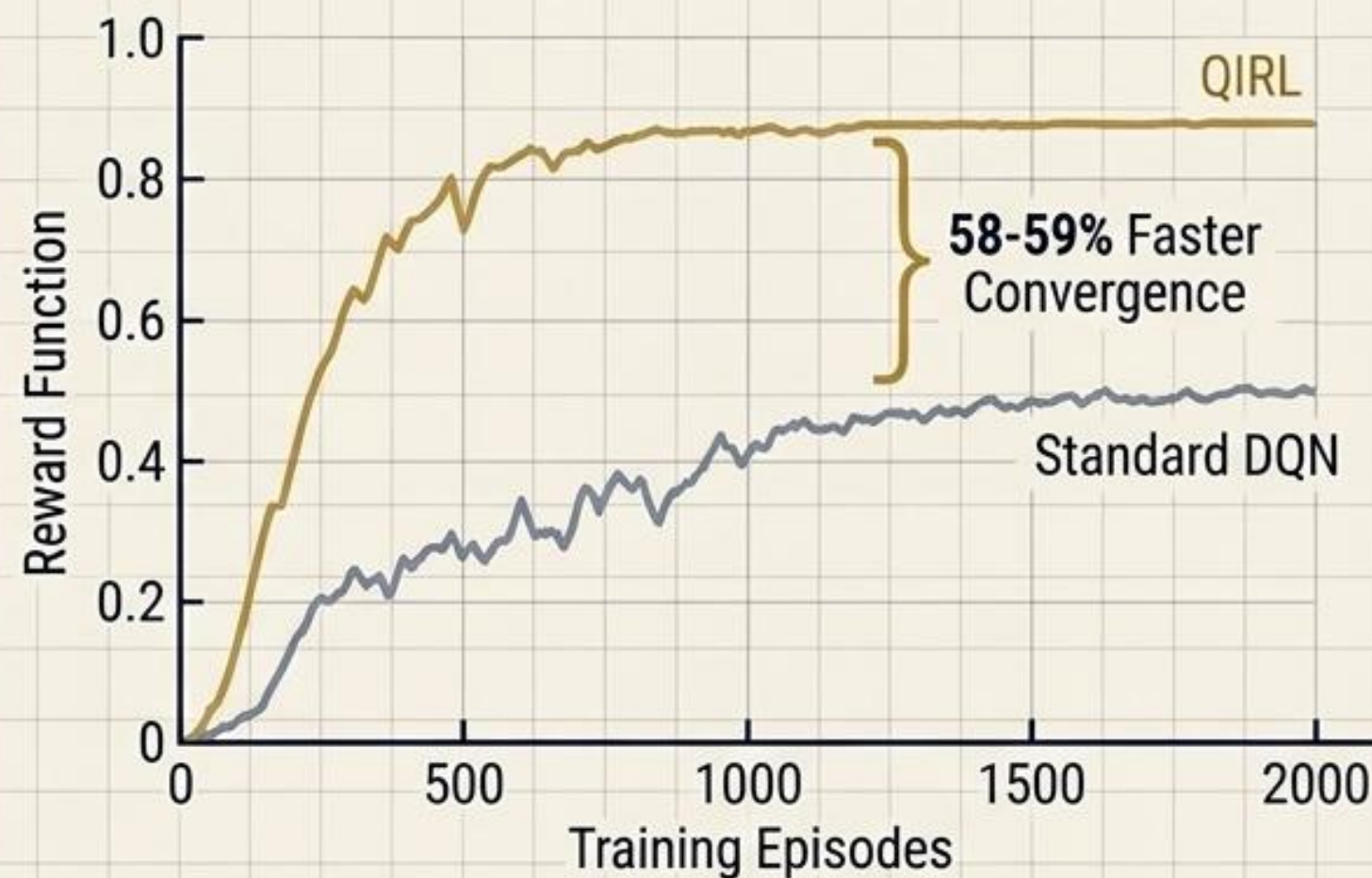


Quantum-Inspired Reinforcement Learning Tunnels Out of Local Optima

Conceptual Diagram

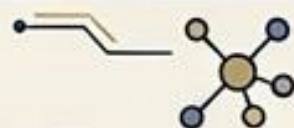


Convergence Speed Graph



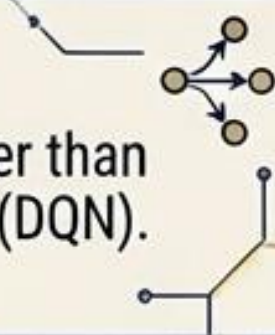
The Mechanism

Uses complex-valued probability amplitudes to evaluate multiple states simultaneously (superposition).



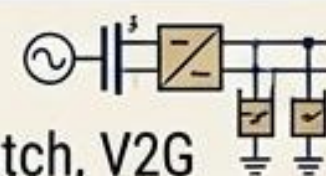
The Result

Converges 58% to 59% faster than standard Deep Q-Networks (DQN).

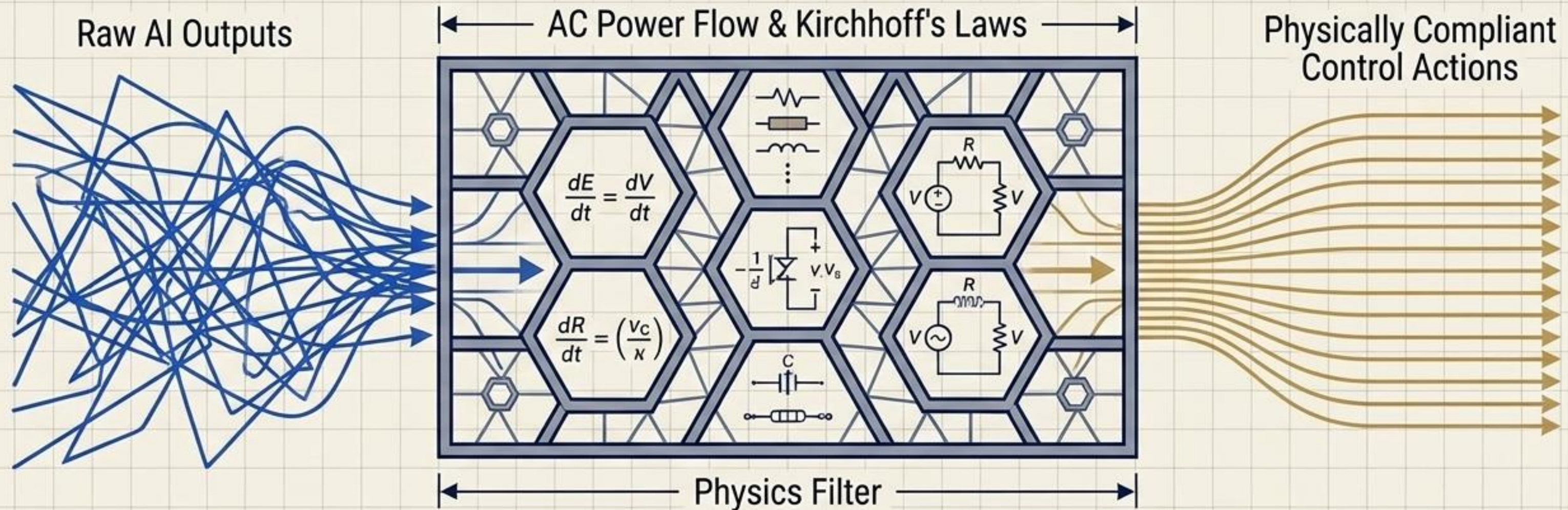


The Application

Handles economic dispatch, V2G scheduling, and coordinated FACTS control in 50ms intervals.



Physics-Informed Neural Networks Guarantee Grid Safety



Embedded Physics

Physical constraints are mathematically embedded directly into the training loss function.

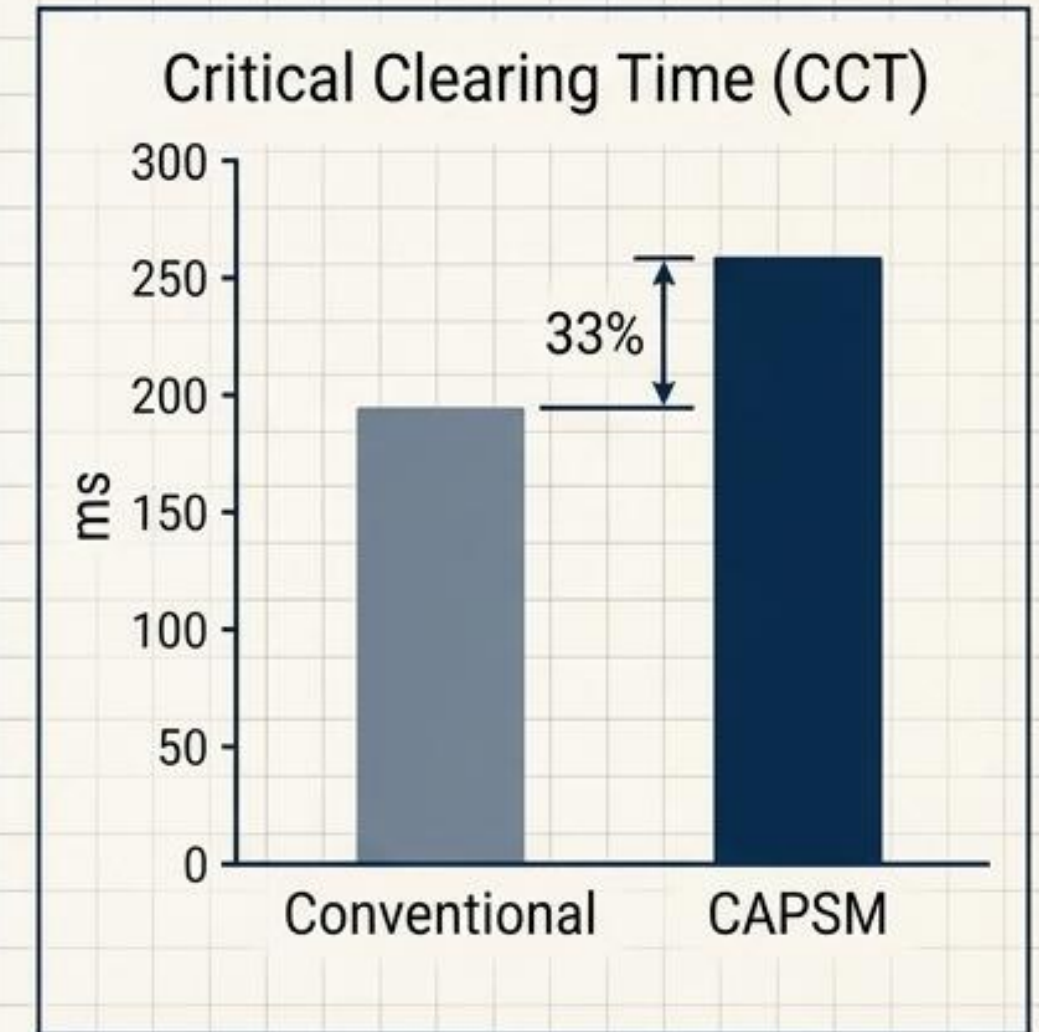
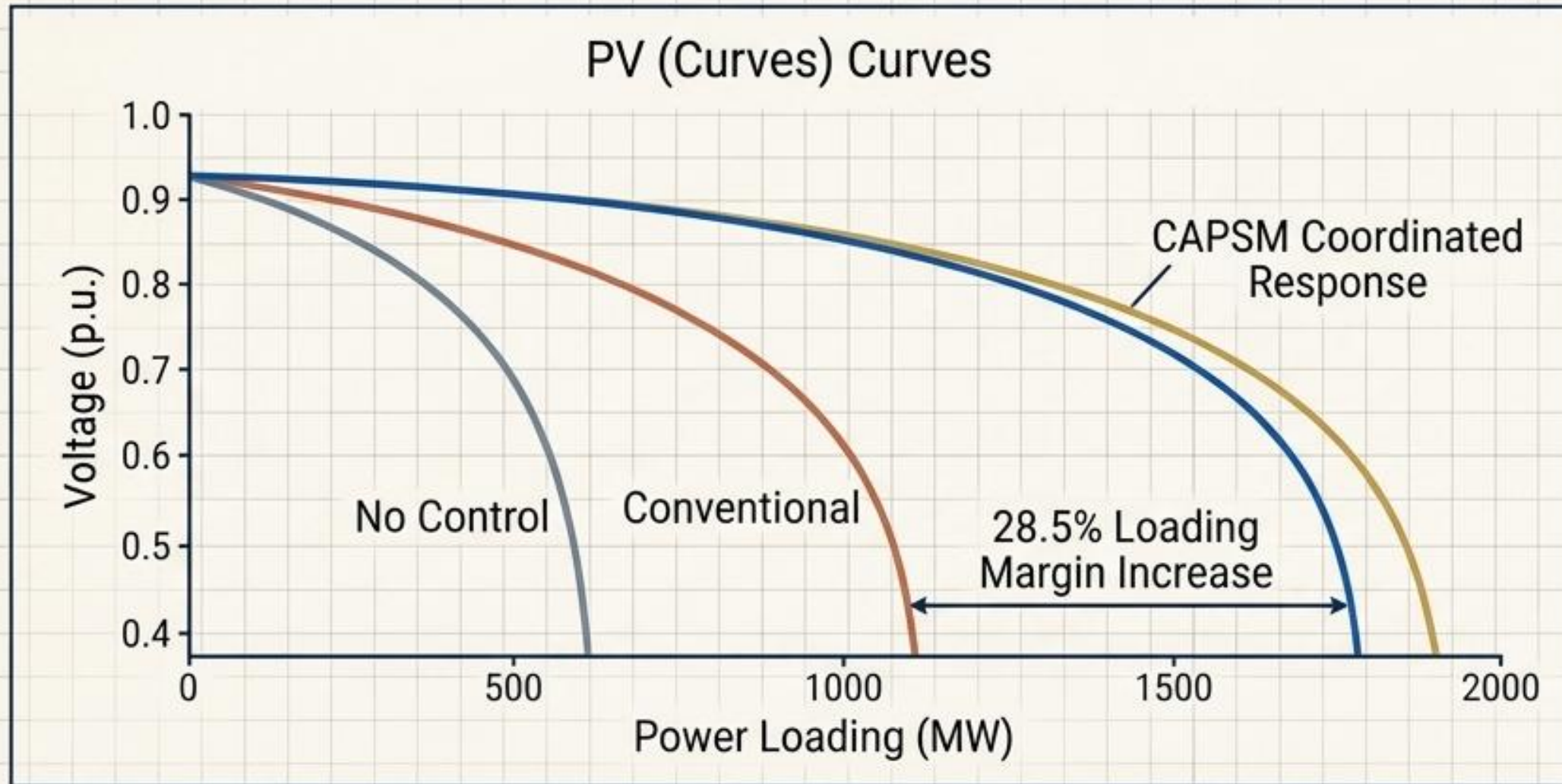
Constraint Compliance

99.97% physics constraint satisfaction rate achieved.

Violation Reduction

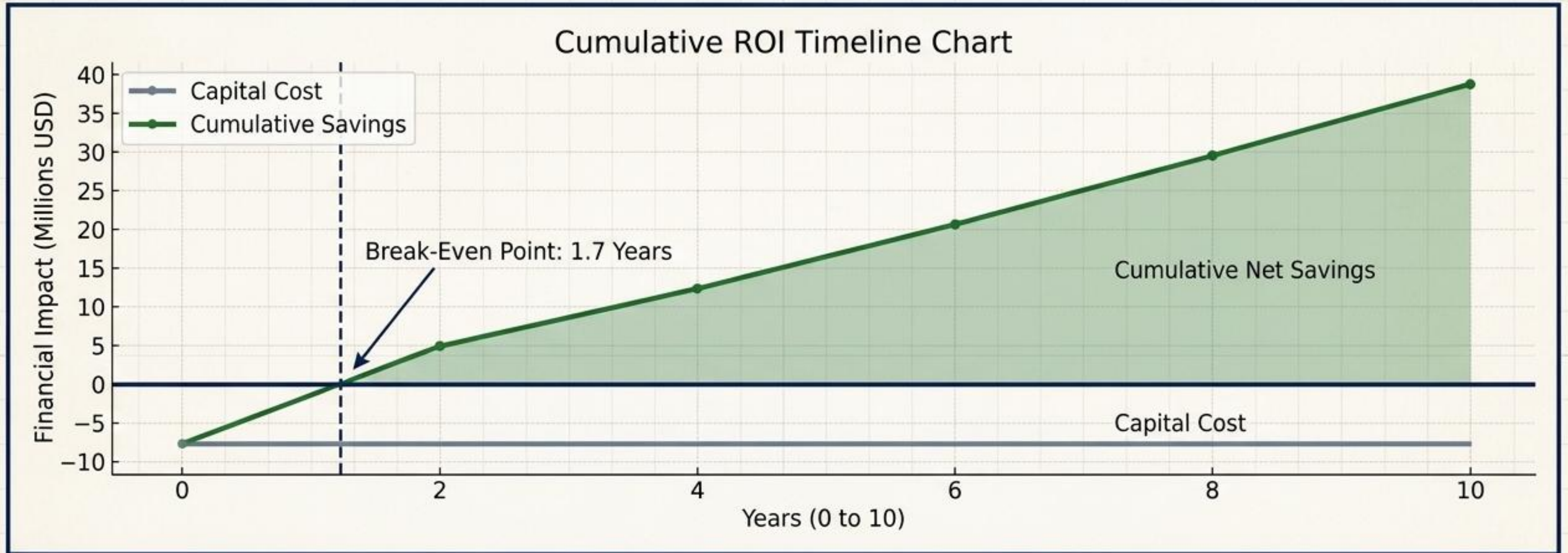
Erroneous control actions dropped from 3,130 to just **11** in 10,000 operational cycles.

Coordinated Control Radically Expands Voltage Stability Margins



- Voltage Stability Coordinated response yields a 42% better overall voltage stability margin.
- Loading Margin 28.5% increase in loading capacity before collapse.
- Transient Stability 20-33% improvement in Critical Clearing Time during severe grid faults.

Bidirectional EV Integration Drives Immediate Economic Returns



V2G Coordination

Synchronizes Vehicle-to-Grid charging to shave peaks and provide frequency regulation.

Operational Savings

Drives a 12.4% total operational cost reduction.

Financial Metrics

Achieves a 1.7-year break-even ROI and an 83.7% Internal Rate of Return (IRR).

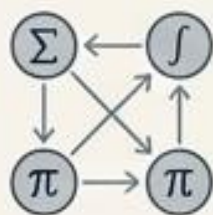
CAPSM Outperforms All Conventional and State-of-the-Art Methods

	Response Time	Adaptability to Uncertainty	Physics Constraint Compliance	Real-Time HIL Readiness
Conventional SCADA				
Rule-Based Expert Systems				
Standard Deep RL (DRL)				
CAPSM Framework				

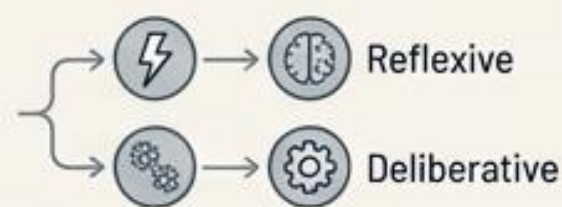
Takeaway: CAPSM is the only solution that bridges theoretical AI performance with fully verified HIL real-time physical constraints.

Eight Core Contributions Advance the Field of Cognitive Grids

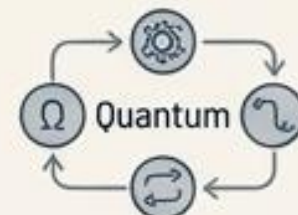
C1: Unified Math Framework



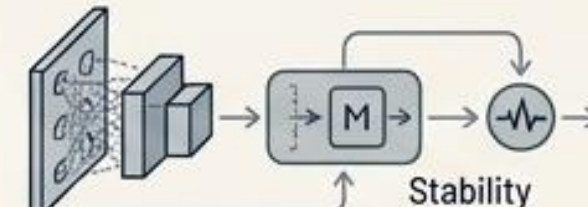
C2: Dual-Process Architecture



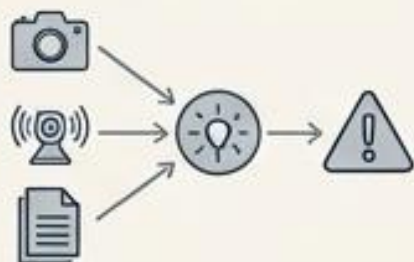
C3: Quantum-Inspired RL (QIRL)



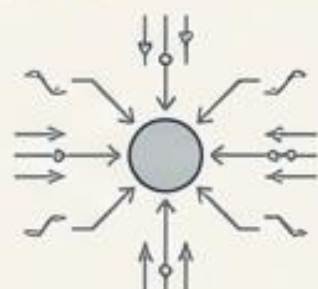
C4: CNN-LSTM Stability Model



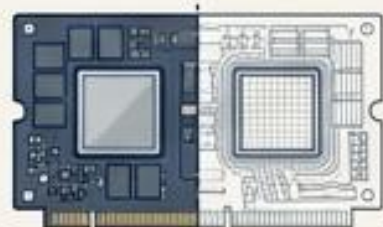
C5: Multi-Modal Fault Detection



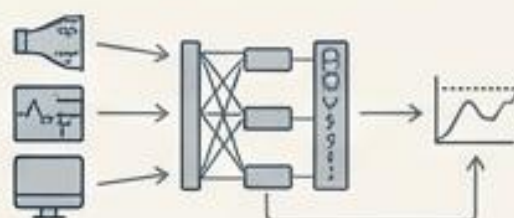
C6: Coordinated Control



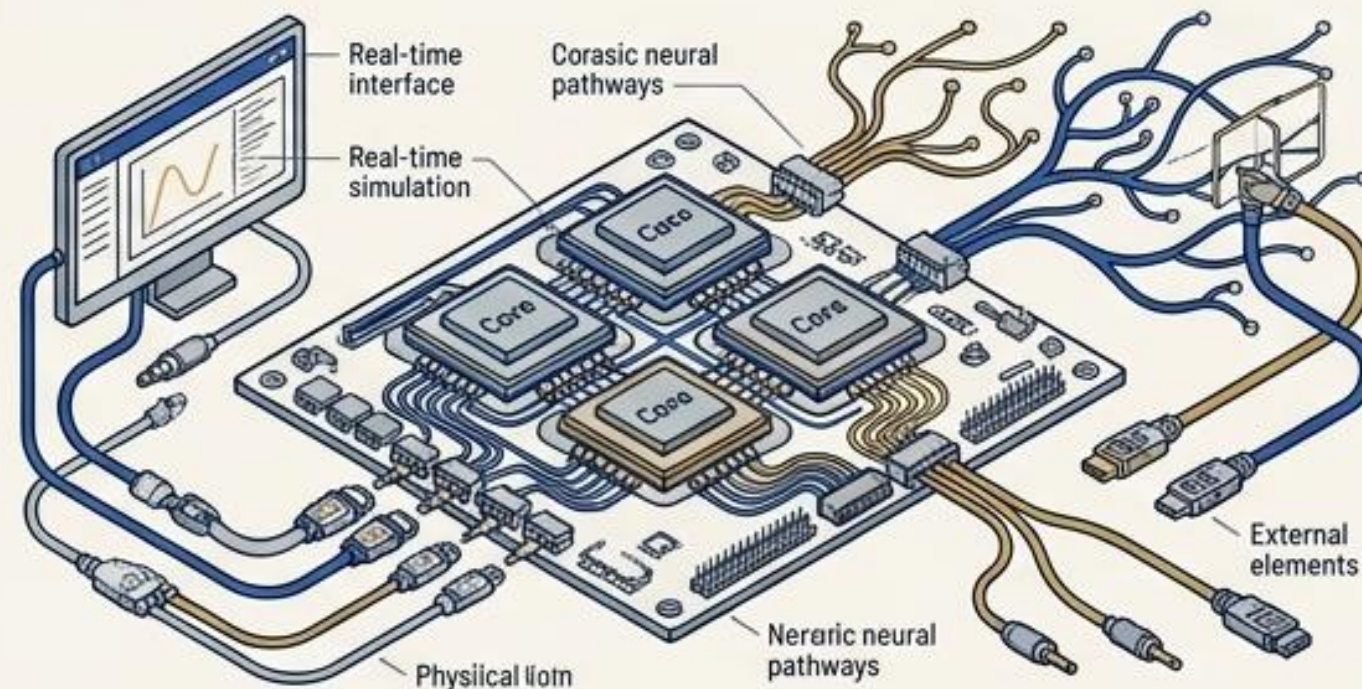
C7: Digital Twin Methodology



C4: CNN-LSTM Stability Model



C8: Hardware-in-the-Loop Validation on 4-Core OPAL-RT



The critical bridge from simulation to practical implementation, validating real-time constraint compliance and sub-10ms response times.

Dissemination

Journals

Conferences

Open Source Data

Transparent Acknowledgment of Current Limitations

Known Constraints

Physical & HIL Constraints

- Current OPAL-RT validation is limited to selected components.
- EV fleet models require further unaggregated physical testing to validate complex localized thermal dynamics.

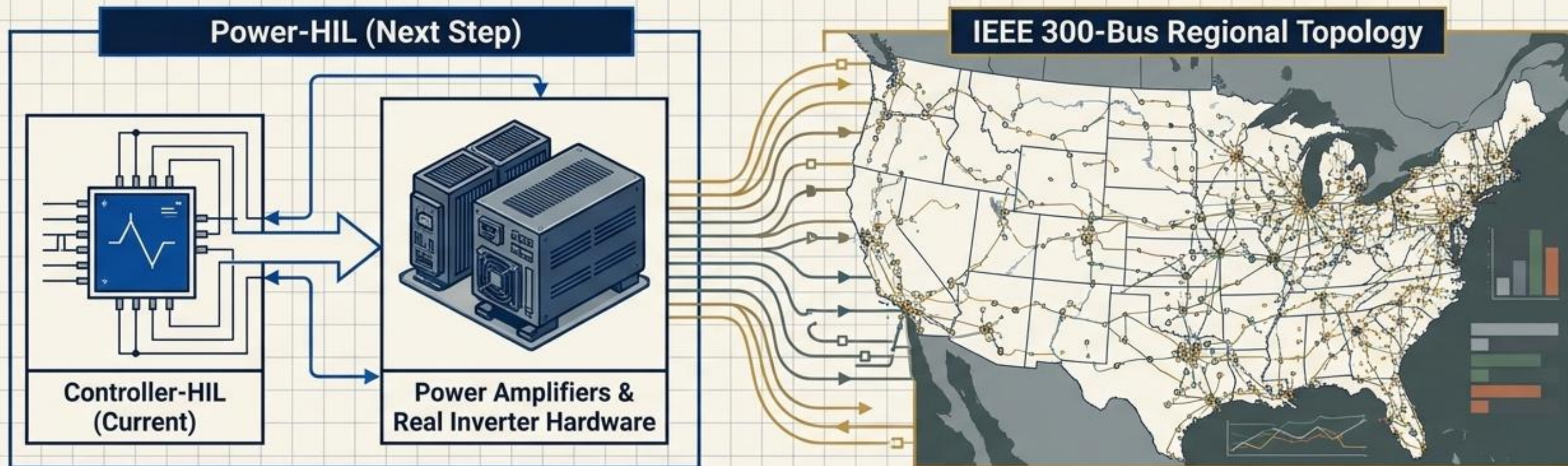
Computational Bottlenecks

- Scaling ePHASORSIM integration for 1000+ node EMT constraints remains processor-bound on current 4-core hardware architectures.

Regulatory Barriers

- Autonomous AI dispatch without human-in-the-loop oversight faces strict, current grid code and liability barriers.
- Certification pathways for non-deterministic AI controllers do not yet exist.

The Horizon Extends to Power-HIL and Regional Grid Deployment



Immediate Next Step

- Transition from testing control algorithms to testing closed-loop physical inverter interactions.

Scaling Up

- Bringing the IEEE 300-bus system into real-time OPAL-RT simulation.

Industry Roadmap

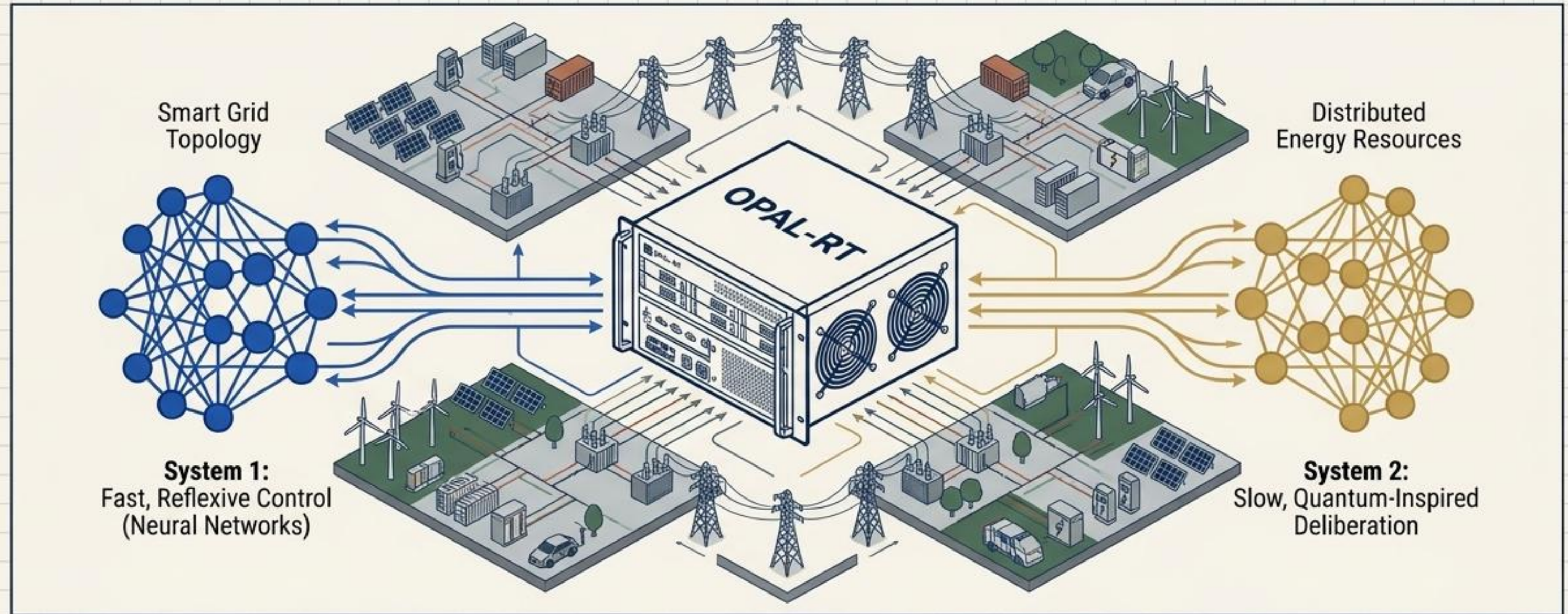
Phase 1:
Advisory Mode

Phase 2:
Limited Closed-Loop

Phase 3:
Integrated Control

Phase 4:
Advanced Autonomous

A Validated Blueprint for Autonomous Energy Communities



Timeline of Control: Shifting from Centralized/Deterministic to Distributed/Cognitive

CAPSM successfully bridges the theoretical AI-to-practice gap.

By combining the speed of reflexive neural networks with the depth of quantum-inspired deliberation, we provide a fast, safe, and economically viable brain for the modern renewable grid.

Final Thesis Statement:
The journey toward truly intelligent power systems—validated on real hardware—has officially begun.